Sharing the caring? Dynamic interaction between siblings in the provision of care to parents

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Abstract

Adult daughters provide nearly three times as many hours of care to elderly parents as adult sons do. I analyse the role of strategic interaction between siblings in exacerbating this gap in care provision. To do so, I build and estimate a dynamic discrete-choice game in which siblings make location, work and care choices. I find that the opportunity for strategic play increases gender differences in caring responsibilities: sons in particular strategically shirk providing care, as they believe their sibling is relatively likely to provide care in their absence. Counterfactual experiments show that if strategic interaction were eliminated by each child instead taking their sibling's care provision as fixed at the population average, the gender care gap in son-daughter families would be 18-20% smaller. Unobserved preference differences between sons and daughters are far more important in driving the gender care gap than differences in earnings, suggesting that policies to close the gender earnings gap may make little difference to the gender care gap. Cooperation between siblings would raise care provision but widen the gender care gap in absolute terms, as daughters would take on more of the additional care.

JEL classifications: D64, J14, J16

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1 Introduction

Children are a key source of care for their elderly parents: in the US, around 17% of all adult children will provide care to a parent at some point in their lives and conditional on providing positive care hours an adult child will provide on average 77 hours of care per month (Wettstein and Zulkarnain 2017). Studies have shown the adverse effects of caregiving on labour force attachment (Van Houtven, Coe, and Skira 2013) and mental and physical health (Bom et al. 2019), and there is little evidence of caregivers receiving direct contemporaneous cash transfers in exchange for their care (Barczyk and Kredler 2018; Groneck 2017). It is generally daughters and not sons who select into caregiving (Grigoryeva 2017), with my calculations in this paper showing daughters providing around 3 times the care hours of sons even though all children plausibly benefit from the public good of having their parent cared for. Why do we observe such a gender gap in selecting into this role?

The existing literature has tried to analyse the issue of children providing care to elderly parent(s) either from the perspective of an individual adult child (Ko 2022; Mommaerts 2024; Skira 2015) or with multiple adult children in a static setting (Byrne et al. 2009; Fontaine, Gramain, and Wittwer 2009), imposing different preferences or costs of caring for different types of children to match gaps in care provision. While this literature has yielded many rich insights, it has not accounted for the fact that care provision by children is fundamentally a problem of dynamic interaction. Modelling dynamics is crucial because children's decisions in response to parental health shocks will affect their ability to provide care in the future; modelling interactions is crucial because children make these decisions with the knowledge that their siblings are making similar calculations and because each child's decision affects the optimal behaviour of all their siblings. This creates a space for strategic play over time between siblings to amplify gaps arising from any difference in preferences or costs of caring.

A simple example will illustrate. Suppose Adam and Beth are siblings. Suppose there is something about Beth's preferences or costs for providing care that make her more likely to provide care (she has a lower opportunity cost of providing care, she finds it less burdensome to provide care, their common parent prefers Beth's care, and so on). If Adam and Beth made decisions independently, without regard for what the other was doing, then Beth would on average provide more care because of these differences in preferences and costs. However, if Adam knows Beth is likely to provide care, then other things being equal he will reduce his care effort; Beth, knowing this, will increase her care effort to compensate, meaning Adam could cut his care effort more, and so on. Thus, the possibility for strategic interaction can amplify existing gender care gaps. This amplification could be particularly important in dynamic games where agents' costs and preferences of providing care are not fixed exogenously but are instead endogenously

generated by past decisions: Adam will be even less likely to provide care today if he sees today's care decision as locking him in to the caring role in future.

In this paper I explore this type of interaction by situating the decisions of adult children over the care provided to their elderly parents in a rich dynamic model, recognising a) that children may have different preferences over their parent's care arrangements and b) that children can take strategic actions to make it easier for them to provide (or avoid the burden of providing) care in future. In doing so, I make two contributions to the literature on informal care provision within families. Firstly, I present the first estimates of the importance of strategic interaction in driving the gender care gap: in families with a son and a daughter, the gender care gap would be 18-20% smaller, depending on specification, if there were no strategic interaction between siblings. Secondly, I provide new insight into the drivers of the gender care gap: differences in care provision between sons and daughters are driven much more by differences in preferences than by differences in earnings.

These findings have important policy implications. Given that differences in income processes explain only around a quarter of the gender care gap, policies to reduce the gender earnings gap in isolation are unlikely to change the division of care between sons and daughters by much. However, given the role of strategic interaction, policies which do manage to substantively change agents' incentives to provide care could significantly change the level and distribution of overall care provision as other agents strategically alter their caregiving in response.

The paper proceeds as follows. To motivate my structural approach, I first use the US Health and Retirement Study to document some stylised facts about the provision of care for elderly parents by their adult children. In keeping with previous results in the literature, I confirm that it is generally daughters who end up providing the care, and I exploit variation in the gender composition of sibling groups to offer some prima facie evidence that children's caregiving is not independent of the caregiving of their siblings, consistent with care provided by sons being crowded out by care provided by daughters. I also examine the compensation of child caregivers by parents. As in Groneck (2017), I find that caregiving children receive larger bequests than their non-caregiving siblings on average, but I also show that this average is driven by the children of the wealthiest parents, and that children in the bottom 90% of the parental-estate distribution can expect far less in the way of bequests. The typical caregiver can expect little compensation, either in the form of bequests or non-cash inter vivos transfers, which raises the question of what drives these children into the caregiving role.

I then use these stylised facts to motivate a dynamic model of interaction between two adult siblings. Each period, each sibling chooses whether to live close to their parent, whether to work full-time, part-time or not at all, and, if their common parent needs care, whether to provide no care, moderate care or intense care. Sons and daughters differ in

their preferences for providing care and face different income processes. Children find it costly to provide care themselves but each child receives a benefit if any child provides care, meaning that the model resembles a public good provision problem. Care needs that the children do not meet are covered by formal care, paid for out of the parent's wealth until it falls to the Medicaid asset limit. When the parent dies, the children inherit the remaining wealth, with the split of the bequest depending on the model specification. Otherwise, children face no financial incentive to provide care and cannot transfer money between themselves.

In keeping with the finding in Hiedemann, Sovinsky, and Stern (2018) about the importance of state dependence in caring arrangements, the model incorporates direct or implicit transition costs in changing one's location, labour force participation or caregiving, which in turn imply transition costs associated with changing caring arrangements. In particular, the children suffer a wage penalty if they have been absent from the labour market and they suffer a utility penalty from changing location or starting to provide care. These transition costs are the key feature distinguishing this dynamic model from static models of care provision because care, location and work decisions today have a bearing on the optimal decisions in the next period.

I estimate the model by maximising the pseudo log likelihood of the observed decisions (Aguirregabiria and Mira 2007). In particular, to make the estimation tractable, I use forward simulation to construct agents' value functions using the process set out in Bajari, Benkard, and Levin (2007) and employed in Ko (2022). Key estimated parameters include preference differences for providing care between sons and daughters and parameters capturing the extent to which adult children benefit from their siblings' care provision. I show that the model is able to match key patterns in the data, including for one-child families not included in the estimation sample.

I use the estimated model to conduct counterfactual experiments to quantify the drivers of the gender care gap: I examine the relative contributions of income and preferences to this gap, and I assess the role of strategic interaction in exacerbating it. These exercises shed light on what we might expect to happen to the gender care gap in future given broader trends such as a shrinking gender earnings gap. I find that preferences are far more important than income in driving the gender care gap: giving sons and daughters identical preferences, while holding the overall care rate fixed, removes 86-92% of the gap, whereas giving them identical income processes removes only around a quarter. I also find that if each child instead took their sibling's care provision as fixed at its population average, so that there were no strategic interaction, the gender care gap in son-daughter families would be 18-20% smaller.

I also assess the inefficiencies associated with non-cooperation, by re-solving the model under the assumption that both children act to maximise the sum of their utilities. Under cooperation, both sons and daughters provide more care, and the share of periods in which

a sick parent receives care from neither child falls from 45-46% to 31-32%, depending on the specification. Daughters, who find caregiving less costly, take on more of the additional care, so the gender care gap widens in absolute terms, and sons gain more from cooperation than daughters. For children whose parent has severe care needs at age 81, the median age for developing care needs for the first time in the data, the gain from cooperation, split equally between the siblings, is equivalent to a permanent increase in consumption of 0.28-0.50%, depending on the bequest specification. Splitting the gains equally in son-daughter families requires a transfer from sons to daughters equivalent to 0.13-0.18% of permanent consumption.

Finally, I assess a directly policy-relevant counterfactual in which informal carers are paid the hourly wage of a home health aide. This substantially increases informal care, raising daughters' care rate by around 10-12 percentage points (pp) and sons' by around 8-10pp, so that the gender care gap widens in absolute terms but narrows in percentage terms. The subsidy also deepens the fall in full-time work when a parent first needs care, and mainly draws into caregiving children with lower incomes and the children of poorer parents.

This paper contributes to the literature on informal care arrangements within families (Byrne et al. 2009; Checkovich and Stern 2002; Fontaine, Gramain, and Wittwer 2009; Hiedemann, Sovinsky, and Stern 2018; Mommaerts 2024; Stern 2023), and in particular to the literature on dynamic strategic interaction over care provision within a family (Barczyk and Kredler 2018; Ko 2022).[1] Previous papers on dynamic strategic interaction have focussed on interaction between a parent and child, rather than on strategic interaction between children; instead, to the best of my knowledge this has only been studied in a static setting (e.g. Byrne et al. (2009)), or with strong assumptions about identical preferences between children such that the family can be treated as a unit (e.g. Hiedemann, Sovinsky, and Stern (2018)). By modelling interaction between children in a rich dynamic setting I am able to capture the full importance of strategic play in driving the unequal distribution of the care burden.

Another important methodological contribution of this paper is the modelling of location choice along with caring and labour choices in a dynamic model of care provision. Previous papers have treated child location as fixed (e.g. Ko (2022)) or have modelled the location decision of adult children as a one-shot decision taken early on in life (e.g. Stern (2023)). In practice, adult children can change their location in response to parental health shocks by moving back close to their parents in order to provide care and there is some evidence that they do so (e.g. Hiedemann, Sovinsky, and Stern (2018)). It is important to allow for this channel of behaviour to properly model a child's ability to alter their own cost of providing care and thus their ability to strategically shirk caregiving[2].

[1] Sovinsky and Stern (2016) review the literature on dynamic models of caregiving within families.

[2] For instance, suppose Adam lives close to his mother but his sister Beth lives far away, and that

This paper is closest in approach to Ko (2022), which models dynamic interaction between a parent and child and uses this to explain low long-term care insurance (LTCI) take-up. I use many of the same modelling assumptions. For instance, I use a similar dynamic framework of non-cooperative interaction over caregiving decisions, combined with a labour participation decision on the part of the child. As suggested above, the key differences are that I model interaction between children, rather than between a parent and a child, in order to work out what drives differences in provision between children, and that I allow children's location to change endogenously. On the other hand, to make the model tractable, I abstract from modelling LTCI, and in particular the interaction between LTCI demand and the availability of informal care, a key focus of Ko (2022).

The inclusion of the labour force decision for children, and the emphasis on the costs associated with leaving the labour force to provide care, links this paper to Skira (2015), which presents a dynamic model of women's labour force participation and caregiving for an elderly parent. This paper expands on that paper by modelling the interaction between multiple children and allowing children choice along more dimensions, though this extra complexity comes at the cost of a simpler labour market and wage model than in Skira (2015).

The paper also draws from Hiedemann, Sovinsky, and Stern (2018), which similarly provides dynamic models of care provision within a family and finds important results about the state-dependence of care provision over time. In Hiedemann, Sovinsky, and Stern (2018), because the focus is on a sophisticated treatment of unobserved heterogeneity versus state dependence in a dynamic setting, the authors abstract from within-family strategic interaction. Instead, family members are modelled as having the same preferences over care outcomes, so the family can be treated as a unit. The key contribution of the current paper relative to Hiedemann, Sovinsky, and Stern (2018) is that children in the model are separate decision-makers with their own preferences and interact non-cooperatively, allowing me to quantify the importance of strategic interaction over time.

Finally, with its emphasis on endogenous location choice and strategic shirking of a caregiving role the paper builds on work by Konrad et al. (2002), Maruyama and Johar (2017), Rainer and Siedler (2009), and Stern (2023). These papers all examine one-shot location decisions by adult children early in life seeking to avoid the burden of providing care to elderly parents in future[3]. The contribution of this paper relative to these others is that it allows for adult children to change location multiple times over their lives, meaning

their mother suddenly falls ill. If Adam and Beth cannot change location, then both Adam and Beth know that the probability of Beth providing care is very low, so Adam is effectively making decisions in isolation over whether to care for his mother. If, on the other hand, Beth is able to move back, or Adam is able to move away, then Adam will see the probability of Beth providing care as higher, so might be more tempted to shirk in the hope that Beth moves back and provides care.

[3]In Konrad et al. (2002), children make a one-shot location decision early in life and then parents can make a decision when they are elderly and need care to live close to a particular child. As discussed in Appendix B, in my data it is mainly children not parents who relocate when the parents need care.

they can respond to parental health shocks. As outlined above, it is important to model this channel of behaviour in order to better capture the scope for strategic interaction between siblings over caring.

The rest of this paper is organised as follows. In Section 2 I set out some descriptive evidence about caregiving by adult children of elderly parents. In Section 3 I write down a dynamic model of interaction between siblings in the provision of care. Section 4 discusses data and estimation, providing intuitive arguments for identification and setting out results and model fit. Section 5 shows the results of counterfactual experiments and Section 6 concludes.

2 Descriptive evidence

I use data from the Health and Retirement Study (HRS) between 1998 and 2018[4]. The HRS asks respondents a battery of questions about their health, financial situation and relationships. A key advantage of the HRS, relative to other surveys of the older population in different countries, is its extensive use of proxies and following respondents into nursing homes, meaning that it is more successful at mitigating sample attrition for respondents who develop care needs (Sonnega and Weir 2014; Weir, Faul, and Langa 2011).

In addition, HRS respondents are asked to provide information on any children they have. Notably, the HRS records for each child their income (in brackets), whether they are working, their education, their marital status[5], whether they live within 10 miles of the parent and how much care they provide to the parent. This allows me to link each parent with their possible child caregivers.

For calculating descriptive statistics and establishing general empirical patterns I use the full sample of respondents to these waves of the HRS who have at least one child record associated with them. This comprises 35,105 unique respondents over the 11 biennial waves, with 200,385 respondent-wave observations, meaning that each respondent appears for an average of 5.7 waves. In Appendix A.1 I show some descriptive statistics on the HRS sample.

In the structural model of this paper, I focus on two-child families of single parents to be able to address strategic interaction between children in the simplest possible setting, and I then validate the model by comparing its predictions for one-child families to the data. While this means that the model is estimated only on a subset of all families, relatively many HRS respondents have only one or two children: Figure 17 in Appendix

[4]I use the RAND HRS Longitudinal File and the RAND HRS Family Data File and combine these with data from the RAND HRS Fat Files for each of the relevant waves (*HRS* 2024; *RAND* 2024).

[5]Throughout this paper, I will use coupled/married interchangeably to refer to all children who are classed as partnered, married or other by their parents.

A.1 shows that conditional on having a child record associated with them, 12% of HRS respondents report having one child and 32% report having two children (the modal category), meaning that 44% of respondents with at least one child would be included in either the model estimation or validation. The corresponding figures for single respondents only are 17% and 31%. Table 9 in Appendix A.1 sets out differences in key demographic variables in the single-parent, two-child subsample, suggesting that there are no major differences relative to the full sample apart from the fact that women are overrepresented and respondents are less likely to own a home.

I now use the HRS sample to present some key facts about the provision of care: namely, that there is a gender gap in the provision of care, the size of this gender gap depends on family composition, and most children do not receive substantial compensation for providing care.

2.1 There is a gender gap in the provision of care

Children are an important source of care to parents. Defining a parent as having care needs if they report difficulty with any Activity of Daily Living (ADL) or Instrumental Activity of Daily Living (IADL)[6], in the HRS sample I consider the average parent with care needs receives 17 hours of care per month from any daughters they have and 6 hours of care per month from any sons they have, together amounting to around a third of the 77 care hours per month that a parent with care needs receives from all sources[7]. For parents without a spouse, these figures are 27 hours and 10 hours respectively, making up 51% out of a total of 72 hours. In both cases, therefore, daughters provide nearly 3 times the care hours of sons. Appendix A.2 discusses in more detail the breakdown of care receipt by source.

To identify some predictors of provision of care I use the child records associated with each respondent in the HRS. I run an OLS regression of a dummy for whether a given child provided any care to a parent with care needs on a set of child and parent demographics[8]. I also regress the log of hours of care provided on the same set of explanatory variables for the subsample of children who provided positive hours of care. These regressions are carried out for the full sample of HRS respondents with children, and are displayed in Table 1 below.

[6]ADLs are very basic activities required for independent life such as basic mobility or dressing oneself. IADLs are more complex activities like preparing food or managing money. See Cleveland Clinic (2024) for more details.

[7]As shown in Appendix A.2, the average parent with care needs receives 17 hours of care from daughters, 6 hours from sons, 33 hours from a spouse, 10 hours from other family members and 12 hours from non-family sources (not including employees of institutions).

[8]Note that if two members of a couple respond to the HRS, then any children they have will appear twice in the regressions below, once for each parent. In other words, the observations in this regression are unique parent-child links, rather than unique children.

Table 1: Predictors of care provision and the amount of care provided

	Dependent variable:		
	I(Hours help p.m. > 0)		log(Hours help p.m)
	(1)	(2)	(3)
Kid is daughter	0.009*	0.009*	−0.069
	(0.004)	(0.004)	(0.084)
Parent is mother	0.002		−0.030
	(0.003)		(0.068)
Daughter x Mother	0.050***	0.051***	0.227**
	(0.004)	(0.004)	(0.081)
Kid is eldest	0.001	0.002	−0.003
	(0.002)	(0.003)	(0.047)
Kid is youngest	0.008**	0.005	0.000
	(0.003)	(0.003)	(0.046)
Kid lives less than 10 miles away	0.115***	0.127***	0.508***
	(0.003)	(0.003)	(0.038)
Kid has sister	−0.013***		−0.082
	(0.003)		(0.046)
Parent in couple	−0.042***		−0.108*
	(0.003)		(0.045)
Kid in couple	−0.027***	−0.029***	−0.620***
	(0.003)	(0.003)	(0.062)
Daughter x Kid in couple	0.011*	0.015**	0.330***
	(0.004)	(0.005)	(0.075)
Kid went to college	0.009***	0.014***	−0.177***
	(0.002)	(0.003)	(0.039)
No. of children of parent	−0.005***		−0.021
	(0.001)		(0.012)
Kid cared in prev. period	0.344***	0.364***	0.435***
	(0.008)	(0.010)	(0.038)
Observations	118 900	113 694	14 022
Adjusted R2	0.238	0.385	0.134
Parent x wave FEs	N	Y	N
Mean dep. var.	0.101	0.101	2.974

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: estimation via OLS. Only children of sick parents (ADL or IADL >0) are included. Regression weights are HRS respondent-level weights. In all regressions, other controls are dummies for number of ADL and IADL difficulties, education of parent, total number of grandkids of the kid, quadratics in age for kid and parent, a dummy for parent being White Caucasian, the inverse hyperbolic sine of parent wealth and homeownership status of kid. For the fixed effects regression, the parent-level controls drop out. Standard errors clustered at the parent level.

In the Table, Columns 1 and 2 capture the extensive margin of care provision. In each case I regress a dummy for care provision on a set of parent and child characteristics. The difference between Columns 1 and 2 is that in Column 2 I include fixed effects for parent interacted with wave. As such, I consider the role of each child's characteristics relative to his or her siblings in a given wave in determining whether that child provides care to a particular parent[9].

Several important empirical patterns emerge. Note that as I include dummies for the kid being a daughter, the parent being a mother and the kid being in a couple, as well as interactions of the daughter dummy with the other two, the omitted category is single sons caring for fathers. The coefficient on "Parent is mother" is very close to zero, suggesting that sons are just as likely to provide care to mothers as they are to fathers. The dummy for "Kid is daughter" is small but positive and significant, so single daughters provide slightly more care than single sons to fathers, and the interaction between "Kid is daughter" and "Parent is mother" is large and significant, so daughters provide much more care than sons to mothers[10].

Youngest children are slightly more likely to provide care than either eldest children or children who are neither youngest nor eldest, though this is much less of a predictor of care provision than the child being a daughter.

Kids are unsurprisingly much more likely to provide care if they live close to their parent. However, it is unclear whether these are children who were living within 10 miles of their parent before the health shock occurred (e.g. they never left the local area) or whether these are children who move back to provide care[11]. If the latter, then this suggests the welfare costs of providing care are greater, because they involve the cost in time and money of relocating, possibly changing jobs, and so on. Also, children moving back to provide care would point to a greater role for strategic interaction because children have more control over whether they are in a position to provide care. I examine this question in Appendix B, considering kids' location choices around their parent falling sick, exploiting the panel nature of the data to conduct an event study. The results offer

[9]Because the fixed effects capture any variation common to all children of a particular parent in a given wave - e.g. the gender or marital status of the parent - these explanatory variables drop out of Column 2. I also drop "Kid has sister" as a regressor for the fixed effect regression because I am already controlling for a given child's gender relative to that of their siblings through a combination of controlling for the child's gender and the parent $\times$ wave fixed effect.

[10]This interaction between parent and child gender suggests that at least part of the aggregate care gap is driven by the fact that women are overrepresented among the elderly needing care because fathers tend to die younger than mothers. Indeed, in the estimation sample for Table 1, 60.3% of parents are mothers, and 66.8% of children are in a couple. As a rough calculation, the results in Table 1 suggest that if instead there was a 50:50 split of mothers and fathers in the sample then, other things being equal, daughters would be 4.1pp more likely to provide care in the aggregate rather than 4.7pp, after controlling for all other regressors. This is because $0.009 + 0.6026 \times 0.050 + 0.668 \times 0.011 \approx 0.047$ but $0.009 + 0.5 \times 0.050 + 0.668 \times 0.011 \approx 0.041$.

[11]Hiedemann, Sovinsky, and Stern (2018) consider the potential endogeneity of location, finding some evidence that location is endogenous, although the authors argue that any ensuing bias will be small in magnitude.

some evidence that daughters in particular do move closer to parents following a parental health shock: on average, daughters are 1.0pp more likely to live within 10 miles of their parent in the (two-year) period where the parent is first observed with care needs relative to the period before the parent develops care needs, compared to a 0.6pp increase for sons; overall, 38.9% of daughters and 38.2% of sons live within 10 miles of their parent.

Kids are less likely to provide care if they have at least one sister[12], a phenomenon to my knowledge first noted by Grigoryeva (2017). The fact that children provide less care if they have a sister hints at the role of strategic interaction between siblings, discussed in more detail in the next section.

Both the parent having a spouse and the kid having a spouse are negatively associated with the probability of the kid providing care. The former is presumably due in part to there being a better "outside option" of care receipt for parents who have spouses. The negative association with the kid having a spouse is stronger for sons than for daughters: being in a couple reduces a son's probability of providing care by 2.7pp, compared with 1.6pp for a daughter. This may be because partnered kids have more demands on their time and these demands may fall differently on sons and daughters.

More educated kids are more likely to provide care. Interestingly, this result holds even in Column 2, i.e. even controlling for unobserved heterogeneity at the parent-wave level through a fixed effect. In other words, kids who are more educated than their siblings are more likely to provide care than their siblings. This is striking because this provides some prima facie evidence against the hypothesis that children with a lower opportunity cost of providing care in terms of foregone wages will provide more care. Instead, it seems that within a particular family it is the more educated, and thus likely the higher earning, child who ends up providing more care. This point will be revisited when discussing the results of the estimation of the structural model.

Column 3 captures the intensive margin: for those providing positive hours of care, I regress the log of hours of help provided per month on the same explanatory variables as the first column. I do not include FEs here because many parents have only one child providing positive care hours to them in a given wave.

Similar patterns emerge. Conditional on providing care, daughters provide more care than sons to mothers, though single daughters do not provide more hours of care than single sons to fathers. Kids provide much more care if they live close, and slightly less care if they have a sister. Conditional on providing care, having a spouse is associated with a large reduction in hours of care, particularly for sons: partnered sons provide around 46% fewer hours than single sons, and partnered daughters around 25% fewer hours than single daughters. This negative association is much stronger than the negative association

[12]Note that one of the controls in this regression, as set out in the note beneath Table 1, is the number of children, so this dummy does not just capture the effect of the kid in question having at least one sibling.

between the parent having a spouse and hours of care provided by the kid, and may point to the role of grandchildren in creating extra demands on children's time. Also, although more educated kids are more likely to provide positive hours of care, they provide less care conditional on positive provision than less educated kids.

In Table 10 in Appendix A.3, I present the equivalent of Table 1 limited to two-child, single parent families only, the focus of the structural model. The coefficients on the gender and couple terms have the same signs as in Table 1 but, given the much smaller sample, are imprecisely estimated; the main difference is that in this sample the parent being a mother is itself associated with significantly more care. Notably for this paper, the coefficient on the sibling being a sister is significant and in fact larger in absolute value than in Table 1.

2.2 The gender care gap depends on family composition

The fact that kids provide less care when they have at least one sister hints that they might reduce their care effort if they know a sister will "step up" in their absence. Here I consider this point in more detail.

In principle one could examine the role of strategic interaction between children by making use of data on only children, i.e. children without brothers or sisters. If the ratio of care hours by only daughters to only sons is less than the ratio of care hours of daughters to sons within multiple child families, then this suggests that agents change their behaviour depending on the presence of siblings. Table 11 in Appendix A.3 makes this comparison and finds that there is indeed a substantial difference.

The problem with using this approach to identify the role of strategic interaction in driving the gender care gap is that the number of children that a family has is likely endogenous. Parents have a significant degree of control over how many children they have, and thus the number of children they have will likely be correlated with observed and unobserved heterogeneity at the level of the parent.

Instead, I exploit variation in the gender composition of children, holding fixed the total number of children. In particular, I consider care provision by sons and daughters in two-child families. While gender composition could itself be endogenous[13] this is plausibly much less of a concern than for the number of children.

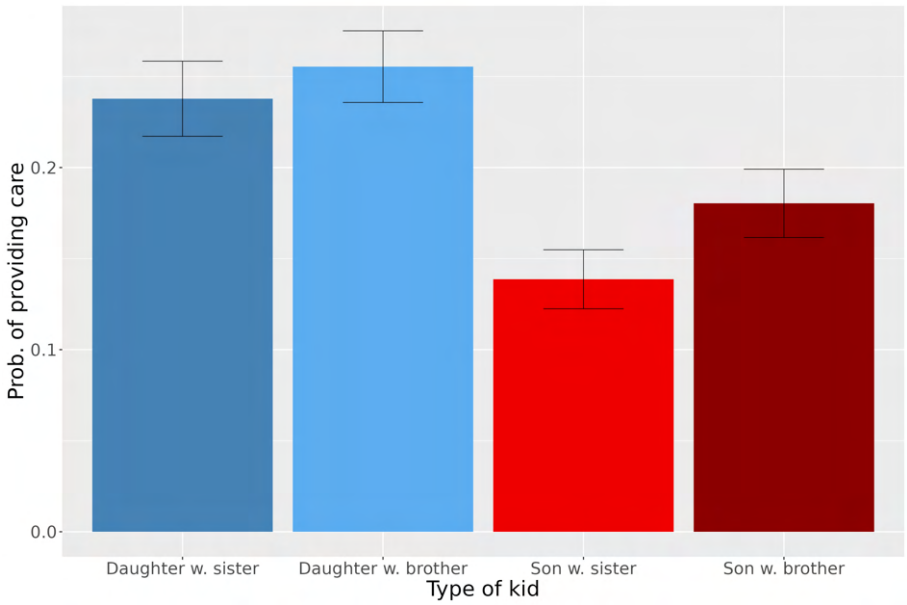
Figure 1 shows the probability of providing any care to a sick single[14] parent for

[13]For instance, if parents decide to keep having children until they have a daughter, then they are in some sense choosing the final gender composition of their children. Angrist and Evans (1998) use the sex mix of the first two children to predict whether a woman will have a third child, exploiting parents' preferences in general for having children of both sexes. Preferences for having children of both sexes - or alternatively preferences for having at least one daughter - might be plausibly correlated with e.g. expected care need in old age.

[14]I focus here on single parents because in married couples the partner tends to be the primary caregiver and because children of single parents will be the focus of my model.

different children in two-child families. The first bar is the probability of providing care by a daughter who has a sister, the second bar is the probability of providing care by a daughter with a brother, and so on.

Figure 1: Probability of providing care by type of kid



Notes: probabilities of providing care to a sick single parent for daughters whose only sibling is a sister, daughters whose only sibling is a brother, sons whose only sibling is a sister and sons whose only sibling is a brother.

Figure 1 suggests that daughters provide more care when their other sibling is a brother than when their other sibling is a sister; similarly, sons provide more care when their other sibling is a brother than when their other sibling is a sister. In other words, the graph suggests that relative to the case when their sibling is of the same gender as them, daughters increase their care and sons decrease their care when the sibling is of the opposite gender. An interpretation of this would be that sons shirk caregiving when they know their sisters will "step up" to fill the breach; conversely, daughters put in more effort to compensate for their brothers' shirking.

In Figure 20 in Appendix A.3, I present further information on the implications of this for total child-provided care hours received by the parent, and in Figure 19 in Appendix A.2 I show the proportion of parents receiving care from various other sources when their children are not providing care. In particular, I show that for low levels of care need, the dominant outside option to child-provided care for single parents is no care receipt, but for more severe care need (closer to how care need is defined in the structural model), receiving formal care is the most common outside option. In Appendix A.3 I also discuss the issue of the influence of family composition of a person's family-in-law on their caring decisions, arguing that it is not of first-order importance to consider interaction between

nuclear families as well as within nuclear families when it comes to the provision of care[15].

2.3 Most child caregivers do not receive substantial compensation

There is little prima facie evidence of child caregivers in the US receiving contemporaneous cash compensation from their parents (Barczyk and Kredler 2018; Groneck 2017). It is possible, however, that children are compensated by their parents in other ways. I consider here whether caregiving children receive higher bequests. In Appendix C, I discuss two other candidate mechanisms of compensation, namely that children are compensated through the provision of rent-free accommodation and through the provision of childcare, and argue that in these two cases there is only limited evidence of compensation for caregiving for most children.

2.3.1 Compensation through bequests

Groneck (2017) uses 2002-2012 HRS Exit data on the bequests of single parents to argue that children who provide care receive more bequests than their non-caring siblings. He finds that, using family fixed effects to control for heterogeneity at the level of the family, a child who provides care is 5.4pp more likely to receive a bequest and, considering only children of parents with a positive estate to bequeath, children who provide care receive on average \$19.6k more than their siblings who do not provide care[16].

To assess this mechanism, I carry out a set of fixed effects regressions of measures of bequests on child characteristics, using 2002-2018 HRS Exit data, i.e. using three more waves of HRS Exit data[17]. In each case, the regression takes the form:

$$Y_i = \alpha_f + X_i\beta + \gamma CARE_i + \epsilon_i \tag{1}$$

where Y_i is the outcome variable[18], α_f is a fixed effect for family f, X_i is a vector of controls (see notes of Table 2), and $CARE_i$ is a dummy for whether child i provided care with ADLs or IADLs before the parent's death[19].

[15]More broadly, parents who only have sons see a drop off in the total hours of care they receive from their family, so daughters-in-law are clearly not generally stepping in and compensating for their husbands' lack of care. For sick single parents with two children, those with two sons receive 31 hours of care from all family sources per month on average, compared to 44 hours for parents with one son, one daughter and 47 hours for parents with two daughters.

[16]As well as the family fixed effects model, Groneck (2017) also presents results from a 2SRI model and a Tobit model.

[17]As far as possible I use the same methods of sample selection and of construction of measures of bequests as in Groneck (2017). I would like to thank Max Groneck for sharing his replication code. Any errors are my own.

[18]This is either a dummy for receiving any bequest or the size of the bequest that child i receives in 2012 dollars.

[19]In particular, whether the child was included in the roster of helpers with ADLs or IADLs in the

The main results of these regressions confirm the findings of Groneck (2017): using the longer HRS 2002-2018 exit sample, I find that a child who provides care is 4.4pp more likely to receive a bequest and considering only children of parents with a positive estate to bequeath, children who provide care receive on average $13.1k more than their siblings who do not provide care.

However, as pointed out in Groneck (2017), many parents do not have an estate to bequeath in the first place: in my 2002-2018 HRS Exit sample, only 53% of children have parents who die with a positive estate to bequeath. As such, it is difficult to infer from these headline results how much extra the *typical* child would expect to receive relative to their siblings if they provide care[20]. To assess this, I carry out a series of further regressions of the size of a bequest that a child receives on the same RHS variables but in each case changing the subsample of observations used.

Table 2: Bequest received by care provided

	Dependent variable:				
	Bequest received ($1000 in USD2012)				
	(1)	(2)	(3)	(4)	(5)
Child provided care	13.11***	7.66***	6.02***	4.10***	3.73***
	(2.78)	(1.51)	(0.92)	(0.53)	(0.44)
Parent FEs	Y	Y	Y	Y	Y
Observations	6,672	12,549	12,425	11,996	11,437
Adjusted R2	0.85	0.86	0.82	0.75	0.60
Mean Dep. Var	62.85	33.41	24.70	14.06	8.65

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: estimation via OLS. Column 1 considers only those children of parents with a positive estate at death. Column 2 considers all children. Column 3 drops children of parents with estates above the 99th percentile. Column 4 drops children of parents above the 95th percentile. Column 5 drops children of parents above the 90th percentile. Controls are age of child, child education, child income, whether the child owns a home, whether the child lives within 10 miles of the parent, whether the child is co-resident with the parent, frequency of contact with the parent and number of children of the parent. Parent fixed effects throughout.

Column 1 shows the baseline result, using only the children of parents with a positive estate to bequeath. In Column 2, I run the regression for all children, including the 47% of children whose parents die without a positive estate to bequeath. As such, rather

parent's exit interview.

[20]Groneck (2017) does examine how bequests change along the wealth distribution by carrying out the regression in Equation 1 individually for each quintile of total family bequests, finding that in each case caregiving children receive higher bequests than their non-caregiving siblings. However, the regressions still condition on the parent having a positive amount to bequeath.

than only capturing the intensive margin of bequest receipt, I capture the intensive and extensive margins together. The coefficient on $CARE_i$ is cut almost in half as a result of expanding the sample.

The baseline result is also significantly driven by extreme results towards the top of the distribution of wealth. In Columns 3, 4 and 5, I restrict the sample of Column 2. In Column 3, I drop children of parents with estates above the 99th percentile in size; in Column 4, I drop children of parents above the 95th percentile; and in Column 5, I drop children of parents with estates above the 90th percentile.

As more and more of the children of richer parents are dropped, the coefficient on $CARE_i$ unsurprisingly gets smaller. What is notable, however, is that even considering the bottom 90% of children in terms of parental wealth (i.e. looking at Column 5), the coefficient is much smaller (around 3.5×) than the headline result[21].

As such, it is interesting and important that on average, the mean caregiving child receives a significantly larger bequest than the mean non-caregiving child within a particular family. However, the results above suggest that the typical (e.g. median) caregiving child can expect much less in the way of compensation for providing care. In the model I check how results change with different assumptions on bequests, ranging from an equal split of bequests regardless of care effort to having the division of bequest depend entirely on care effort, and carry out welfare and counterfactual experiments for these different specifications.

3 Model

3.1 Model overview

The model is a finite-horizon dynamic game between two agents: Child A (elder) and Child B (younger). The two agents have a common single elderly parent. The parent is entirely passive in the sense that their decisions are entirely determined by state variables and their children's decisions.

At the start of each period, the parent experiences health and wealth shocks. After these shocks are realised, if the parent is still alive, the two agents make (discrete) location, labour and care provision decisions and flow payoffs are realised. In the main specification the children move sequentially, with the order determined each period by an *iid* fair coin flip. I also consider alternative specifications with a fixed or choice-dependent order, or with simultaneous moves.

[21] According to Genworth (2008), a full-time (44 hours per week) home health aide costs \$3.8k-\$4.0k per month in 2012 USD, depending on whether the parent lives in a rural or urban area. If we interpret the extra bequest for caregiving children as payment for their care, these children are therefore paid a very small fraction of the market rate for care work, given that children often provide care for months or years at a time.

If the parent is sick and child i provides care but child j does not then both children potentially derive a benefit from care being provided but only child i bears the cost of the provision of care. In this sense, the game resembles a public good provision problem: it is costly to provide the public good (informal care to a parent), yet both children derive a benefit if this good is provided. Any care needs that the children do not meet are met by formal care, which the parent pays for out of their wealth, subject to a Medicaid asset floor, decreasing the potential bequest that the children receive.

When the parent dies, their remaining wealth is split between the children as a bequest. The nature of this split again depends on the model specification: in the main specification the wealth is divided equally, and in alternative specifications it is divided according to whether each child provided care, in proportion to each child's care hours, or so as to equalise the children's remaining lifetime resources. The children consume this bequest over a set number of remaining periods and the game ends[22].

3.2 Environment

Time is discrete, with each period lasting 2 years, matching the gap between HRS interview waves. The parent's health in period t is given by $h_t \in \{0, 1, 2, 3\}$, where $h_t = 0$ denotes that the parent is healthy, $h_t = 1$ denotes that the parent has moderate care needs, $h_t = 2$ denotes that the parent has severe care needs and $h_t = 3$ denotes that the parent is dead. At the start of the game, the parent is healthy and aged 55, and has a maximum age of 100. The parent's health follows a first-order Markov process, with transition probabilities varying by current health, age, gender and whether any of their

[22]The model just described is evidently a model of non-cooperative interaction between siblings. The empirical literature on caregiving by adult siblings has generally assumed that siblings behave non-cooperatively (Byrne et al. 2009; Checkovich and Stern 2002; Fontaine, Gramain, and Wittwer 2009; Hiedemann and Stern 1999), as have dynamic models of interaction between parents and children (Barczyk and Kredler 2018; Ko 2022). Exceptions include Hiedemann, Sovinsky, and Stern (2018), who treat the family as a single decision-making unit with common preferences, Pezzin and Schone (1999), who allow for bargaining between adult daughters and their parents, and Mommaerts (2024), who models a parent and child as cooperating with limited commitment.

I follow the majority of the literature and model siblings as non-cooperative, for two main reasons. First, the usual justifications for a cooperative model do not generally apply to adult siblings. For instance, Mommaerts (2024) justifies cooperation between parents and adult children by the extent of their interaction, such as co-residence and help with the parent's finances, whereas adult siblings typically live in separate households and do not pool resources. Second, papers that directly compare the two approaches find that non-cooperative models better describe siblings' care decisions: Knoef and Kooreman (2011) find that a non-cooperative model fits the data better for 71% of two-child families in Europe, and Bergeot (2024) finds that the non-cooperative model outperforms a cooperative model in predicting care by children in families in France. In both papers, the cooperative model predicts more care than parents actually receive. Since each child benefits when either child provides care but ignores the benefit their own care confers on their sibling, non-cooperation implies that care is under-provided relative to the cooperative outcome, as Maruyama and Johar (2017) also find for siblings' decisions to live near their parents. I therefore maintain the assumption of non-cooperation in all specifications, and in Section 5 I quantify the welfare gains from a switch to cooperation and the transfers between siblings needed to share these gains equally.

children has a college education[23]. I assume health transitions are exogenous and do not depend on informal care receipt, following the findings of Byrne et al. (2009) that care receipt does not substantively change health.

Each family has a set of fixed characteristics: for the parent, their gender and whether they live in an urban/rural area, and for each child, their gender, education and marital status. Each family also has a set of time-varying characteristics throughout the game: parental age, health and wealth (inclusive of the current period wealth shock), and for each child, a summary of their choices in the previous period, namely whether they provided any care, whether they worked full-time, and whether they lived near the parent. The collection of these fixed and time-varying state variables at the start of period t is given by s_t.

3.2.1 Choices

Each period the two children make location, labour market and care decisions. Each child i chooses a location $far_t^i \in \{0, 1\}$, corresponding to respectively living within 10 miles of one's parent and living further than 10 miles from one's parent[24]. Each child also chooses a labour market status $l_t^i \in \{NW, PT, FT\}$, corresponding to *No work*, *Part-time* or *Full-time*. Finally, each child chooses a level of care $k_t^i \in \{0, 1, 2\}$, i.e. *No care provision*, *Moderate care provision* or *Intense care provision*: if their parent is healthy ($h_t = 0$), children choose no care ($k_t^i = 0$) by default, while if their parent is sick ($h_t = 1$ or $h_t = 2$) they can choose any of the three. Thus in general in each period t each child i makes a single discrete choice $d_t^i = (far_t^i, l_t^i, k_t^i)$ out of the feasible set of choices F_t, where F_t has $2 \times 3 \times 1 = 6$ elements if the parent is healthy and $2 \times 3 \times 3 = 18$ elements if the parent is sick.

I assume that *Moderate care provision* implies one normalised hour of care and *Intense care provision* implies $\kappa > 1$ normalised hours of care.

The children's move order depends on the model specification. In the main specification, children move sequentially, and at the start of each period the move order within that period is determined by an *iid* fair coin flip. As such, on average, each child is equally likely to be the first mover in any given period and choices of the children do not affect their likelihood of being the first mover in subsequent periods. In Appendix G, I present and discuss results for alternative specifications under which either i) move order is fixed throughout the game, ii) move order is a fully deterministic function of children's

[23]For reasons of tractability neither parental education nor parental permanent income is included as a state variable in the model. I instead use whether any of the parent's children went to college as a proxy for the parent's socioeconomic status, which may have a bearing on the parent's health transition probabilities.

[24]I assume all moving is done by children, and parental location is held fixed. As set out in Appendix B, in my HRS descriptives sample approximately four times out of five when a child lives far in one wave and near in the next it is because the child has moved rather than the parent moving.

choices or iii) children move simultaneously.

3.2.2 Preferences

Child i's per period flow utility when the parent is alive is:

$$u^i(d_t^i, d_t^j, s_t, \epsilon_t^i) = g(c_t^i, l_t^i) + \omega(d_t^i, k_t^j, s_t) + \phi(far_t^i, s_t) + \epsilon_t^i(d_t^i) \tag{2}$$

The first component of utility is $g(\cdot)$, which captures utility from consumption c_t^i and the disutility of labour supply l_t^i. The other components of utility are $\omega(\cdot)$, which captures the (dis-)utility associated with the provision and receipt of care, $\phi(\cdot)$ which captures the (dis-)utility associated with the child's location choice, and a Type 1 Extreme Value preference shock ϵ_t^i associated with child i's discrete choice d_t^i. The (dis-)utility associated with the provision and receipt of care depends on the set of common states s_t, both children's care decisions and the child's own location and labour choices. The preference shock ϵ_t^i is *iid* across choices, children and time and has scale one.

The $\omega(\cdot)$ function takes the form:

$$\omega(d_t^i, k_t^j, s_t) = \omega_{pg}(k_t^i, k_t^j, s_t) + \omega_{warm}(d_t^i, s_t) \tag{3}$$

Here, $\omega_{pg}(\cdot)$ ("public good") captures the benefit to child i of their parent receiving informal care when sick, regardless of who provides the care. In other words, this is the benefit the child derives from the public good of having a cared-for parent[25], which is why it depends on child j's care choice as well as child i's.

The other term in the equation for $\omega(\cdot)$, namely $\omega_{warm}(\cdot)$, captures the "net warm glow" to child i of they themselves being the child who provides care - in other words, the net utility benefit associated with the act of providing care[26]. This part of utility does not depend on the other child's care choice. Note that it is plausible that the "net warm glow" of caring for a parent is negative, once the mental and physical burden of providing the care is taken into account. I defer the exact parameterisation of $\omega_{pg}(\cdot)$, $\omega_{warm}(\cdot)$ and $\phi(\cdot)$ to Section 3.4.

[25] In particular a parent who receives care from a family member, rather than from a formal source.

[26] The reason this term depends on the child's general discrete choice d_t^i, rather than only the care choice k_t^i, is that warm glow care utility depends on the current location choice (it is less costly to provide care when living closer to a parent) and the current labour supply choice (it is more costly to provide care when working), as discussed in Section 3.4.

3.2.3 Resources

As in Ko (2022), children consume their income every period[27], so their budget constraint for consumption is simply:

$$c_t^i = w(l_t^i, s_t) \tag{4}$$

In this constraint, $w(\cdot)$ is the equivalised income function[28], determining child i's equivalised income (and hence consumption) as a function of their labour decision l_t^i and the set of common states s_t. In particular, equivalised household income depends on age, gender, education, marital status and whether the child worked full-time in the previous period.

3.2.4 Parental wealth and bequests

As set out above, the parent's decisions in the model are entirely determined by state variables and their children's choices.

The parent starts the model with assets a_0. Their budget constraint takes the form:

$$a_{t+1} = a_t - (c_t^P - y^P) - ltc_t + b_t + \xi_{t+1} \tag{5}$$

Thus, assets tomorrow are equal to assets today minus consumption net of permanent income ($c_t^P - y^P$), minus out-of-pocket long-term care costs ltc_t, plus government benefits (including Medicaid) b_t and an *iid* wealth shock ξ_{t+1}, with distribution $N(\mu_\xi, \sigma_\xi^2)$. These wealth shocks are included in the model to capture variation over time in parental wealth for reasons other than long-term care costs[29].

I assume the parent always consumes their permanent income, so $c_t^P - y^P = 0$.

Long-term care costs are determined by the parent's residual care need after the children have supplied care. Let $\overline{ltc}(h_t)$ denote the cost of meeting the parent's care need

[27] An implication of this assumption is that children do not save in the model. This is partly for computational reasons, since saving would add a continuous wealth state variable for each child, and partly because of data limitations, since the HRS records nothing about children's wealth beyond whether they own a home. Abstracting from saving removes a mechanism by which children with a greater ability to save (e.g. those with higher incomes) might be better placed to leave the labour force to provide care when a parent develops care needs, because they have an asset buffer. This could lead the model to understate the role of incomes, and hence opportunity costs, in determining care provision, and so the share of the gender care gap attributed to differences in income processes. Reassuringly, the regressions in Table 1 control for child homeownership as a proxy for wealth, so the pattern there that children with apparently higher opportunity costs of providing care (the more educated) provide more care is not driven by omission of any controls for wealth.

[28] Here and throughout, equivalisation divides the relevant quantity by the square root of household size, where household size is 1 for single children and 2 for coupled children.

[29] If there were no such shocks in the model, children would be able to completely prevent any reduction in the parent's overall potential bequest through their caregiving, meaning that they would have an unrealistic degree of control over the parental bequest. I estimate the wealth shock process directly from the data in order for the stochasticity of the parent's wealth trajectory, and thus the children's degree of control, to be driven by the data instead.

entirely through formal care, which differs between urban and rural areas (see Appendix E). If the parent receives no informal care, they pay $\overline{ltc}(h_t)$ in full. If $h_t = 1$, i.e. the parent has moderate care needs, then any child providing care ($k_t^i \geq 1$) meets that need, so there are no residual long-term care costs. Likewise, if $h_t = 2$ and at least one child provides intense care ($k_t^i = 2$), there are no residual long-term care costs. If $h_t = 2$ and no child provides intense care, the parent pays $(\kappa - 1)/\kappa$ of $\overline{ltc}(2)$ if one child provides moderate care, and $\max(0, (\kappa - 2)/\kappa)$ of $\overline{ltc}(2)$ if both children provide moderate care, where κ is the ratio of care hours when providing intense care to providing moderate care. In other words, if intense care meets the entire care need, moderate care meets only $1/\kappa$ of it.

I assume that government benefits b_t (including Medicaid) take two forms. First, they cover any shortfall in the parent's resources, so that the parent always consumes their permanent income and their assets cannot fall below 0. Second, Medicaid pays any out-of-pocket long-term care costs that would take the parent's assets below a floor $\underline{a}$, so that long-term care costs cannot run the parent's assets down below $\underline{a}$, or below their current level if this is already lower. This acts as a (simplified) Medicaid means-tested asset floor[30].

Formally, letting $\tilde{a}_t$ denote the parent's assets after long-term care costs are deducted but before the wealth shock ξ_{t+1} is realised:

$$\tilde{a}_t = \begin{cases} \max\left(\min(a_t, \underline{a}),\ a_t - ltc_t\right), & \text{if } ltc_t > 0 \\ a_t, & \text{otherwise,} \end{cases} \tag{6}$$

and overall assets evolve according to

$$a_{t+1} = \max(0,\ \tilde{a}_t + \xi_{t+1}). \tag{7}$$

Thus, when the parent is healthy or their care need is fully met by their children, $ltc_t = 0$, so $\tilde{a}_t = a_t$ and $a_{t+1} = \max(0, a_t + \xi_{t+1})$, i.e. assets follow a random walk (with drift, if $\mu_\xi \neq 0$), subject to being non-negative.

When the parent dies, their assets are split among their two children. The nature of this split depends on the model specification, and in each case depends on the children's choices in the last period in which the parent was alive. I consider four cases:

- *Equal split:* the bequest is split equally, regardless of care effort (the main specification).
- *Proportional bequest (binary):* if child i provided any care and child j did not, then child i receives all the bequest, and child j receives nothing. Otherwise, the bequest

[30] In practice, Medicaid eligibility also depends on income, and recipients must contribute most of their income towards their care. For tractability I abstract from these features, holding the parent's consumption fixed at their permanent income.

is split equally.

- *Proportional bequest (hours):* child i's share of the bequest is equal to their share of the care hours provided by the two children. If neither child provided care, the bequest is split equally.

- *Redistributive split:* the bequest is split so as to equalise as far as possible the two children's remaining lifetime resources (the sum of their remaining equivalised income and their equivalised bequest), so that children with lower incomes receive a larger share.

The precise operationalisation of these four bequest specifications is discussed in Appendix D.2.

After the parent dies, I assume that children work full-time until the age of 65, after which they retire and do not work. They split the bequest across the rest of their lifetime in such a way as to equalise as far as possible the marginal utility of consumption in each period, and value this consumption with the same utility function as consumption while the parent is alive. For them, the game ends 5 periods (10 years) after the period in which the parent would have reached their maximum possible age of 100. More details are given in Appendix D.2.1.

In summary, apart from the direct utility benefit and cost to them of doing so, children are incentivised to provide informal care when their parent is sick to reduce the parent's formal care costs, and hence stop their potential bequest being run down[31]. This matches the mechanism in other papers like Ko (2022) and Barczyk and Kredler (2018).

3.3 Equilibrium

Recall that s_t denotes the state at the start of period t, and let I_t^i denote child i's information set at the point they make their period-t choice. If child i moves first in period t, or if both children move simultaneously, $I_t^i = s_t$; if child i moves second, they additionally observe their sibling's realised choice this period, so $I_t^i = (s_t, d_t^j)$. Let $\sigma^i(I_t^i, \epsilon_t^i)$ be child i's strategy, i.e. a mapping from their information set I_t^i and their vector of preference shocks to an action in their feasible set F_t, which depends on the health of the parent[32]. Then, let $\sigma = \{\sigma^i, \sigma^j\}$ be the strategy profile across the children.

Let $V^i(s_t, \sigma)$ denote child i's expected value at the start of period t, before the move order (in the main specification) and the preference shocks are realised. Child i's choice-specific value function for d_t^i is their expected flow payoff from choosing d_t^i, excluding the preference shock, plus their expected discounted continuation value:

[31] Under some specifications, there is the added incentive to provide care to secure a higher share of the bequest.

[32] This is because children can provide care only if their parent has care needs.

$$v^i(d_t^i, I_t^i, \sigma) = E\left[\pi^i(d_t^i, d_t^j, s_t) + \beta V^i(s_{t+1}, \sigma) \mid I_t^i, d_t^i, \sigma\right] \tag{8}$$

where $\pi^i(d_t^i, d_t^j, s_t) = u^i(d_t^i, d_t^j, s_t, \epsilon_t^i) - \epsilon_t^i(d_t^i)$. For a second mover, d_t^j is known. For a first mover, the expectation is over their sibling's response to d_t^i, so a first mover anticipates how their choice will affect their sibling's choice in the same period.

Then, the strategy profile σ^* is a Markov Perfect Equilibrium iff:

$$\sigma^{*i}(I_t^i, \epsilon_t^i) = \text{argmax}_{d_t^i \in F_t}\left\{v^i(d_t^i, I_t^i, \sigma^*) + \epsilon_t^i(d_t^i)\right\} \tag{9}$$

for all t, for all I_t^i, for all ϵ_t^i and for all $i \in \{A, B\}$. This equilibrium condition captures the idea that in an equilibrium, each player's strategy is a best response to all other players' strategies.

The game has a finite horizon, perfect information[33] and (given continuous preference shocks) a unique optimal action every time a player takes an action. Thus, under the main specification, and all other specifications involving sequential moves, the game will have a unique equilibrium in principle obtainable through backwards induction. For the alternative specification with simultaneous moves, in which multiple equilibria are possible, I assume that the same equilibrium is played by all families in the data.

3.4 Functional forms and parameterisation

In this subsection I discuss the exact functional forms I choose to provide structure to the model.

3.4.1 Utility over consumption and labour supply - $g(\cdot)$

I assume that agents have log utility over consumption. I assume their utility from the labour choice is captured by coefficients on the dummies for part-time and full-time work respectively. Thus, $g(c_t^i, l_t^i)$ is given by:

$$g(c_t^i, l_t^i) = \theta_c \log(c_t^i) + \theta_{PT} I(l_t^i = PT) + \theta_{FT} I(l_t^i = FT) \tag{10}$$

The scale of the parameters is normalised by the scale parameter of the *iid* Type 1 Extreme Value preference shock in Equation 2, which I set equal to 1. Note that in the specification of $\omega_{warm}(\cdot)$, I also allow for an interaction between care effort and the labour choice in order for the model to capture differences in the marginal cost of providing care depending on hours of work.

[33]Strictly speaking, for the model to be one of perfect information, we would require the second mover's preference shock to be drawn after the first mover's choice, and for the second mover to observe the first mover's preference shock. As the shocks are *iid* across children and time, knowing or not knowing a sibling's preference shock, conditional on knowing the action that they took, will make no difference to optimal choices, and hence this assumption is without loss of generality.

3.4.2 "Public good" component of care utility - $\omega_{pg}(\cdot)$

As set out above, $\omega_{pg}(\cdot)$ captures the benefit to a child of their parent receiving some form of informal care, regardless of which child provides it. Recalling that $k_t^i \in \{0, 1, 2\}$ denotes child i's care choice at t (no, moderate and intense care respectively) I write it as:

$$\omega_{pg}(\cdot) = \begin{cases} \alpha_{h1}, & \text{if } h_t = 1 \ \ \& \ \ k_t^i + k_t^j > 0. \\ \alpha_{h2}, & \text{if } h_t = 2 \ \ \& \ \ k_t^i + k_t^j > 0. \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

Thus, for instance, if the parent has health $h_t = 1$ and at least one child provides care ($k_t^i + k_t^j > 0$) then both children receive benefit α_{h1}, regardless of who actually provided the care. The benefit they derive potentially varies with the severity of the health condition of the parent. Note that $\omega_{pg}(\cdot)$ does not depend on the intensity of the care provided: the children derive the same benefit from their parent receiving moderate or intense care.

3.4.3 "Net warm glow" component of care utility - $\omega_{warm}(\cdot)$

The function ω_{warm} captures the utility net benefit to child i of they themselves providing care (i.e. regardless of what their sibling does). It comprises three parts: a fixed cost (or benefit) of starting to provide care, having not done so the previous period; a baseline warm glow level depending on the intensity of care provided and the health of the parent; and a linear index of other child and parent characteristics, which I assume scales proportionately with care intensity. Formally:

$$\omega_{warm}(\cdot) = \begin{cases} \gamma_{start} I(k_{t-1}^i = 0) + \delta_{h_t,mod} + X_t^i \gamma, & \text{if } h_t > 0 \ \ \& \ \ k_t^i = 1 \\ \gamma_{start} I(k_{t-1}^i = 0) + \delta_{h_t,int} + \kappa X_t^i \gamma, & \text{if } h_t > 0 \ \ \& \ \ k_t^i = 2 \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

where $\delta_{h_t,mod}$ and $\delta_{h_t,int}$ are constants for each combination of the parent's health severity and the child's own care intensity (moderate or intense), and X_t^i is a set of other characteristics of the child and parent. In particular, X_t^i contains controls for:

- Whether the child lives further than 10 miles from the parent at the time of providing care ($far_t^i = 1$). This is to capture the extra time and money spent in travelling to provide care;

- Whether the child is a son. Allowing the net benefit of providing care to depend on child i's gender means that the model allows for there to be unobserved preference differences between men and women when it comes to the provision of care to their parents. In principle, this simple preference difference could capture lots of different types of differences in the situations of sons and daughters. For instance, if the coefficient on son^i is negative, this could capture the fact that daughters genuinely derive more enjoyment from providing care than sons; or that daughters are "better" at providing care than sons in the sense that for a given unit of costly effort daughters will provide more effective care than sons; or that parents prefer receiving care from daughters rather than sons, and it is costly to go against parents' wishes; or that it is costly to go against some norm in society regarding daughters' role as primary caregiver; or some combination of these explanations. For instance, Byrne et al. (2009) argue both that daughters are better at providing care than sons and experience less burden from doing so. For the sake of tractability of the model I do not distinguish between these separate drivers of preference difference between sons and daughters.

- Whether the parent is a father, and an interaction between parent and child gender. This is to allow for the stylised fact from Section 2 that the gender gap in care provision between sons and daughters is greater when the parent is a mother.

- Whether the child is the younger of the two children ("seniority"), and an interaction between seniority and child gender, to capture any social norms or preference differences connected with being the eldest/youngest child, and in particular with being an eldest/youngest daughter.

- Whether the child has a college education. This is to capture that college-educated children may differ notably from non-college-educated children when it comes to preferences for providing care (in particular, the reduced form evidence of Table 1 suggests that more educated children are more likely to provide care, which cuts against a basic story where the only relevant difference between college and non-college educated children is their opportunity costs of providing care).

- Whether the child is married, and an interaction between marital status and child gender. Married children are more likely to have children of their own, and so competing demands on their time, and these demands may fall differently on married sons and married daughters.

- Whether the parent is Medicaid-eligible, i.e. has wealth at or below the Medicaid asset floor $\underline{a}$ so that their formal care cost would be covered in its entirety, to allow for the fact that agents' net warm glow of providing care could depend on the parent's formal care outside option.

- The hours of labour market work that the child is carrying out. This is to allow for the fact that plausibly marginal utility of leisure will be higher, hence the marginal cost of providing care will be higher, the less residual leisure time is left after the labour market choice.

The fixed cost of starting to provide care, γ_{start}, is included to capture an important dynamic aspect of care provision, namely the inertia of care provision arrangements: families tend to have a primary caregiver which does not change over time (Hiedemann, Sovinsky, and Stern 2018). It is important to include this term for the sake of capturing the nature of strategic interaction between children. If providing care in one period causes a child to be "locked in" to the caregiving role, in the sense that the cost of them providing care in subsequent periods will be much lower than that of their siblings, then this makes the initial strategic interaction between the children at the point of a parent falling sick more significant in explaining the distribution of care roles into the future. I have kept this term as a separate standalone term rather than letting it scale with care hours because the model fit in terms of the persistence of care arrangements is notably worse if disutility from starting care scales with care hours, encouraging its inclusion as a fixed cost instead.

I write Equation 12 out in full in Appendix D, where I also discuss the issue of unobserved heterogeneity in preferences for providing care.

3.4.4 Location choice utility - $\phi(\cdot)$

The flow utility from location choice takes the following form:

$$\begin{aligned}\phi(\cdot) = \big(&\phi_{far} + \phi_{col} \cdot col^i + \phi_{son} \cdot son^i + \phi_{urban} \cdot urban \\ &+\phi_{urban \times col} \cdot urban \cdot col^i\big) \cdot far_t^i + \phi_{move} \cdot I(far_t^i \neq far_{t-1}^i)\end{aligned} \tag{13}$$

Thus, children derive (dis-)utility from living far from their parents, with this (dis-)utility depending on their characteristics and whether they were living far from their parents the previous period, i.e. a moving cost. The reason for allowing for different preferences by gender is to allow for possible drivers of living close to parents between sons and daughters other than through the care channel. I also allow for different preferences for living near parents by education, whether the parent lives in an urban area ($urban$), and an interaction of these terms, to capture the fact that more educated children may find it more difficult to find jobs close to their parents, and that this difficulty could be more acute if their parent lives in a rural area.

4 Estimation

4.1 Estimation sample

For the estimation sample I use data from HRS Waves 4 to 14 (1998-2018). I select those families with one parent and exactly two (biological) child records in a given wave, where the parent is observed at least once between 55 and 85 in the data[34].

I drop any observations in which the parent holds LTCI or LTCI information is missing[35]. I drop any cases where in the first period where the family is observed the parent is sick and either child provided care in the previous period. I do this so at the start of the sample I avoid an "initial conditions" problem of children already having selected into caregiving, as discussed in Appendix D. Finally, in cases where data on work, location, education or marital status of child, or health of the parent, are missing for a single wave I impute this from the child's data in the previous wave but if the data are missing for more than one consecutive wave I drop the family from the first wave with missing data. I am left with an estimation sample of 9,075 family-wave observations, consisting of 2,542 families who are in the sample for an average of 3.6 waves each.

For the children's work choices, I use the information provided by HRS respondents as to whether a given child is working full-time, working part-time, or not working. For location choices, I likewise use information provided by HRS respondents as to whether a given child lives within 10 miles or not. For the care choices, I use the care hours data that HRS respondents provide: if a given child is recorded as providing fewer than 8 hours of care per month, or lacks information on hours, I assume that child is providing no care. If a child is recorded as providing between 8 and 60 hours of care per month, I assume that the child is providing moderate care, and if the child is providing over 60 hours of care per month, I assume the child is providing intense care[36]. For the parent's urban/rural location, I use the parent's 2013 Beale Rural/Urban Continuum Code at their first observation, classifying them as living in an urban area if they are classified as living in an urban or suburban area according to that scale, and rural otherwise.

In the model, children take their and their sibling's gender, education and marital status as fixed going forward. In the estimation, I use the values recorded in the data for

[34]If the family is observed at parent ages before 55, I drop all observations until the parent is 55.

[35]I drop these observations because I do not model LTCI choice or coverage. Only 11% of the full HRS sample has LTCI.

[36]The reason for treating those providing below 8 hours of care as not providing care in the model is in order to focus on those children who are providing substantial amounts of care, rather than conflating children who are regular caregivers with those who visit their parents for a handful of hours per month. The hours data, presented in Figure 22 in Appendix E, shows significant spikes in reported care hours at multiples of 30, corresponding to respondents who report hours in terms of hours per day, which are then multiplied by 30. I select a cutoff of 60 hours so that those who provide one or two hours a day on average, corresponding to those who provide hours of care less than a part-time job, are captured in the moderate care bin, whereas those who provide more are captured in the intense care bin. Qualitative results are robust to different choices of the threshold.

each wave.

4.2 Parameters from outside the model

Many of the parameters of the model I set using values from the literature or directly from the data. The choices are summarised in Table 3 below.

Table 3: External parameters

Parameter	Value	Source
Health transition probabilities	-	HRS
Formal care cost p.a., $h_t = 1$, rural	\$6.9k	Genworth (2008)
Formal care cost p.a., $h_t = 1$, urban	\$7.2k	Genworth (2008)
Formal care cost p.a., $h_t = 2$, rural	\$66.3k	Genworth (2008)
Formal care cost p.a., $h_t = 2$, urban	\$72.9k	Genworth (2008)
Care hours per month, moderate / intense	30 / 90	Author's choice
κ (intense/moderate care hours ratio)	3	Author's choice
Full-time/part-time work hours ratio	2	Author's choice
$\underline{a}$ (Medicaid asset floor)	\$2.0k	Author's choice
μ_ξ (mean of wealth shock, per period)	−\$1.2k	HRS
σ_ξ (SD of wealth shock, per period)	\$76.3k	HRS
Income process	-	PSID
Initial age	55	Author's choice
Terminal age	100	Author's choice
Age gap to elder child	24	HRS
Age gap to younger child	29	HRS
β (discount factor, per period)	0.93	Author's choice

Notes: see Appendix E for detail on how these values are chosen or estimated. Dollar values in 2010 USD.

4.2.1 Income process and work hours

Central to the question of the opportunity cost of providing care is the income process for children in the model. The HRS provides information on child income only in broad bins, so is less useful for capturing e.g. the gender wage gap. Instead, I estimate these income processes outside the model using data from the PSID.

In particular, using PSID Family File data from 1999-2019, for men and women separately, I regress log equivalised total household income per (two-year) period, $\log y_t^i$, on demographics and labour market choices:

$$\begin{aligned}\log y_t^i = \eta_0^g + \eta_1^g age_t^i + \eta_2^g (age_t^i)^2 + \eta_3^g col_t^i + \eta_4^g couple_t^i \\ + I(l_t^i \neq NW)\left[\lambda_0^g + \lambda_1^g age_t^i + \lambda_2^g (age_t^i)^2 + \lambda_3^g col_t^i + \lambda_4^g I(l_{t-1}^i = FT)\right] \\ + \lambda_5^g I(l_t^i = FT) + \nu_t^i\end{aligned} \tag{14}$$

where $g \in \{\text{man}, \text{woman}\}$ is child i's gender, $l_t^i \in \{NW, PT, FT\}$ is their labour market choice, col_t^i is a dummy for any college education and $couple_t^i$ is a dummy for having a spouse or partner in period t. The η^g terms capture income that does not depend on the child's own labour supply (including any partner's income). The λ^g terms capture the return to working, which varies with age and education. λ_4^g captures the return to having worked full-time in the previous period, so that leaving full-time work to provide care carries a cost in future as well as current income, as in Skira (2015). λ_5^g is the additional return to full-time over part-time work. In the model, child i's consumption is the predicted value $c_t^i = w(l_t^i, s_t) = \exp(\widehat{\log y_t^i})$.

As such, the model allows for different age trajectories, part-time work penalties, college premia, penalties for being out of the labour force and dependence on couple status for equivalised household income depending on gender.

The results of the estimation of Equation 14 are given in Appendix E. The notable gender differences are that men have a higher estimated income penalty for not working FT the previous period and the return to FT work above and beyond PT work is higher for men. I also discuss the modelling of post-parental death income, which is necessary for assessing the utility benefits of bequests.

As for work hours, I assume that full-time workers work twice the hours of part-time workers, motivated by (for instance) an assumption that full-time workers work 40 hours and part-time workers 20 hours a week. The only role for this ratio in the model is in the interaction with care hours in the utility function.

4.3 Parameters estimated inside the model

I estimate the rest of the model by maximising the pseudo likelihood of agents' choices (Aguirregabiria and Mira 2007). There are 27 parameters to be estimated, summarised in Table 4 below.

I follow Ko (2022) in using the insight of Bajari, Benkard, and Levin (2007) regarding exploiting the linearity of the flow payoff function, thus the value function, to speed up estimation.

I describe here how I estimate the main specification, where move order is determined by an *iid* coin flip every period.[37] I start by estimating agents' conditional choice prob-

[37]For the specifications with a fixed or deterministic move order, the estimation is simpler, because move order is by assumption directly observable from the data.

Table 4: Parameters estimated inside the model

Parameter	Description
Consumption and labour supply	
θ_c	Weight on consumption
θ_{PT}	Utility dummy for part-time work
θ_{FT}	Utility dummy for full-time work
Child's "public good" utility of providing care given that...	
α_{h1}	At least one child provides care, parent has moderate care needs
α_{h2}	At least one child provides care, parent has severe care needs
Child's "net warm glow" utility of providing care given that...	
$\delta_{h1,mod}$	Parent has moderate care needs, child provides moderate care
$\delta_{h1,int}$	Parent has moderate care needs, child provides intense care
$\delta_{h2,mod}$	Parent has severe care needs, child provides moderate care
$\delta_{h2,int}$	Parent has severe care needs, child provides intense care
γ_{son}	Child is a son
γ_{start}	Child starts providing care this period
γ_{far}	Child currently lives far from parent
$\gamma_{youngest}$	Child is youngest sibling
$\gamma_{youngest \times son}$	Child is youngest sibling and is a son
γ_{dad}	Parent is a father
$\gamma_{dad \times son}$	Parent is a father and child is a son
γ_{col}	Child has a college education
$\gamma_{married}$	Child is married
$\gamma_{married \times son}$	Child is married and is a son
$\gamma_{medicaid}$	Parent is at or below Medicaid asset floor
$\gamma_{lab \times cg}$	Interaction of labour hours and care hours
Child's utility from location choice given that...	
ϕ_{far}	Child lives far from parent
ϕ_{col}	Child lives far from parent and has a college education
ϕ_{son}	Child lives far from parent and is a son
ϕ_{urban}	Child lives far from parent, parent lives in urban area
$\phi_{urban \times col}$	Child lives far from parent, has college education, and parent in urban area
ϕ_{move}	Child moved this period

Notes: see Section 3 and Appendix D for discussion of how these parameters enter into a child's utility function.

abilities (CCPs) in each state using a logit regression of observed choices in the data on state variables. As move order is not observable or derivable from the data under the main specification, I assume that in each case each agent is equally likely to be a first or second mover. I then use these CCPs to construct the value functions via simulation. Specifically, let the true probability of child i choosing discrete choice d_t^i given information set I_t^i and strategy profile σ be:

$$P^i(d_t^i|I_t^i,\sigma) = \int I\left(\sigma^i(I_t^i,\epsilon_t^i) = d_t^i\right) f(\epsilon_t^i)d\epsilon_t^i \tag{15}$$

i.e. the probability that d_t^i is the option "chosen" by i's strategy given the preference shocks and information set.

I estimate the sample analogue of Equation 15 from the data using a logit regression, regressing observed choices on the variables in the assumed set I_t^i (which will be the state variables s_t for first-movers or simultaneous movers, and s_t and the first-moving sibling's care choice for second-movers). This produces estimates of the form $\hat{P}^i(d_t^i|I_t^i)$[38].

Using the fact that preference shocks are *iid* Type 1 Extreme Value, the choice-specific value function for choice d_t^i, relative to some reference choice d_t^{i0}, will be (Hotz and Miller 1993):

$$v^i(d_t^i,I_t^i,\sigma) - v^i(d_t^{i0},I_t^i,\sigma) = \ln P^i(d_t^i|I_t^i,\sigma) - \ln P^i(d_t^{i0}|I_t^i,\sigma) \tag{16}$$

and thus I can recover an estimate of the strategy of child i using:

$$\hat{\sigma}^i(I_t^i,\epsilon_t^i) = \text{argmax}_{d_t^i\in F_t}\left\{\ln \hat{P}^i(d_t^i|I_t^i) - \ln \hat{P}^i(d_t^{i0}|I_t^i) + \epsilon_t^i(d_t^i)\right\} \tag{17}$$

which is derived by substituting the sample analogue of Equation 16 into the definition of equilibrium in Equation 9[39].

Given the policy function estimates $\hat{\sigma}$, I estimate the value functions by using $\hat{\sigma}$ to simulate forward and summing up flow payoff in each period, taking the mean across simulated value functions as my estimate. As in Ko (2022), because the flow payoff functions, hence the value functions, are linear in parameters, this simulation procedure has only to be done once for a given set of policy function estimates. This is because the choice-specific value function $v^i(d_t^i,I_t^i,\sigma)$ can be written as $W^i(d_t^i,I_t^i,\sigma)\theta$, where θ is the vector of parameters to be estimated and where $W^i(d_t^i,I_t^i,\sigma)$ does not depend on unknown parameters, so I only need to estimate $W^i(d_t^i,I_t^i,\sigma)$ once and then scale by the candidate parameter value θ. I then choose the θ to maximise the pseudo likelihood of

[38]There is no equivalent of σ in this expression because strategies are not directly observed. Recall that (by construction in the sequential-move case, and by assumption in the simultaneous-move case) there will be a single equilibrium played in the data, so all children will be playing the same strategy, conditional on being an older or younger child.

[39]To do this one must recognise that $\text{argmax}_{d_t^i\in F_t}\left\{v^i(d_t^i,I_t^i,\sigma) + \epsilon_t^i(d_t^i)\right\} = \text{argmax}_{d_t^i\in F_t}\left\{v^i(d_t^i,I_t^i,\sigma) - v^i(d_t^{i0},I_t^i,\sigma) + \epsilon_t^i(d_t^i)\right\}$ for reference choice d_t^{i0}.

the observed choices, as in Aguirregabiria and Mira (2007), taking into account the fact that under the main specification move order is unobserved yet the choices of agents in the model change the conditional probability that they were a first or second mover. This process is discussed in more detail in Appendix F.

Finally, with the estimated parameters from this first pass, I solve the model through backwards induction and generate model-consistent conditional choice probabilities for every information set I_t^i through the solved model using these first-pass parameters. In other words, rather than inferring choice probabilities from the data and assuming that in every observed family either child was equally likely to be first- or second-mover, I generate the choice probabilities for first- and second-movers in every state implied by the first run of the model. I then take these CCPs and run through the same process of estimating value functions through forward simulation and then estimating a new set of parameters to maximise the pseudo likelihood. I take this second set of estimated parameters as my final estimates[40].

4.4 Identification

I here provide some informal arguments for how the parameters of the model can be identified by variation in the data. Note that all parameters are scaled relative to the Type 1 Extreme Value shock scale term of 1.

The consumption and labour supply parameters are pinned down by children's work choices. The utility costs of part-time and full-time work, θ_{PT} and θ_{FT}, are common to all children, so they are identified by the overall propensity to choose each work status: the less often children choose part-time or full-time work, given the income gains from doing so, the greater the estimated costs of part-time or full-time work. The income gain from working, by contrast, varies across children with age, gender, education, couple status and whether the child worked full-time in the previous period, and since consumption is equal to the child's equivalised household income, this variation shifts work choices only through the weight on consumption. The more strongly work choices respond to these differences in the income gain from working, the larger the estimated weight on consumption, θ_c.

[40]The reason for iterating on the initial set of estimated parameters is to ensure that the future behaviour used to generate the value functions through forward simulation is the kind of future behaviour that would be generated by the model. This is important to allow the model to match dynamics in the data over time. In the main specification with unobserved move order in the data, there is an additional advantage that it is possible to make use of separate choice probabilities for first- and second-movers, generated from the model, rather than pooling together observed choices from the data without observing the first mover. In principle, rather than stopping after one iteration, it would be possible to iterate many times to convergence - in my setting, due to slow convergence and a very large state space this is prohibitively costly in terms of computation. Note that in the alternative specification with simultaneous moves, because the model cannot be solved through backwards induction I leave out this subsequent pass step and use the first set of estimated parameters as the final estimated parameters.

The preference for living far from one's parent, and its variation by demographics, is pinned down by the proportion of children who live far from their parent by demographics, and the transition cost of changing location is pinned down by the rate of transition between the two location statuses.

The overall motivation to provide care - i.e. the sum of the "public good" and "net warm glow" parameters - is pinned down by different rates of providing (different intensities of) care according to different demographics in the data. In particular, the interaction term for the care hours and labour market hours is identified by the degree to which agents in the data work less when providing care.

Finally, the relative sizes of the "public good" and "net warm glow" parameters are pinned down by variation in a child's sibling's care decisions. If child i is just as likely to provide care if their sibling is providing care (or is likely to provide care), other things being equal, this suggests that there is no role for the public good care parameters in explaining behaviour. Conversely, the greater the reduction in child i's care effort when child j is providing care (or is likely to provide care), all else equal, the greater the role of the public good care parameter. Note that in this way the public good care parameters will also absorb any *benefit* to each child of providing care at the same time as their sibling (reflecting perhaps the reduced stress of carrying out this exercise together), because any such benefit would instead incentivise children to provide care at the same time as their sibling. As such, the estimated public good terms will likely be a lower bound on the true public good benefit of having a sibling provide care.

4.5 Results

The results of the estimation are given in Table 5 (care parameters) and Table 6 (non-care parameters) below. I estimate standard errors using 20 bootstrap replications. The objective function reported in each table is the minimised negative pseudo log likelihood per family-period.

For the main specification (Column 1: random sequential moves, bequest always split equally), the key results are as follows. Starting with the care parameters, the estimates of the public good utility parameters are positive for both levels of care need, at 0.421 for parents with moderate care needs and 0.456 for parents with severe care needs, though only the latter is (marginally) statistically significant. The value of 0.456 suggests that, in the absence of care provision by the child themselves, a sibling providing care to a parent with severe care needs brings a utility benefit equivalent to a 24.5% increase in consumption for one period[41]. The corresponding figure for a parent with moderate care needs is 22.4%.

[41] With log utility over consumption, a utility gain of α is equivalent to multiplying consumption by $\exp(\alpha/\theta_c)$; here $\exp(0.456/2.083) \approx 1.245$.

Table 5: Estimation results - care parameters

	(1) Main (equal beq.)	(2) Proportional beq. (binary)	(3) Proportional beq. (hours)
Public good care params			
α_{h1} - P in moderate bad health	0.421	0.691	0.684
	(0.293)	(0.395)	(0.393)
α_{h2} - P in severe bad health	0.456	0.851	0.837
	(0.231)	(0.334)	(0.335)
Net warm glow care params			
$\delta_{h1,mod}$ - P moderate bad health, K moderate care	-0.198	-0.876	-0.870
	(0.240)	(0.330)	(0.327)
$\delta_{h1,int}$ - P moderate bad health, K intense care	-0.064	-1.110	-1.190
	(0.286)	(0.392)	(0.396)
$\delta_{h2,mod}$ - P severe bad health, K moderate care	-0.033	-0.882	-0.861
	(0.192)	(0.292)	(0.293)
$\delta_{h2,int}$ - P severe bad health, K intense care	-0.013	-1.334	-1.451
	(0.238)	(0.337)	(0.343)
γ_{son} - K is a son	-0.031	0.037	0.042
	(0.079)	(0.076)	(0.077)
γ_{start} - K just starting caring	-1.037	-0.929	-0.940
	(0.128)	(0.142)	(0.143)
γ_{far} - K currently lives far	-0.647	-0.649	-0.643
	(0.075)	(0.074)	(0.076)
$\gamma_{youngest}$ - K youngest sibling	0.038	0.063	0.067
	(0.050)	(0.055)	(0.057)
$\gamma_{youngest \times son}$ - K youngest & K son	-0.035	-0.057	-0.054
	(0.089)	(0.091)	(0.092)
γ_{dad} - P is father	-0.228	-0.300	-0.311
	(0.107)	(0.121)	(0.122)
$\gamma_{dad \times son}$ - P father & K is son	-0.029	-0.118	-0.149
	(0.131)	(0.176)	(0.188)
γ_{col} - K has a college education	0.075	0.120	0.130
	(0.058)	(0.070)	(0.073)
$\gamma_{medicaid}$ - P at/below Medicaid asset floor	-0.048	0.165	0.184
	(0.052)	(0.066)	(0.068)
$\gamma_{married}$ - K is married	-0.119	-0.022	-0.014
	(0.074)	(0.081)	(0.082)
$\gamma_{married \times son}$ - K married & K is son	-0.222	-0.271	-0.276
	(0.070)	(0.073)	(0.072)
$\gamma_{lab \times cg}$ - labour hours × care hours	-0.151	-0.150	-0.150
	(0.033)	(0.033)	(0.033)
Obj. function	2.407	2.413	2.413

Notes: "P" refers to the parent and "K" refers to the kid. Estimation via maximisation of the pseudo likelihood. Standard errors calculated using 20 bootstrap replications.

The net warm glow intercepts $\delta_{h,mod}$ and $\delta_{h,int}$ are small, negative and not statistically significant, and do not by themselves differ much by the parent's health or by care intensity. Note, however, that these intercepts are not the full net warm glow of providing care: for intense care the index of child and parent characteristics $X_t^i\gamma$ is scaled up by $\kappa = 3$, and since most of its components are negative (for example, living far from the parent, caring for a father, or combining care with work), intense care is considerably more costly than moderate care for the typical child.

As for the other net warm glow parameters, these have plausible directions. Sons derive less net warm glow from providing care than daughters, with the gap being slightly larger for younger sons versus younger daughters and for sons of fathers versus daughters of fathers, and much larger for married sons versus married daughters. The coefficient of -0.031 for γ_{son}, which is not statistically significant, suggests that for the reference category of unmarried elder sons of mothers versus unmarried elder daughters of mothers there is little difference in preferences for providing care. However, with the additional coefficient of -0.222 for $\gamma_{married\times son}$, a married son's net warm glow from providing moderate care is 0.25 utility points lower than that of an otherwise identical married daughter.

College educated children and younger children derive a slightly greater net warm glow from providing care, though neither coefficient is statistically significant, and it is costlier to provide care to a father than to a mother ($\gamma_{dad} = -0.228$). It is much costlier to provide care if the child lives more than 10 miles from the parent: the coefficient of -0.647 for γ_{far} is the largest of the characteristics in the net warm glow index. Children are slightly less likely to provide care if their parent has wealth at or below the Medicaid asset floor, though this coefficient is small and not statistically significant. There are very significant start-up costs for providing care, with γ_{start} being large and negative: a child who did not provide care in the previous period faces a cost of 1.04 utility points of starting to provide care, relative to an equivalent child who did provide care in the previous period.

Finally, the coefficient for $\gamma_{lab\times cg}$ is -0.151. Moderate care hours are normalised to 1 (with intense care hours being 3) and part-time work hours are also normalised to 1 (with full-time work hours being 2), so this coefficient implies an extra utility cost of 0.15 (0.45) of providing moderate (intense) care when working part-time, and of 0.30 (0.90) when working full-time, relative to providing the same care while not working.

The utility penalty for part-time work ($\theta_{PT} = -1.851$) is considerably larger than the utility penalty for full-time work ($\theta_{FT} = -0.627$), suggesting that, over and above the lower income from part-time work, there are disadvantages of part-time work that are not explicitly captured by the model[42].

[42]One of the factors driving this is that in my estimation sample, part-time work is observed relatively rarely. Of those children working part-time or full-time, only 10.6% work part-time. This compares to

Table 6: Estimation results - non-care parameters

	(1) Main (equal beq.)	(2) Proportional beq. (binary)	(3) Proportional beq. (hours)
Consumption and labour			
θ_c - weight on consumption for K	2.083	1.903	1.897
	(0.090)	(0.075)	(0.072)
θ_{PT} - weight on PT work dummy for K	-1.851	-1.777	-1.774
	(0.046)	(0.042)	(0.042)
θ_{FT} - weight on FT work dummy for K	-0.627	-0.475	-0.470
	(0.087)	(0.077)	(0.075)
Location choice params			
ϕ_{far} - K currently lives far	-0.009	-0.008	-0.008
	(0.014)	(0.014)	(0.014)
ϕ_{col} - K lives far, K college educ.	0.103	0.103	0.103
	(0.023)	(0.024)	(0.024)
ϕ_{son} - K lives far, K is son	0.042	0.042	0.042
	(0.016)	(0.015)	(0.015)
ϕ_{urban} - K lives far, P is urban	0.001	0.003	0.003
	(0.020)	(0.020)	(0.020)
$\phi_{urban \times col}$ - K lives far, K college educ., P is urban	-0.036	-0.036	-0.036
	(0.029)	(0.029)	(0.029)
ϕ_{move} - K moved location	-1.814	-1.816	-1.816
	(0.023)	(0.024)	(0.024)
Obj. function	2.407	2.413	2.413

Notes: "P" refers to the parent and "K" refers to the kid. Estimation via maximisation of the pseudo likelihood. Standard errors calculated using 20 bootstrap replications.

As for the location choice parameters, the estimates suggest that college educated children have a higher preference for living far from their parents; this preference is somewhat weaker for children of parents who live in an urban area, though this difference is not statistically significant. Sons are slightly more likely than daughters to live far from their parent, while whether the parent lives in an urban area makes essentially no difference by itself. Finally, there are very large costs of relocating, matching low rates of moving in the data.

4.5.1 Alternative specifications

Columns 2 and 3 present the results from alternative specifications of the model. Here I focus on differences in specifications along the dimension of how bequests are treated. In Appendix G, I present results for other differences in the specification, such as differences in the move order or differences in the parameterisation of the model.

In Column 2, the bequest-sharing rule is that if child i provided care in the last period in which the parent was alive and child j did not, then i receives all the bequest and j receives nothing; otherwise the bequest is split equally. In Column 3, the parent divides the bequest in proportion to the care hours provided in that period (so, for example, if child i provided intense care and child j provided moderate care, then child i would receive $3/(1+3) = 75\%$ of the bequest).

In both columns, the most notable change relative to the main specification is that the public good parameters become larger, at 0.69 and 0.85 for moderate and severe care needs respectively in Column 2, with the parameter for severe care needs clearly statistically significant, and the net warm glow intercepts become considerably more negative. This reflects the fact that when bequest shares respond to care effort, there is an increased incentive to provide care when one's sibling is providing care, to avoid them receiving a larger share of the bequest. To carry on matching the degree of joint caregiving in the data, the model therefore puts greater weight on the public good care parameters, and correspondingly makes the net warm glow parameters more negative, so that children are less likely to provide care when their sibling is providing care. The estimates in Columns 2 and 3 are very similar; the main difference is that the intercepts for intense care are somewhat more negative in Column 3, where intense care secures a larger share of the bequest than moderate care.

Most other parameters are similar across specifications. The notable change is in the parameter for the parent being at or below the Medicaid asset floor, which is slightly negative and not statistically significant in the main specification but positive and sig-

21.2% in the US as a whole (Bureau of Labor Statistics 2026). Children's work choices are derived from parents' responses about their children's work situations: plausibly, parents with limited information about their children's working hours will be more likely to classify their children into the no work or full-time work categories.

nificant in Columns 2 and 3. Such parents offer very little in the way of bequests, so in specifications where the promise of a higher bequest share is a greater motivator to provide care, children need an extra incentive to match care provision in the data when parents are at these low levels of wealth.

In both cases, the value of the pseudo likelihood objective function is worse than in the main specification. However, for the counterfactual and welfare comparisons to follow, I present results for the model in Column 2 as well as the main specification in Column 1, in order to show how sensitive the findings are to assumptions about bequests.

4.6 Model fit

To assess model fit I generate simulated data on care, work and location choices for families using the model. For each simulation, I draw a family's initial observation from the data, assign that family the health states of the parent observed in every period in the data, and simulate the choices they would make according to two estimated models: the main model (with equal bequest splitting) and the proportional bequest (binary) model, whereby the bequest is given entirely to the child who provided care the previous period if only one child provided care, and otherwise is split equally. I will refer to these models henceforth as Model 1 and Model 2. The family drops out of the simulation in the same period they drop out of the data.

4.6.1 Caregiving

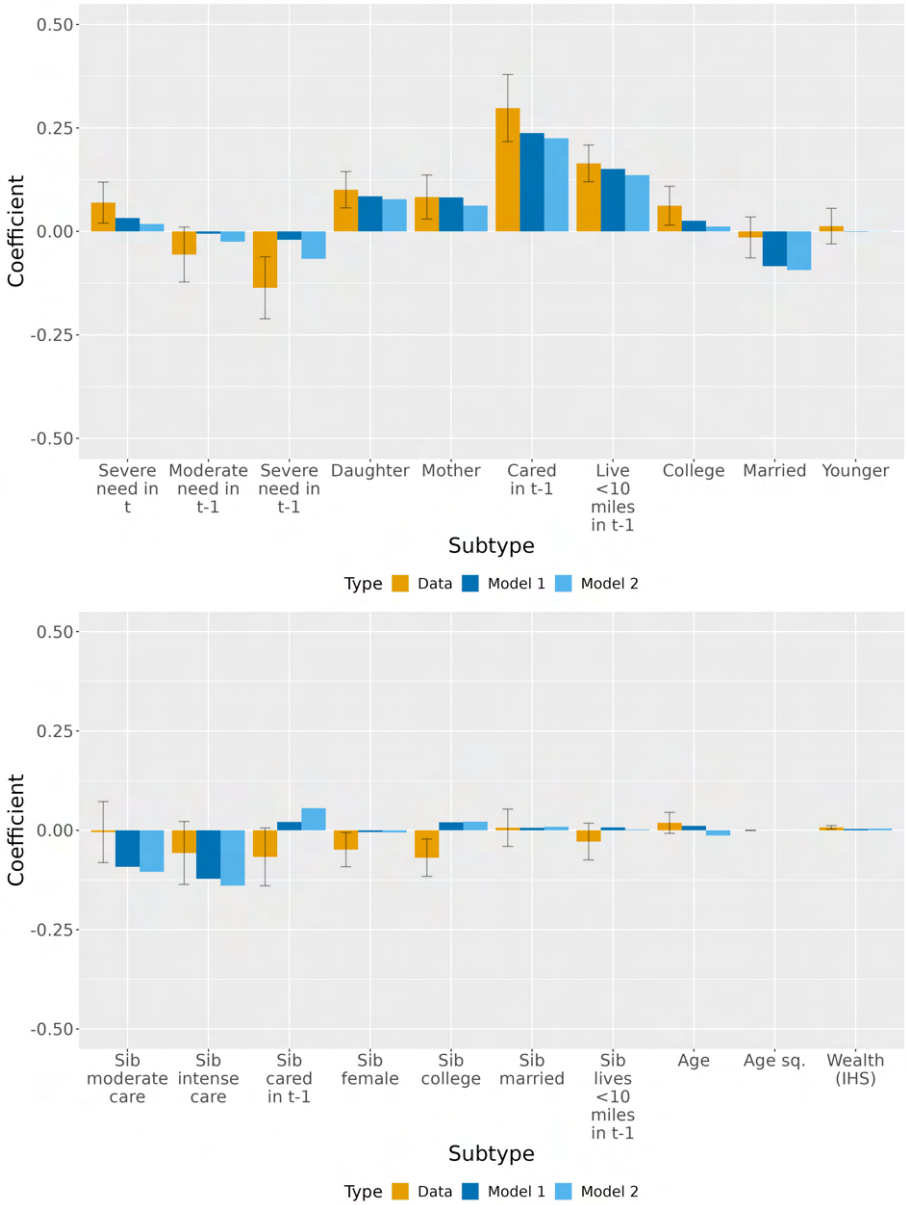
Figure 2 shows the results of a regression of a dummy for a child providing any care, conditional on their parent having care needs, on a set of observables, for both the real data and the simulated data from the two estimated models.

The fit is broadly good. In particular, the coefficients on the dummy for the child being a daughter, the coefficient on the dummy for the parent being a mother and the coefficient on the child living within 10 miles in the previous period match well across the real data and the simulated data from the two models. Also, although the models do not match the size of the correlations with reference child college status and marital status, directionally the models are able to match the data[43].

There are two notable failures of model fit, however, pointing to simplified elements of the model. The first relates to the health process. In the model, by the nature of the Markov process for health, it is only current health that should matter for care provision, other things being equal. However, in the data, current care provision depends (negatively) on previous period's health status, even controlling for current health status, which in turn means the coefficient on whether the reference child provided care in the

[43] Also, the fit by college status and marital status is better when the outcome variable is care hours rather than a care dummy, as shown in Figure 25 in Appendix H.

Figure 2: Results from regression of care dummy on observables - data versus model



Notes: estimation via OLS. Standard errors clustered at parent level. Error bars represent 95% confidence intervals. "Sib" refers to the child's sibling, so e.g. "Sib cared in t-1" is a dummy for whether the child's sibling provided care in the previous period.

previous period is higher in the data than in the model (because the reference child was more likely to provide care in the previous period the worse was the parent's health in that period). The data thus suggest that parents whose care need started earlier are less likely to receive care in the current period, other things being equal, potentially pointing to a caregiver burnout channel not included in the model.

The second failure of model fit relates to the correlation between the reference child's care provision and their sibling's characteristics. In the model, conditional on a given level of care provision by a sibling, the characteristics of the sibling are not very relevant to the reference child's care decision: in the current period, they only affect the reference child's utility through their effect on the sibling's propensity to provide care, so conditional on that sibling care level the only remaining channel through which they can influence the reference child's care decision is through expectations of future care provision. For this reason, in the model, it is the sibling's care decisions themselves which correlate with the reference child's care decision. In the data, the situation is different: even conditional on the sibling's care decision, the sibling's characteristics continue to predict the reference child's care decision, with (for instance) the reference child being less likely to provide care if their sibling cared in the previous period, even conditional on current sibling care provision.

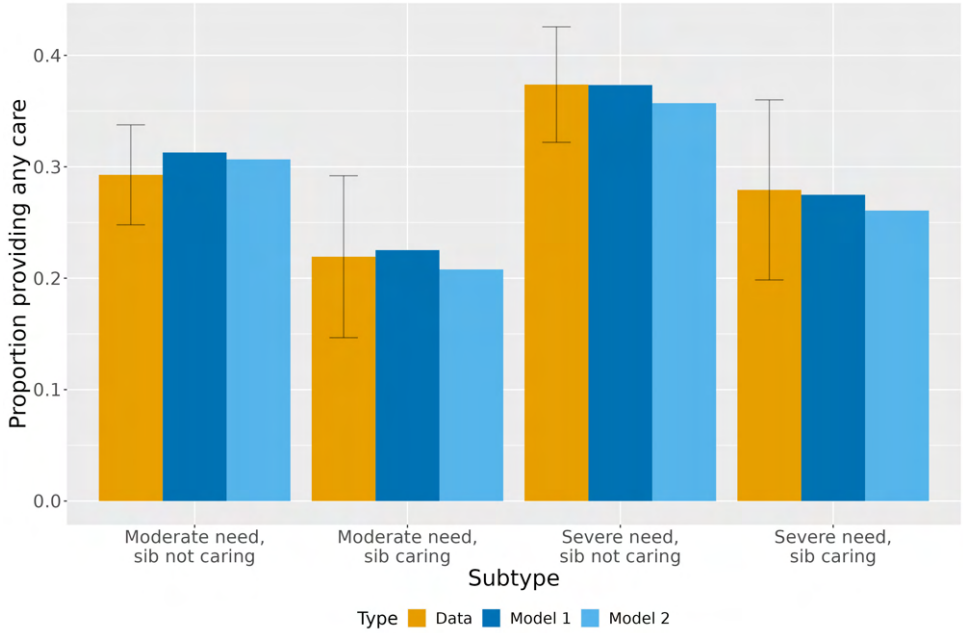
One explanation for this relates to move order. Under the sequential-move specification used here, the second mover always perfectly observes the first mover's choice, hence the significant role for the sibling's actual care decisions over and above their characteristics. This sequential-move assumption is theoretically motivated, insofar as it permits a unique equilibrium for the game and hence much more straightforward counterfactual comparisons, but it may exaggerate the extent to which agents in the data are able to base their decisions on their sibling's decisions. In a simultaneous-move model, by contrast, the sibling's realised choice would by construction be irrelevant conditional on observables, because the probability of the sibling's choice would be fully determined by observables and its realisation would be due to an *iid* preference shock; it would then be the sibling's characteristics, and not their current choices, that would matter for the reference child's current choices. As such, imposing a sequential move order, and thus observability and irreversibility of actions, means the model may overstate the degree to which caring behaviour responds to realised sibling behaviour rather than to the ex ante probabilities of sibling care.

A second explanation is that care decisions are shaped by norms about who should provide care, for instance that a daughter should care where there is one, or that a child who has cared before should continue to do so, independently of who actually provides care in a given period. Neither move order specification can represent such norms directly. In the sequential specification they appear as the failure of fit described above. A simultaneous-move specification can match these patterns only by attributing

them to the public good motive: a child who cares less when their sibling is a sister must be responding to the sister's higher probability of providing care. Appendix G shows that the simultaneous-move specification produces public good parameters that are much larger and much less precisely estimated than in the main specification, consistent with this. This frames the sequential-move specification, which I take forward to the counterfactual analysis in Section 5, as the more cautious specification for modelling the welfare consequences of interaction between siblings insofar as it uses a more conservative estimate of the public good care parameters.

Despite these issues, the model is still able to match patterns in the data relating to correlation between siblings' care behaviour. Figure 3 below presents model fit for care rates by sibling care rate.

Figure 3: Care rates by sibling care rate and parental health



Notes: Standard errors clustered at parent level.

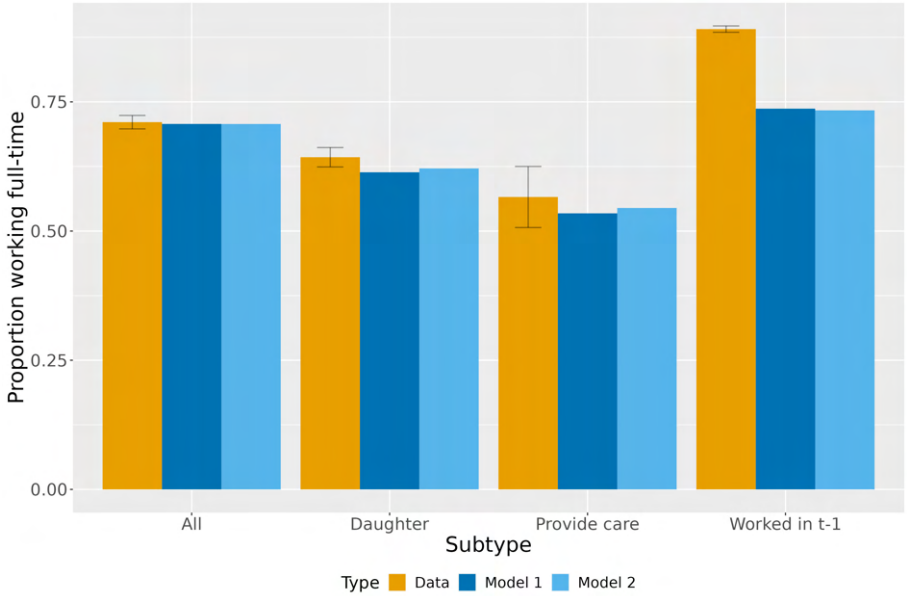
In both the model and the data, children provide less care when their sibling is providing care, controlling for parental health. Thus, the model is able to match broad patterns of crowding-out by sibling's care.

4.6.2 Labour supply

I also assess the model's fit when it comes to the work decision. To do this, I use the simulated data to calculate the proportion of people working full-time in every period and compare these rates to those from the real data. The results are shown in Figure 4.

Again, the fit is reasonably good. The model matches the proportion of people working full-time in the real data well, and also approximately matches the proportion of peo-

Figure 4: Proportion of children working - data versus model



Notes: standard errors clustered at parent level.

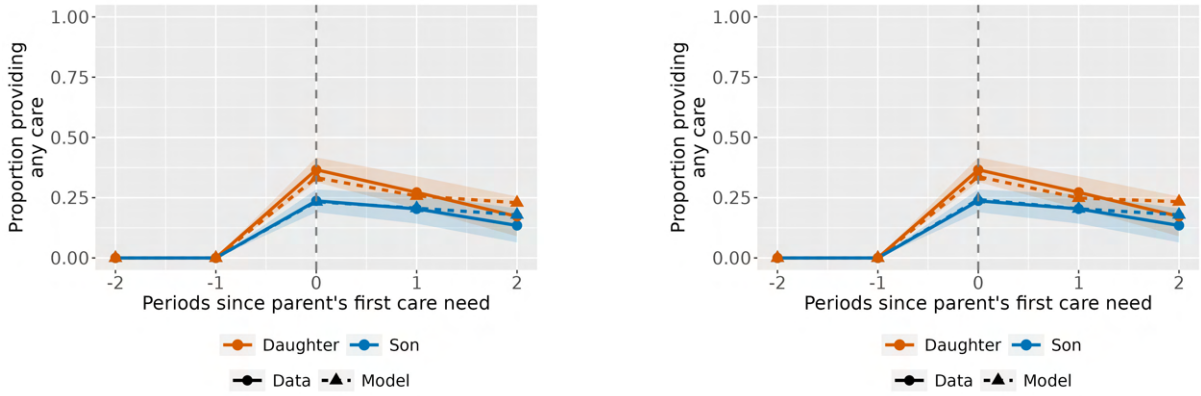
ple working full-time while also providing care, and thus seems to capture the trade-off between working more and providing care to one's parent. The model slightly understates the proportion of daughters who work full-time (and correspondingly overstates the proportion of sons who work full-time) - this holds whether the outcome variable is a dummy for working full-time or whether it is a continuous measure of hours worked, thus capturing the option of working part-time, as shown in Figure 27 in Appendix H. A more pronounced miss is that the model understates the persistence of working arrangements: in the data, conditional on working full-time in $t-1$, the probability of working full-time in t is 89.1%, whereas in the model it is only 73.7% (73.4% in Model 2). One explanation for this is that the only source of persistence in working arrangements in the model is that there is a wage penalty for not having worked the previous period, incentivising agents to not take breaks from working for a period. A more sophisticated model of labour market choices would allow for different preferences for work in the population so that those who are less work-averse select into working, and/or non-wage penalties for leaving the labour market (e.g. difficulties of finding a new job having left the labour market), adding different forms of persistence into the labour market choice.

4.6.3 Dynamics

As the model is a dynamic model, it is important that it is able to match the dynamics of responses to parental health shocks. In Figures 5, 6 and 7 below I present mean rates of caregiving, living close to one's parent and working FT, respectively, for the periods immediately around the first time the parent is observed with care needs in the data. In

each case, I calculate the means only across children of parents who are observed without care needs for at least one period before they are observed with care needs.

Figure 5: Caregiving dynamics

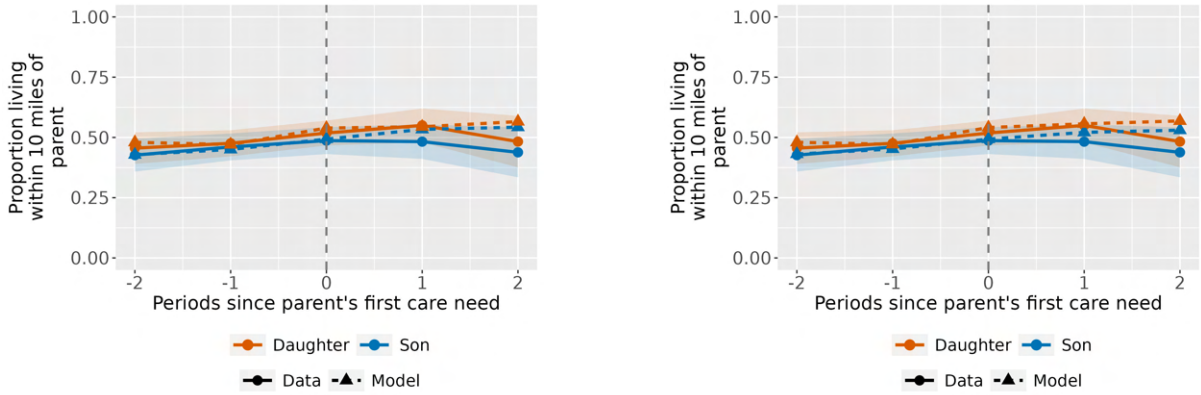


(a) Data vs Model 1 (b) Data vs Model 2

Notes: figures show means of the outcome variable in periods around the event of the parent developing care needs ($h \geq 1$) for the first time. Shaded bands are 95% confidence intervals.

For the caregiving dynamics the model gives the best match, being able to replicate both the initial response to a parent developing care needs, differentially by gender, as well as the persistence in caring behaviour over time following the shock.

Figure 6: Location dynamics

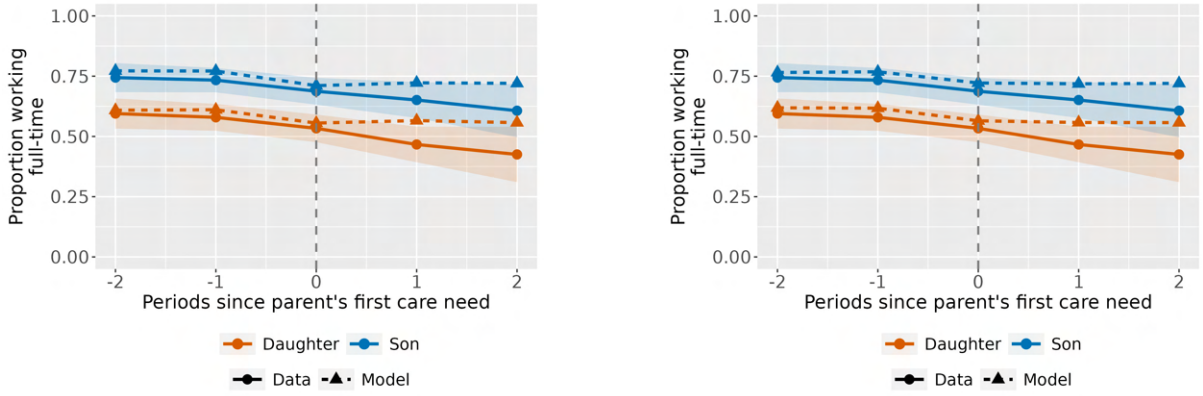


(a) Data vs Model 1 (b) Data vs Model 2

Notes: figures show means of the outcome variable in periods around the event of the parent developing care needs ($h \geq 1$) for the first time. Shaded bands are 95% confidence intervals.

For location, the model matches the small rise in the share of children living near their parent around the onset of care needs. However, in the model this move is permanent, whereas in the data it partly reverses within two periods, particularly for sons. This is consistent with the large estimated moving cost, which makes location choices very persistent in the model.

Figure 7: FT work dynamics



(a) Data vs Model 1

(b) Data vs Model 2

Notes: figures show means of the outcome variable in periods around the event of the parent developing care needs ($h \geq 1$) for the first time. Shaded bands are 95% confidence intervals.

The dynamics of full-time work are where we observe the biggest departures from the data. The model is able to match the initial response to the onset of care needs well: in the data, the FT work rate decreases by 4.6pp (4.6pp) for daughters (sons) between periods -1 and 0, when the care need first develops, compared to 5.5pp (6.1pp) for Model 1 and 5.2pp (4.6pp) for Model 2. However, the model diverges from the data in underplaying the persistence of this switch out of FT work. In the data, by period 2, the FT work rate is 15.4pp (12.7pp) lower than it was in period -1, compared to 5.4pp (5.1pp) for Model 1 and 6.0pp (4.8pp) for Model 2. This follows from the point discussed in relation to Figure 4, that given the simple model of the labour supply choice here the model does not match persistence in labour supply choices over time[44].

4.6.4 Validation against one-child families

The two-child model can be thought of as nesting a simpler one-child model, where one of the two children (for instance, child B) never provides care, and is known to never provide care. In such a case, B's choices never affect child A, so child A will act as if they are an only child.

Given this, it is possible to validate the estimated model against data from one-child families, not used in the estimation, to check the plausibility of its predictions. In particular, this is relevant for assessing the plausibility of the estimated public good/net warm glow split in the incentives to provide care: an only child will receive the public good benefit only if they themselves provide care, because they have no sibling who could provide it, so other things being equal will be more incentivised to provide care

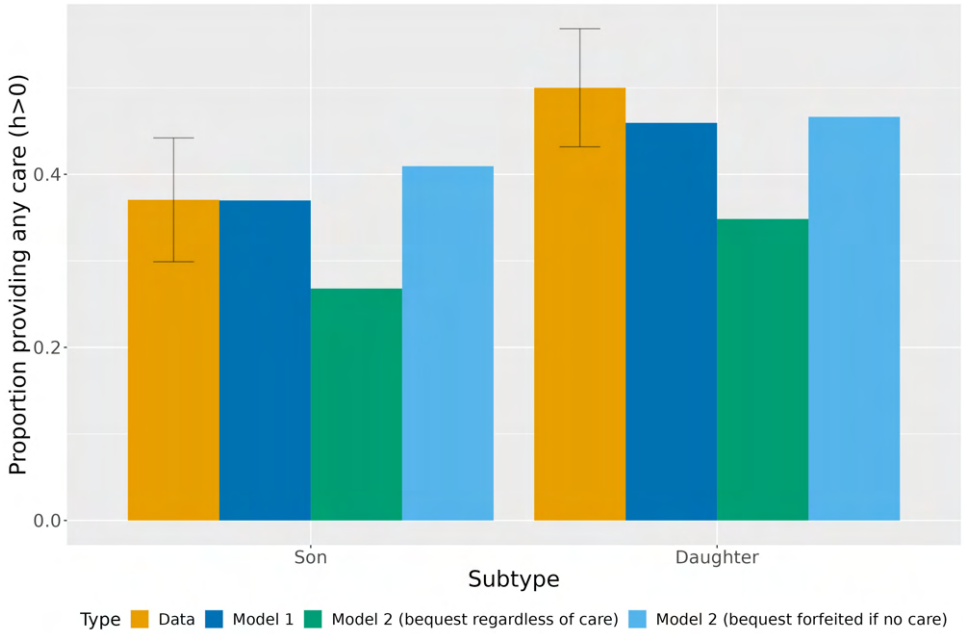
[44] A similar point applies when the outcome variable is a continuous measure of labour hours, as shown in Figure 29 in Appendix H.

than a child in a two-child family. The greater the relative weight on the public good terms versus the net warm glow terms, the bigger this gap between care rates for only children versus children in two-child families. Hence, a predicted only-child care rate much bigger than that shown in the data would suggest that this public good benefit is being overestimated.

A complication here relates to Model 2, i.e. the binary bequest split model. In its original framing for two-child families, if neither child provides care, then the bequest is split equally between them. The most literal extension of this bequest specification to a one-child family would be to say that the only child always receives all the bequest, because whatever level of care they choose is always weakly the most of any child in the family, because they are the only child. However, this would eliminate entirely the strategic bequest mechanism inherent in the two-child specification, under which bequests are awarded in order to reward caregiving. As such, in Figure 8, I offer a separate extension of Model 2 to the only-child case, under which, if the parent had care needs and the child did not provide care in the period before the parent's death, the child receives nothing, otherwise the child receives the whole bequest[45].

Figure 8 below carries out the validation exercise using data and model predictions for one-child families for care rates conditional on child gender.

Figure 8: One-child families: care rates in data versus estimated models



Notes: standard errors clustered at parent level.

The fit here is encouraging. In particular, the model does not overpredict care rates in only-child families, which would be the case, other things being equal, if the public

[45]For Model 1, I assume the only child receives the entirety of the bequest regardless.

good terms were overestimated. The predicted gender gap is also broadly correct though slightly narrower in the model predictions than in the data, particularly for Model 2. Unsurprisingly, the version of Model 2 which eliminates by construction the strategic bequest component while keeping other parameters the same ends up underpredicting care rates.

In Appendix H, I present further figures showing that the model is also able to broadly match levels of children living near parents by gender and parental health status. I also present the equivalents of Figures 2, 3, 4, 5, 7 and 8 using care/labour hours rather than a binary outcome variable for any care/FT work (Figures 25, 26, 27, 28, 29 and 30 respectively). In each case, the results are qualitatively similar.

5 Counterfactuals

In this section I use the estimated model to evaluate some counterfactuals of interest. I focus on four particular questions:

- What proportion of the gender care gap is explained by observed differences in wages versus unobserved differences in preferences?
- What is the role of strategic interaction in exacerbating the underlying gender care gap?
- What would be the efficiency gains of switching to cooperation, and what would the resulting patterns of caregiving/inter-sibling transfers look like?
- What would be the effects of an intervention in the care market through subsidies for informal care?

In each case, I hold all estimated parameters at their baseline values except those determining the mechanism of interest, re-solve the model via backwards induction, then simulate children's decisions in the counterfactual setting. For every counterfactual, as was the case for the model fit, I report the results for both the main model with an equal bequest split (Section 4), i.e. Model 1, and the proportional bequest (binary) model, i.e. Model 2.

5.1 Wages vs. preferences in the gender care gap

In the estimated model, daughters differ from sons in two fundamental ways. First, daughters have different preferences to sons, particularly when it comes to care provision[46]. Second, daughters face lower opportunity costs of providing care as their wages

[46]Though, as discussed in Section 3 above, the parameter on daughter's utility net benefit of providing care captures not only the daughter's attitude to the provision of care but also factors like social norms, so daughters might be just as averse as sons to providing care, but face greater social pressure to do so.

are lower. Both of these drive daughters to provide more care than sons in the model, as in the data. It is interesting to consider which of these is the stronger driver of the gender care gap.

To do this, I consider sons' and daughters' care rates in the counterfactual case where there are no preference differences between sons and daughters, holding fixed total care. In particular, for every child-gender-specific coefficient (γ_{son}, $\gamma_{youngest\times son}$, $\gamma_{dad\times son}$, $\gamma_{married\times son}$ and ϕ_{son}) I set the coefficient to zero and add Δ times its original value to the corresponding gender-neutral term[47]. $\Delta = 0$ gives every child daughters' preferences and $\Delta = 1$ gives every child sons' preferences. I choose the single scalar Δ so that the simulated probability of providing care, conditional on having a sick parent, is the same as in the baseline[48]. I then re-solve the model using backwards induction with the adjusted coefficients and determine choice probabilities at every state for the counterfactual setting.

The results of the counterfactual analysis are shown in Figure 9. The figure shows the care rates of sons and daughters, as well as the percentage gender care gap[49], both in the baseline model (left-hand set of bars) and the counterfactual of interest (right-hand set of bars), for each of Models 1 and 2.

Figure 9 shows that if child-gender-based preferences are removed from the model then sons' care rate rises and daughters' care rate falls, with the overall care rate held at its baseline level by construction. In Model 1, the gender care gap shrinks from 40.4% (i.e. daughters are 40.4% more likely to provide care than sons, conditional on having a sick parent) to 3.3%, a decrease of 37.1pp, or 91.8% of the original gender care gap. In Model 2, the gender care gap shrinks from 37.7% to 5.5%, a decline of 32.2pp, or 85.5% of the original gap. In other words, between 85.5% and 91.8% of the gender care gap is explained by unobserved preference differences.

An alternative counterfactual keeps preferences as in the baseline but gives sons and daughters the same income process. For each coefficient x of the income process (Equation 14 and the post-parental-death income process), I replace the gender-specific values with the weighted average $w\,x^{men} + (1-w)\,x^{women}$, imposed on both sons and daughters. As with the preference counterfactual, I choose $w \in [0,1]$ so that the simulated care rate matches the baseline[50]. The results of this counterfactual experiment are shown in Figure 10.

The changes are smaller: the gender care gap falls from 40.4% to 30.2% in Model 1, a reduction of 10.2pp (25.2% of the original gap), and from 37.7% to 28.6% in Model 2, a

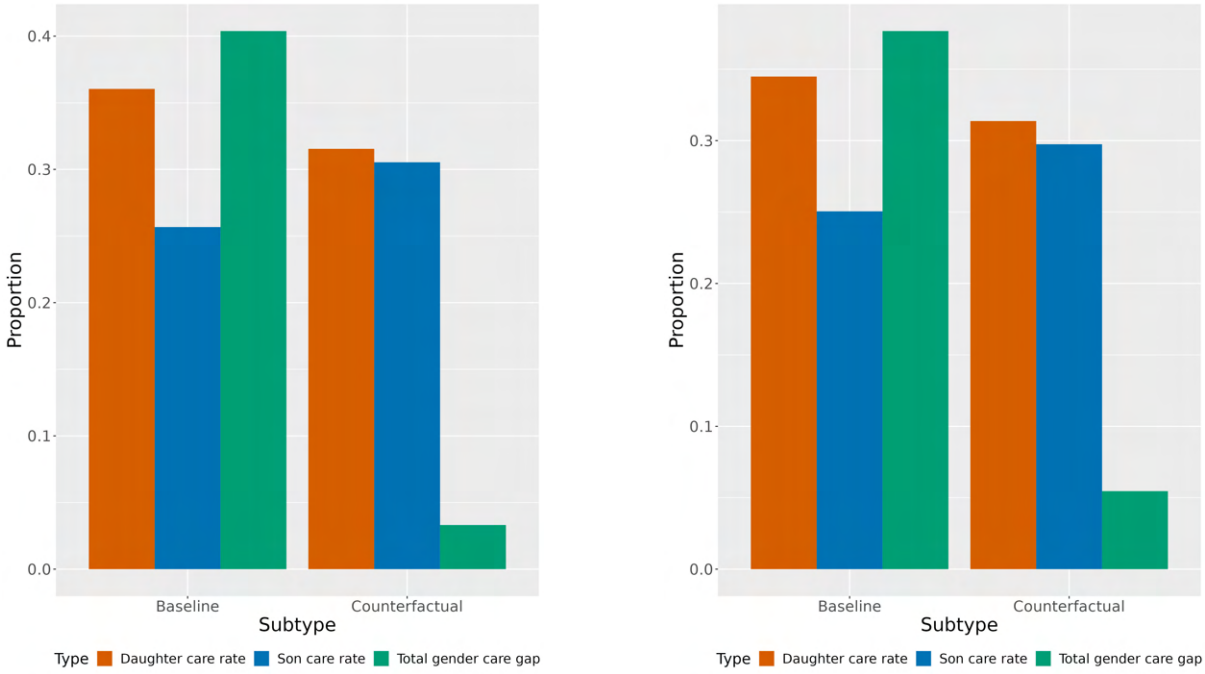
[47] That is, $\Delta\gamma_{son}$ is added to the warm glow level terms $\delta_{h,mod}$, and $\kappa\Delta\gamma_{son}$ to $\delta_{h,int}$, consistent with the linear index $X_t^i\gamma$ scaling with care intensity; $\Delta\gamma_{youngest\times son}$ is added to $\gamma_{youngest}$, $\Delta\gamma_{dad\times son}$ to γ_{dad}, $\Delta\gamma_{married\times son}$ to $\gamma_{married}$ and $\Delta\phi_{son}$ to ϕ_{far}.

[48] The calibrated values are $\Delta = 0.461$ for Model 1 and $\Delta = 0.375$ for Model 2.

[49] Namely, how much more likely daughters are to provide care relative to sons, expressed in percentage terms.

[50] The calibrated values are $w = 0.281$ for Model 1 and $w = 0.344$ for Model 2.

Figure 9: Gender care gap: no preference differences between sons and daughters



(a) Model 1, baseline vs counterfactual

(b) Model 2, baseline vs counterfactual

Notes: "care rate" is the probability of providing care conditional on having a sick parent. "Total gender care gap" is how much larger, in percentage terms, daughters' care rate is relative to sons' care rate.

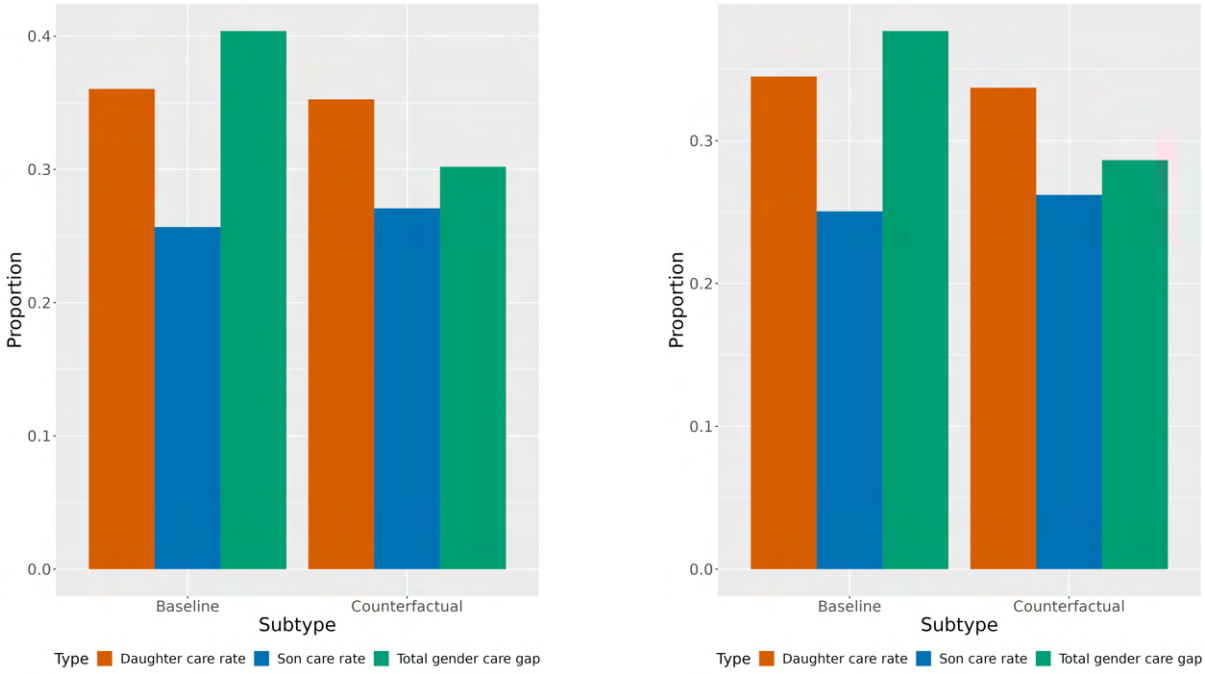
reduction of 9.0pp (24.0%)[51]. Thus, income differences between men and women explain much less of the gender care gap than unobserved preference differences.

This is a striking and somewhat surprising result. Mechanically, it is driven by the low estimate for the interaction between care hours and labour market hours. The wage gap is relevant to the care decision only to the extent that work choices and care choices correlate with each other: if work and care choices were made entirely independently of one another, the wage would have no impact on a child's incentive to provide care. If the child suffered no loss of leisure utility from working and caring at the same time, then the care decision would be independent of the work decision. Thus, the fact that this estimated coefficient is relatively small dampens the importance of the gender wage gap in driving the gender care gap.

This being said, there is some prima facie evidence in the data to suggest that income opportunity cost differentials between children in the same family are not the main driver of differences in care provision. The fixed-effects regressions in Table 1 in Section 2 showed that using parent-wave fixed effects, a child with more education - hence, one might assume, a higher opportunity cost of providing care in terms of foregone wages -

[51]The percentage reductions from the two different counterfactuals, closing the preference gap and closing the income gap, can add up to more than 100% because of the non-linear nature of the model.

Figure 10: Gender care gap: no income process differences between sons and daughters



(a) Model 1, baseline vs counterfactual

(b) Model 2, baseline vs counterfactual

Notes: "care rate" is the probability of providing care conditional on having a sick parent. "Total gender care gap" is how much larger, in percentage terms, daughters' care rate is relative to sons' care rate.

relative to their siblings is in fact more likely to provide care. In this light, the fact that gender income gaps explain so little of the gender care gap is less surprising[52].

Two remarks are in order on this counterfactual. First, the fact that so much of the gender care gap is driven by unobservable preference differences is in one sense discouraging because we have little insight into the nature and causes of this preference difference. However, in a different sense it is important and interesting that so little of the gender care gap is driven by differences in opportunity costs as this suggests that even if the gender wage gap shrinks, there will not necessarily be a significant reduction in the gender care gap, as the causes of the gender care gap are more deep-rooted.

Second, just because the gender care gap is largely driven by differences in preferences for providing care does not mean it is necessarily benign, in the sense that plausibly it is benign for people who have a taste for doing a task to self-select into performing that task. The preference differences here capture not only daughters' enjoyment, or lack of burden, of providing care, but also the extent to which daughters might suffer from going against parental preferences or societal gender norms. These latter two forms of preference difference are plausibly less benign. Thus, just because the gender care gap is

[52]Groneck (2017) similarly notes that there is little evidence that children with higher opportunity costs are less likely to provide care.

largely driven by preference differences does not mean that it is not a matter of policy interest.

5.2 The role of strategic interaction

I also examine how significant a role strategic interaction plays in the gender care gap. If sons' care is crowded out by their sisters' care, it is interesting to consider how much smaller the gender care gap would be - i.e. how much more care sons would provide, relative to daughters - if children did not condition their care behaviour on the characteristics of their siblings.

To do this, I use the model to generate simulated choices while imposing that child i always assumes that child j's probabilities of providing different levels of care are exactly equal to the baseline model population mean probabilities of providing different levels of care, conditional on health status. In other words, child i assumes (correctly) that child j's care level will not depend on j's characteristics nor on j's beliefs about i's care level. Therefore, i will not adjust their play according to j's characteristics, and there is no strategic interaction - the two-player game is effectively reduced to a one-player problem in which the sibling's care is an exogenous random draw from this population distribution. The order of moves is as in the baseline: when the sibling moves first, child i observes the sibling's choice before making their own; when child i moves first, the sibling's care is drawn from the same population distribution, regardless of child i's choice [53].

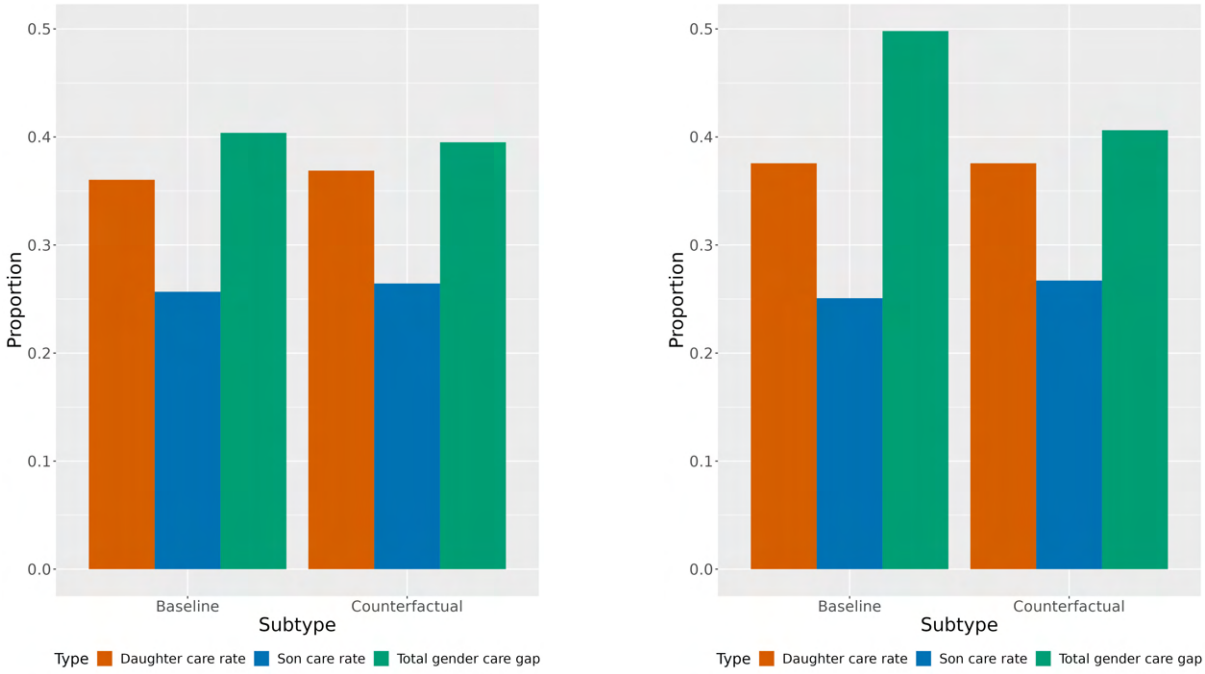
The results are presented in Figures 11 and 12 below, for the two model specifications under consideration. In each graph, the left-hand panel assesses the change in the gender care gap for the whole population. The right-hand panel assesses the change in the gender care gap for son-daughter families, i.e. mixed-gender families.

For the whole population, there is very little difference, which is unsurprising: in the counterfactual, in daughter-daughter families, children now revise down their expectations of sibling caregiving (because they are treating their sibling as a generic child rather than a daughter), so provide more care. Conversely, in son-son families, children provide less care. So, even if in son-daughter families the gender care gap shrinks, in the aggregate, calculating the care gap across families, the effect mechanically roughly cancels out.

The more informative comparison for strategic interaction between brothers and sisters is within son-daughter families, shown in the right-hand panels. There, removing strategic interaction reduces the gender care gap from 49.8% to 40.6% in Model 1, a fall of 9.2pp or 18.4% of the baseline gap, and from 47.5% to 38.0% in Model 2, a fall of 9.5pp or 20.0%. The gap shrinks mainly because sons provide more care: sons' care rate rises

[53] When simulating, I designate one child in each family as the reference child, who makes optimal choices given these beliefs, and draw the other child's choices from the fixed population distribution, so that the reference child's beliefs are correct. I simulate each family twice, once with each child as the reference child. The results below are presented for reference children only.

Figure 11: Gender care gap with no interaction - Model 1



(a) All families

(b) Son-daughter families

Notes: "care rate" is the probability of providing care conditional on having a sick parent. "Total gender care gap" is how much larger, in percentage terms, daughters' care rate is relative to sons' care rate.

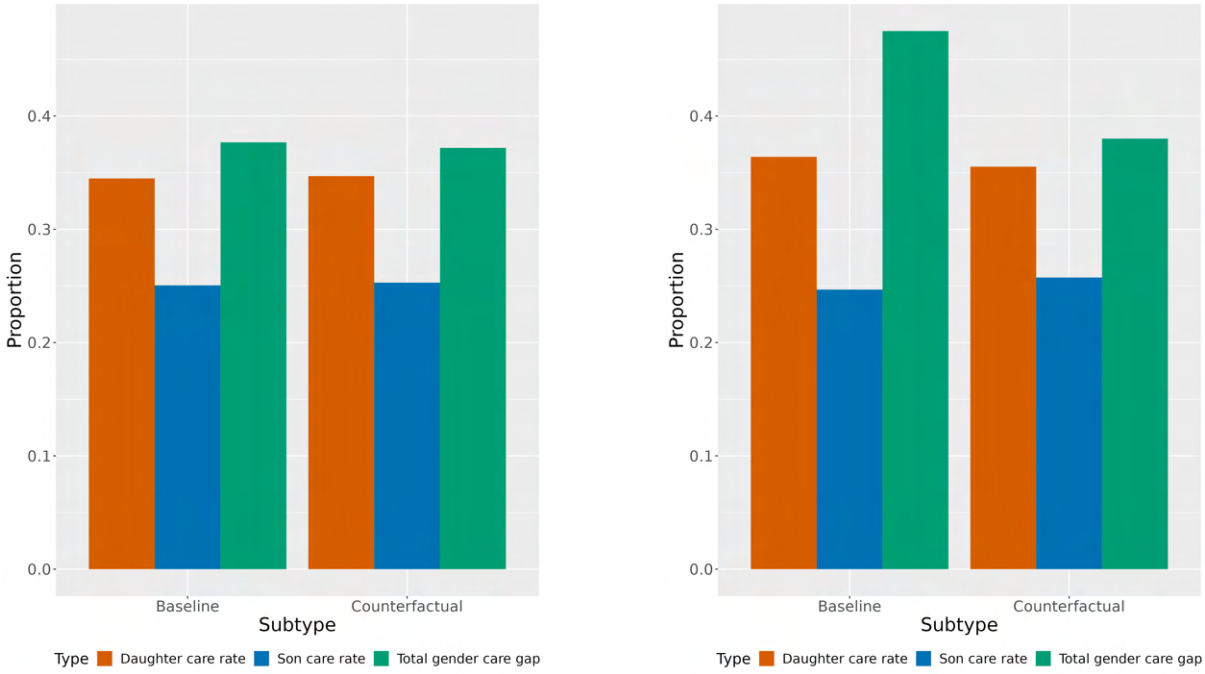
from 25.1% to 26.7% in Model 1 and from 24.7% to 25.7% in Model 2, while daughters' care rate changes much less (37.6% in both cases in Model 1, and from 36.4% to 35.5% in Model 2)[54]. As a benchmark, in the same son-daughter families, removing preference differences reduces the gap by 43.1pp in Model 1 and 39.9pp in Model 2, and equalising income processes reduces it by 12.8pp and 12.3pp. Strategic interaction between siblings thus accounts for roughly three-quarters as much of the gender care gap in mixed-gender families as the gender difference in income processes does.

5.3 The benefits of cooperation

With the estimated model coefficients in hand, it is possible to assess the costs of non-cooperation, by solving the model backwards under the assumption that both children make decisions to maximise the sum of expected utility across children, rather than only their own utility. I carry out this exercise, evaluating the change in caring behaviour under cooperation and the subsequent effects for welfare. In particular, I calculate lifetime expected utilities under non-cooperation and cooperation separately and then calculate the transfer required in order to exactly split the benefits from cooperation between the

[54]The results are similar when care is measured in hours: the gap falls by 11.4pp (17.7%) in Model 1 and 12.1pp (18.1%) in Model 2.

Figure 12: Gender care gap with no interaction - Model 2



(a) All families

(b) Son-daughter families

Notes: "care rate" is the probability of providing care conditional on having a sick parent. "Total gender care gap" is how much larger, in percentage terms, daughters' care rate is relative to sons' care rate.

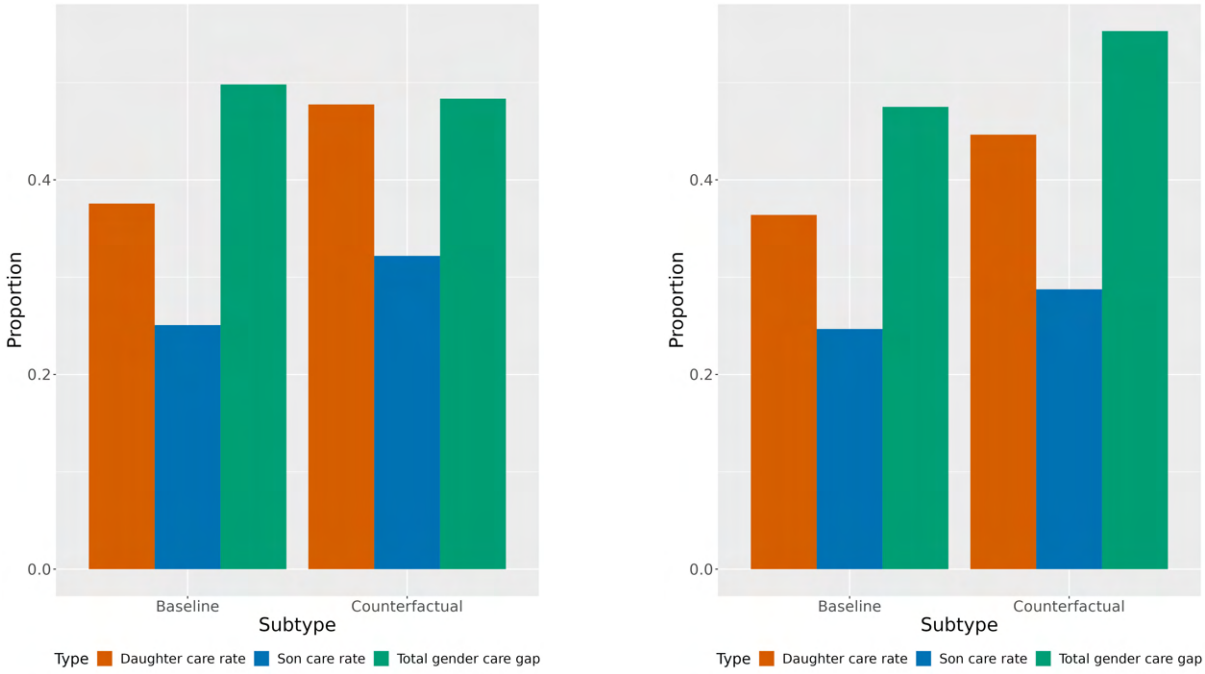
two children in any particular household. Obviously, such an approach abstracts from any benefits to parents from the switch to cooperation that are not already captured indirectly through the children's utility, so will likely be a lower bound on the true welfare gains.

The care rates for sons and daughters in the baseline and counterfactual are presented in Figure 13. I present the results for son-daughter families to focus on the switch from strategic non-cooperation to cooperation between sons and daughters specifically.

The care rates of both sons and daughters increase as they internalise the benefits of their caregiving for their sibling. The effect on the gender care gap depends on whether this gap is measured in percentage or percentage-point terms. In percentage-point terms, the gap increases in both specifications: daughters' probability of providing care rises by more than sons' (the gap rises from 12.5pp to 15.6pp in Model 1, and from 11.7pp to 15.9pp in Model 2). In percentage terms, the effect is more ambiguous: the gap rises from 47.5% to 55.3% in Model 2, but falls slightly, from 49.8% to 48.3%, in Model 1, reflecting the increase in sons' care rate in the denominator. Note, however, that Figure 31 in Appendix H, the hours-equivalent version of Figure 13, shows both the percentage and the percentage-point gaps increasing under cooperation in both models.

The reason why daughters take on more of the additional care under cooperation is specialisation: the model parameters imply that daughters find it less costly to provide

Figure 13: Gender care gap: cooperation, son-daughter families only



(a) Model 1, baseline vs counterfactual

(b) Model 2, baseline vs counterfactual

Notes: "care rate" is the probability of providing care conditional on having a sick parent. "Total gender care gap" is how much larger, in percentage terms, daughters' care rate is relative to sons' care rate.

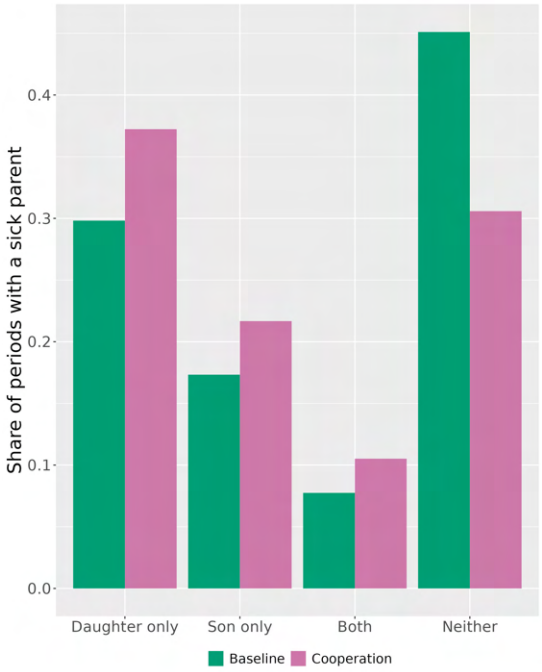
care (through the preference parameters and lower opportunity costs in terms of wages), so aggregate utility across sons and daughters is highest when daughters take on more of the care role[55].

Another feature of the switch to cooperation is the degree to which siblings coordinate their caregiving. Figure 14 shows the share of periods in which a parent with care needs is cared for by their daughter only, their son only, both children, or neither, under the baseline and the cooperative counterfactual.

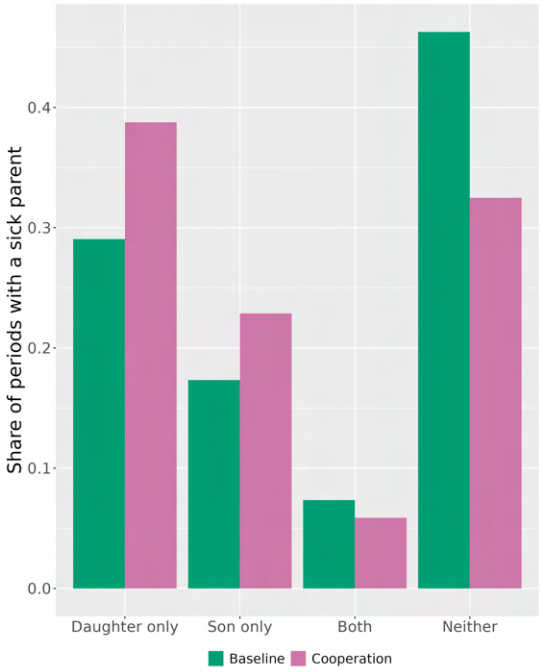
The most notable change in behaviour is that there are far fewer instances when neither child is caring for the parent: the share of periods with a sick parent in which neither child provides care falls from 45.1% to 30.6% in Model 1, and from 46.3% to 32.5% in Model 2. This follows from the modelling of care provision as a public good - under cooperation, children internalise the gain to their sibling of having at least one child provide care, so these instances of neither child providing care are much less common. Note that under Model 1, the proportion of cases where both children provide care increases

[55]The gender care gap widens more under Model 2 than under Model 1 because, in the non-cooperative equilibrium of Model 2, the strategic bequest rule gives each child an approximately equal extra utility gain from providing care. Given logit choice probabilities, this raises sons' lower probability of providing care proportionally more, narrowing the baseline gap. Under cooperation, who receives the bequest is merely a transfer within the family, so this strategic bequest incentive disappears, and the gender care gap widens.

Figure 14: Coordination of care effort: baseline vs cooperative counterfactual



(a) Model 1, baseline vs counterfactual



(b) Model 2, baseline vs counterfactual

Notes: son-daughter families only.

(because of the mechanism in the model by which both children can combine their care hours to prevent a parent having to spend heavily on formal care when they have severe care needs), but in Model 2 the share of cases where both children provide care decreases (because there is no longer a "defensive" incentive to provide care when one's sibling is providing care to protect one's share of the bequest).

The welfare results are presented in Table 7. I present the results for mixed-gender families in order to focus on the extent to which sons and daughters gain differentially from cooperation in these families. I set the reference age to 81, matching the median age of a parent developing care needs in the estimation sample. At parent age 81, the younger child in the family is 52 and the older child is 57, and they both have 30 years remaining in their model lifetimes (i.e. up to 10 years, or 5 periods, after the parent's maximum age of 100, as discussed in Appendix D.2.1). In the table, I convert the differences in lifetime utilities into the equivalents in terms of a one-period increase in consumption and a permanent increase in consumption. For the calculation of transfers required to equally split utility, I simply calculate half the gap between the son's and daughter's gains. In doing so I abstract from differences in the level of income/consumption between sons and daughters, i.e. the transfer is expressed as a percentage of each child's consumption, rather than as a dollar-for-dollar transfer, which would also depend on the siblings' different consumption levels.

Table 7: Welfare gains from cooperation, mixed-gender families.

	Healthy ($h = 0$)	Moderate need ($h = 1$)	Severe need ($h = 2$)
Model 1 (equal bequest)			
A. One-period increase in consumption (%)			
Sons' gain	1.70	3.99	4.58
Daughters' gain	0.46	0.87	0.99
Transfer, son to daughter	0.61	1.50	1.69
Resulting equal gain	1.08	2.41	2.74
B. Permanent increase in consumption (%)			
Sons' gain	0.18	0.41	0.47
Daughters' gain	0.05	0.09	0.10
Transfer, son to daughter	0.06	0.16	0.18
Resulting equal gain	0.11	0.25	0.28
Model 2 (binary bequest split)			
A. One-period increase in consumption (%)			
Sons' gain	2.09	5.75	6.59
Daughters' gain	1.81	3.37	4.35
Transfer, son to daughter	0.10	0.96	0.80
Resulting equal gain	1.89	4.26	4.94
B. Permanent increase in consumption (%)			
Sons' gain	0.21	0.57	0.63
Daughters' gain	0.18	0.31	0.37
Transfer, son to daughter	0.02	0.13	0.13
Resulting equal gain	0.20	0.44	0.50

Notes: The one-period measure is the percentage increase in consumption in the current (two-year) period alone that gives the same utility gain. The permanent measure is the constant percentage increase in consumption in every remaining period of the child's life (15 periods at parent age 81) that gives the same utility gain. In each panel, the transfer from son to daughter is half the difference between the son's and the daughter's gain, and the resulting equal gain is the average of the two gains.

Under cooperation, daughters' bigger absolute increase in care effort means that they benefit less on average from cooperation than sons do, as they take on more of the additional care burden. As such, before any redistributive transfers, sons gain more than daughters across both specifications: the gains from cooperation for the son of a severely sick parent are 4.6% (6.6%) of contemporaneous consumption, and 0.47% (0.63%) of per-period lifetime consumption, under Model 1 (Model 2), compared to gains of 0.99% (4.4%) of contemporaneous consumption and 0.10% (0.37%) of per-period lifetime consumption for daughters. As discussed above, under Model 2, daughters' increased caregiving, and decreased prevalence of joint caregiving, mean that they receive more of the bequest and thus the gap between daughters' gains and sons' gains is smaller - in other words, some of the transfer already takes place through the bequest. Overall, the equal gain across children when the surplus is split equally (i.e. after any transfers between children to equalise gains from cooperation) is 2.8% (5.5%) of contemporaneous consumption and 0.28% (0.50%) of per period lifetime consumption for children of a parent with severe needs under Model 1 (Model 2).

5.4 Subsidies for informal care provision

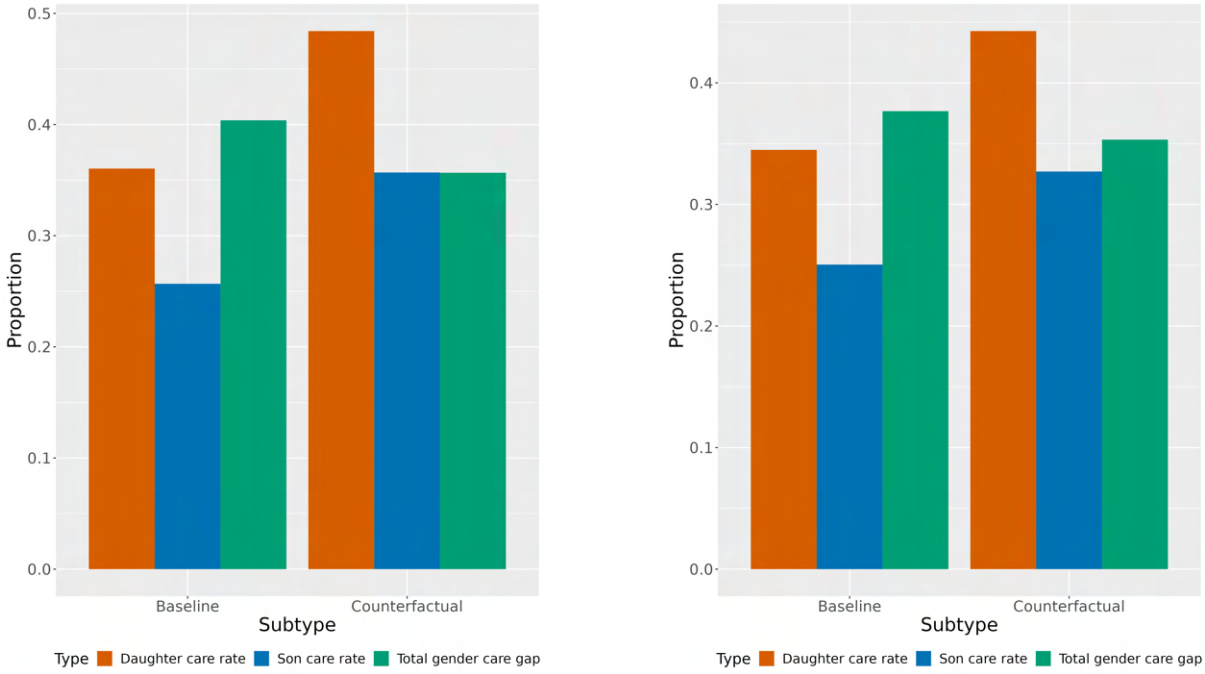
I also consider government intervention in informal care provision in the form of a subsidy for informal care, i.e. paying informal carers a wage for the hours of care they provide. I set this wage equal to the hourly rate of a home health aide, the same rate from which I construct the cost of formal care for a parent with moderate care needs: \$19.90 for the child of a parent in an urban area and \$19.04 for the child of a parent in a rural area. A child providing moderate care therefore receives \$7.2k (\$6.9k) per year, exactly what it would cost the parent to hire a home health aide in their place, and a child providing intense care receives \$21.5k (\$20.6k) per year. The counterfactual abstracts away from how exactly such an intervention would be funded so is necessarily a partial-equilibrium view of the effects of such an intervention.

The change in the gender care gap is shown in Figure 15. The subsidy substantially increases informal care provision by both sons and daughters. In Model 1, daughters' care rate rises from 36.0% to 48.4% and sons' from 25.7% to 35.7%; in Model 2, daughters' rises from 34.5% to 44.3% and sons' from 25.1% to 32.7%. Daughters' care rate rises by more in percentage points, so the gap between daughters and sons widens in absolute terms, from 10.4pp to 12.7pp in Model 1 and from 9.4pp to 11.6pp in Model 2. Sons' care rate rises by more in proportional terms, however, so the gender care gap in percentage terms falls, from 40.4% to 35.7% in Model 1 and from 37.7% to 35.3% in Model 2[56].

As for labour market effects, Figure 16 plots mean proportions of sons and daughters

[56]The same holds when care is measured in hours: the gap in percentage terms falls from 57.8% to 50.7% in Model 1 and from 57.0% to 49.3% in Model 2, while the absolute gap widens.

Figure 15: Gender care gap: informal care subsidy



(a) Model 1, baseline vs counterfactual

(b) Model 2, baseline vs counterfactual

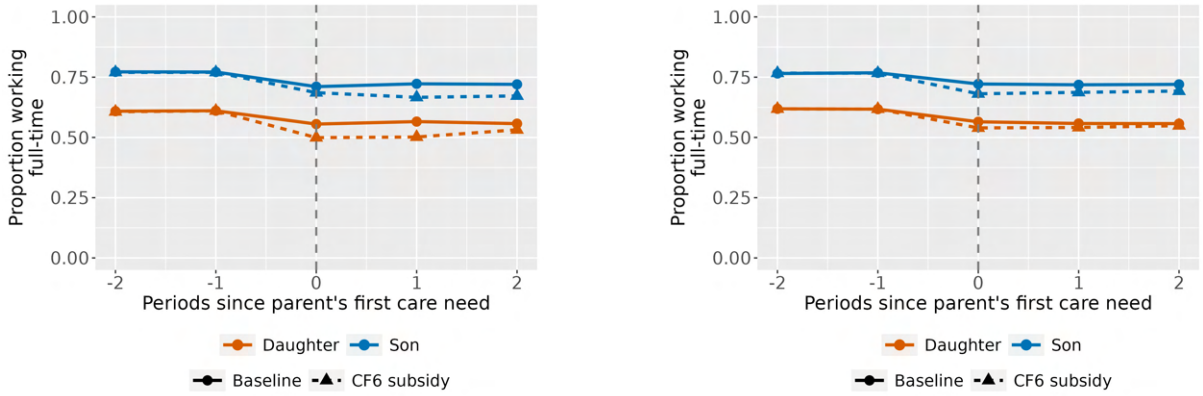
Notes: "care rate" is the probability of providing care conditional on having a sick parent. "Total gender care gap" is how much larger, in percentage terms, daughters' care rate is relative to sons' care rate.

working full-time around the event of their parent developing care needs for the first time, in the baseline and the counterfactual setting. The subsidy deepens the fall in full-time work at the onset of the parent's care needs. In Model 1, the share of daughters working full-time falls by 5.5pp between periods -1 and 0, when the parent first needs care, in the baseline, and by 11.1pp under the subsidy; for sons, the fall increases from 6.1pp to 8.4pp. In Model 2, the corresponding falls increase from 5.2pp to 7.8pp for daughters and from 4.6pp to 8.6pp for sons.

To examine the issue of who switches into care due to the policy, I break down the demographics of those who care in both the baseline and counterfactual scenario, those who care in neither scenario, and those who switch from not caregiving to caregiving between the baseline scenario and the counterfactual scenario[57]. I focus on caregiving in the period t where the parent develops care needs for the first time, corresponding to period 0 in Figure 16, to avoid double-counting children across multiple periods. Table 8 presents the results.

[57]There is of course a fourth potential category, of those who cared under the baseline but do not under the counterfactual. These could be, for instance, second-mover children, who in the baseline saw that their first-mover sibling did not provide care so stepped in to provide care themselves, whereas in the counterfactual scenario their sibling is now motivated to provide care themselves, so the second-mover child no longer needs to step in. However, the proportion of children in this category is very small (0.11% of all children for Model 1 and 0.35% for Model 2), so I exclude them from the Table.

Figure 16: FT work dynamics: informal care subsidy



(a) Model 1, baseline vs counterfactual

(b) Model 2, baseline vs counterfactual

Notes: figures show means of the outcome variable in periods around the event of the parent developing care needs ($h \geq 1$) for the first time.

Table 8: Characteristics of children by caregiving under the informal care subsidy

	Care in both	CF6 only	Care in neither	All
Model 1				
Female	0.616	0.590	0.479	0.529
College educated	0.603	0.440	0.604	0.587
Married	0.592	0.365	0.767	0.676
Lived within 10 miles ($t-1$)	0.607	0.547	0.381	0.462
Worked full-time ($t-1$)	0.664	0.667	0.700	0.687
Parental wealth ($1000s)	243.6	185.8	244.0	238.0
Share of children	0.285	0.102	0.613	1.000
Model 2				
Female	0.608	0.576	0.486	0.529
College educated	0.606	0.469	0.594	0.588
Married	0.592	0.393	0.749	0.676
Lived within 10 miles ($t-1$)	0.588	0.581	0.392	0.464
Worked full-time ($t-1$)	0.679	0.652	0.699	0.689
Parental wealth ($1000s)	307.1	201.5	210.8	238.1
Share of children	0.290	0.078	0.632	1.000

Notes: quantities displayed in the Table are means by child in the first period where the parent develops care needs. "Care in both" refers to children who care in both the baseline and the counterfactual scenario, "Care in neither" refers to children who care in neither scenario. "CF only" refers to those who care in the counterfactual scenario only. Shares of children are shares of the children in these three categories, excluding the small share of children who care in the baseline scenario only.

The subsidy draws a substantial number of children into caregiving: 10.2% of these children in Model 1 and 7.8% in Model 2 provide care in the period their parent first needs it under the subsidy but not in the baseline. In both models, these switchers are disproportionately unmarried (36.5% are married in Model 1, against 76.7% of children who care in neither scenario; 39.3% against 74.9% in Model 2) and less likely to be college educated (44.0% against 60.4% in Model 1; 46.9% against 59.4% in Model 2). They therefore tend to have lower incomes, so a given payment is worth more to them; for married children, the payment is also smaller once equivalised. Switchers are also more likely to be daughters (59.0% and 57.6%, against 47.9% and 48.6% of those who care in neither scenario) and, unsurprisingly, to have lived close to the parent before the parent needed care.

Switchers also have poorer parents than other children: average parental wealth is \$186k for switchers in Model 1, against \$244k both for those who care in both scenarios and for those who care in neither, and \$201k in Model 2. The models differ in the wealth of those who care regardless of the subsidy. Under Model 2, with its strategic bequest component, children who care in both scenarios have much richer parents (\$307k) than those who care in neither (\$211k): children of richer parents are already motivated to provide care by the prospect of a larger bequest. In Model 1, where the bequest does not depend on care, there is no such difference.

6 Conclusion

In this paper I presented a dynamic discrete-choice model of strategic interaction between siblings in providing care to a common elderly parent. The model allowed for endogenous location and work status by letting each child make care, work and location choices every period, and the model isolated the role of interdependence of children's care decisions by separately estimating the utility each child receives from providing care themselves and the utility they receive from having their parent cared for by anyone. By considering non-cooperative interaction between sets of siblings over time, and allowing multiple dimensions of choice for each child in each period, the model goes beyond existing models in the literature and allows a detailed examination of what drives differences in care provision within families, particularly by gender of the child.

The results of the estimation suggest that having a parent receive care is a public good for all children yet it is costly for each child to provide care. Children therefore have an incentive to shirk, relying on their sibling to provide care in their place, and sons do so more than daughters. If children instead took their sibling's care provision as fixed at its population average, so that there were no strategic interaction, the gender care gap in son-daughter families would be 18-20% smaller, mainly because sons provide more care.

The results also show that unobserved preference differences between sons and daugh-

ters, rather than differences in opportunity costs through the gender gap in earnings, are the chief driver of the gender care gap. Giving sons and daughters identical preferences, while holding the overall care rate fixed, removes 86-92% of the gap, while giving them identical income processes removes only around a quarter. Thus, policies which successfully close the gender earnings gap in isolation would leave most of the division of caring responsibilities between sons and daughters unchanged.

Because strategic interaction leads siblings to under-provide care, there are gains from cooperation. If siblings instead chose their actions to maximise their joint utility, both sons and daughters would provide more care, and the share of periods in which a sick parent receives care from neither child would fall from 45-46% to 31-32%, depending on the specification. Daughters, who find caregiving less costly, would take on more of the additional care, so the gender care gap would widen in absolute terms. As a result, sons would gain more from cooperation than daughters, and an equal division of the gains would require a transfer from sons to daughters. For children whose parent has severe care needs, the gain from cooperation, split equally, is equivalent to a permanent increase in consumption of 0.28% or 0.50%, depending on the bequest specification.

I also considered a counterfactual intervention which pays informal carers the hourly wage of a home health aide. This would substantially increase informal care, raising daughters' care rate by around 10-12pp and sons' by around 8-10pp. The gender care gap would widen in absolute terms but narrow in percentage terms. The subsidy would also deepen the fall in full-time work when a parent first needs care, and would mainly draw into caregiving children with lower incomes and the children of poorer parents.

For reasons of tractability I have significantly simplified the decision set of children in the model: they do not save, and they make simplified discrete choices over labour supply, location and caregiving. As discussed in Section 3, the absence of saving in particular could lead the model to understate the role of incomes, and hence of opportunity costs, in the gender care gap. Future work might consider allowing all players a richer choice set, to see whether the conclusions of the more restricted model presented here still hold.

Relatedly, the parent is not included as an active player in the dynamic game, and parents' spouses are excluded by construction by focussing on single parents only. Excluding spouses means that the paper is unable to assess interaction between spouses and children in the provision of care, which would be important to understand in order to analyse the drivers of the overall gender care gap across all caregivers, not just children. Moreover, the absence of a model of parental preferences, or of the effect of caregiving on parental outcomes, means that the paper cannot assess how parents themselves choose between informal and formal care, nor the full efficiency costs of non-cooperative care arrangements, since doing so would require an understanding of the benefits to the parent of different kinds of care arrangements. Developing the ideas of this paper in this direction would allow a more complete analysis of the sources of inefficiency in this area

and of potential policy interventions to alleviate it.

Finally, this paper leaves unresolved the exact drivers of preference differences between sons and daughters when it comes to the provision of care. Future work could decompose this preference difference into (for instance) the enjoyment, or lack of burden, daughters derive from providing care versus the burden daughters experience from going against parental preferences or social norms, while maintaining the setting of dynamic interaction. Relatedly, norms about who within a family should provide care, for instance a daughter where there is one, could make a child's care depend on who their sibling is rather than on what their sibling does. The model does not capture such norms, and the model fit suggests they may matter; separating them from strategic responses to a sibling's care would sharpen the estimates of how much strategic interaction contributes to the gender care gap. Such a decomposition would allow a more sophisticated treatment of what policy approaches, if any, are to be used to reduce the gender care gap.

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Appendices

A Supplementary descriptive figures and tables

A.1 HRS sample descriptive statistics

Table 9 presents some key descriptive statistics on the full sample of HRS respondents for Waves 4 to 14 with at least one child record. This is the sample used to establish the stylised facts in Section 2. I also include statistics for the subsample of single respondents and the subsample of single respondents with exactly two children because the estimation will focus on these groups.

Table 9: Descriptive statistics

	All	Single respondents	Single respondents, 2 kids
Female	0.55	0.72	0.71
Age	66.18	69.89	68.93
Couple	0.68	0.00	0.00
Kids	3.16	3.01	2.00
- of which live within 10 miles	0.90	0.94	0.64
White Caucasian	0.84	0.78	0.83
Homeowner	0.79	0.60	0.64
Some college	0.49	0.40	0.47
Wealth ($1000s)	117	74	107
Observations	200 385	67 123	17 812

Notes: all statistics are weighted means using HRS respondent-level weights, apart from wealth, which is a weighted median to reduce the impact of outliers. "All" refers to all HRS respondents with at least one child record associated with them. As wealth is measured at the household rather than individual level, I allocate half of each couple household's wealth to each member of the couple. Dollar values here and throughout, unless stated otherwise, are expressed in 2010 dollars using the CPI.

The overall HRS sample skews female, given lower life expectancy for men. About two-thirds of respondents are in a couple, about four-fifths are homeowners and about one-half had at least some college education.

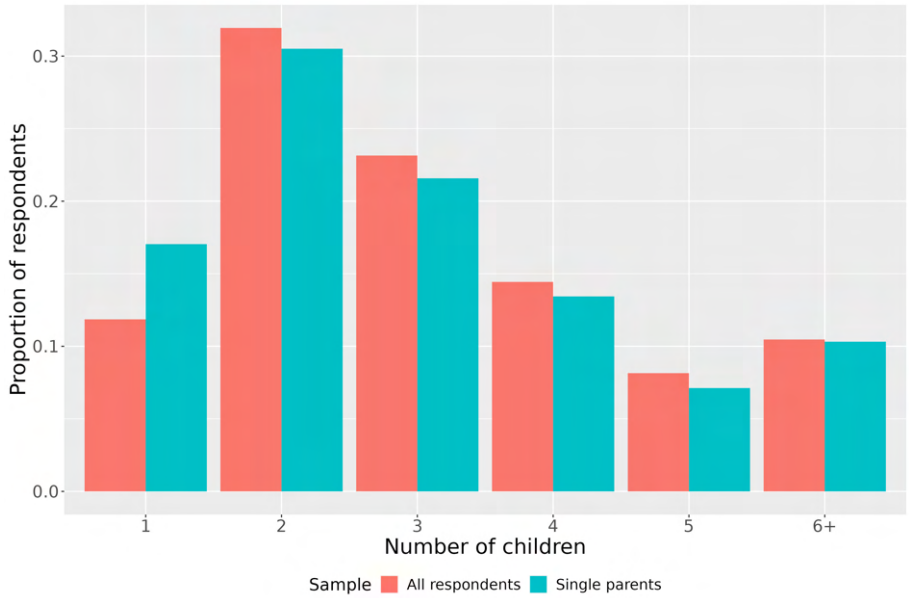
The subsample of singles is notably more female, slightly older, and less likely to own a home or to have gone to college. Median wealth is considerably less than in the full sample. Single respondents are also slightly less likely to be white. It is notable that the average number of kids is lower for single respondents, yet the average number of kids living within 10 miles is higher, suggesting that a higher proportion of the children of single parents live near their parent.

Finally, the subsample of single respondents with exactly two kids is slightly richer and

more educated than the subsample of all single respondents. Indeed, along the wealth, education and race dimensions this subsample more closely resembles the full HRS sample than it does the subsample of singles.

Figure 17 below shows the distribution of number of living children reported by respondents who have at least one child record associated with them, for all respondents and single respondents.

Figure 17: Distribution of number of children of HRS respondents



Notes: respondents who report at least one child and have at least one child record associated with them only. Observations at the respondent-wave level.

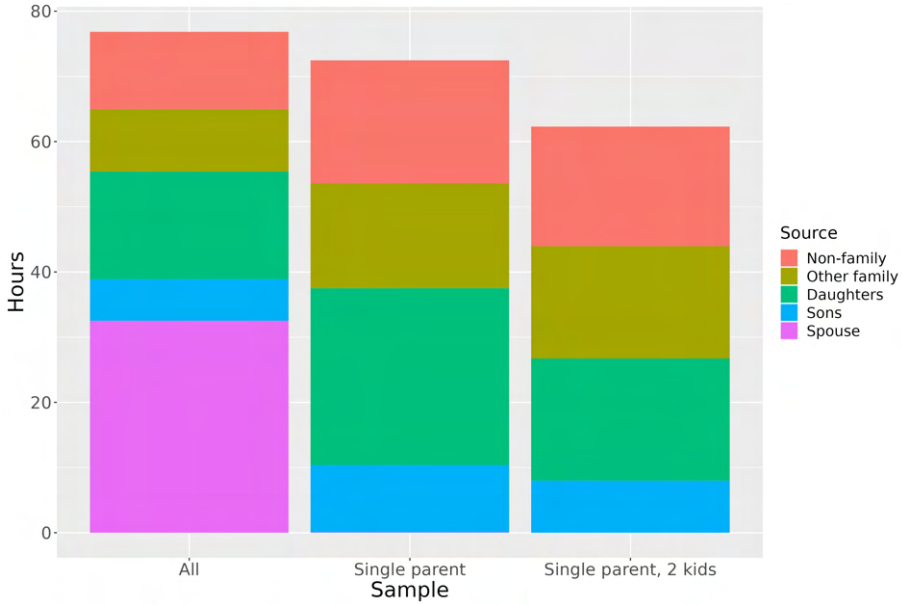
A.2 Hours of care received by care source

Figure 18 presents information on the sources of care to HRS respondents. Each bar shows mean hours of care received by given HRS respondent with care needs[58] by each of five possible sources: spouses, sons, daughters, other family members and non-family sources. I calculate this separately for the whole sample, the subsample of single respondents and the subsample of single respondents with only two children. Note that I do not condition on positive care provision here so the graph captures both the intensive and extensive margins of care provision. Also, employees of institutions are excluded from calculations of total hours of care receipt in the HRS, so the "Non-family" care does not include e.g. care provided by workers in a nursing home.

For the sample as a whole, spouses provided the biggest share of care. A given older person with care needs will receive around 33 hours of care per month from a spouse,

[58]I treat people as having care needs if they report difficulty with at least one Activity of Daily Living (ADL) or at least one Instrumental Activity of Daily Living (IADL).

Figure 18: Hours of care per month by care source



Notes: sick parents (ADLs or IADLs >0) only. "Non-family" care excludes care from employees of institutions. "Spouse" care excludes 1998 (Wave 4) because of a lack of separate recording of spouse hours.

compared to 6 hours from any sons they have, 17 hours from any daughters they have, 10 hours from other family members and 12 hours from non-family sources, adding up to a total of around 77 hours of care.

For the sample of single respondents only, it is notable that the average amount of care hours received stays approximately the same. In other words, all of the other sources of care increase their output to compensate for the lost care from the spouse. In particular, daughters are the biggest single source of care, and provide much more care than sons.

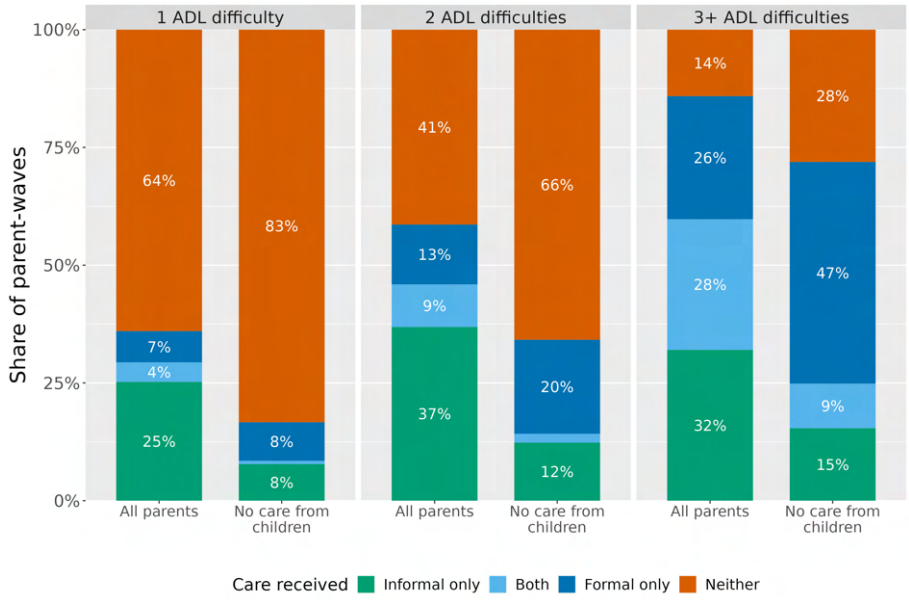
Comparing this to the sample of single respondents with two kids, the sum of care hours from children decreases somewhat, leading to a decrease in the total amount of care received, simply because there are fewer candidate children to provide the care. It is notable that the difference between daughters' care hours and sons' care hours is less pronounced when there are only two children. This is because in many two-child families there are no daughters, hence sons cannot leave it to their sisters to provide the care so must "step up" themselves.

Note that despite the appearance of the graph, the typical older person with care needs does not receive care from many sources in roughly equal amounts. Instead, most older people will have at most one primary caregiver. In the full sample, 76% of people receiving help with IADLs, ADLs or finances from at least one person receive help from exactly one person.

A.2.1 Care sources by children's care provision - single parents, two children

Figure 19 shows the proportion of single parents with two children receiving care, i.e. help with (I)ADLs, from informal sources, formal sources, both, or neither, for different health levels, according to whether their children provide any care. This figure does not account for hours, and instead displays means of dummies for if any care was received from the relevant source.

Figure 19: Care source by health and kid care provision



Notes: HRS waves 1998-2018. Single parents with exactly two child records who have difficulty with at least one Activity of Daily Living (ADL). Formal care is help from an organisation, an employee of an institution, a professional or a paid non-relative, or living in a nursing home. Informal care is help from relatives, paid or unpaid, and from unpaid non-relatives.

For low levels of ADL difficulties, most parents receive no care from any source, particularly if their children do not provide care. However, for parents with greater ADL difficulties, fewer end up receiving no care, and formal care becomes the dominant outside option to child-provided care, with 47% of parents with 3+ ADLs and no care from children receiving exclusively formal care, and a further 9% receiving both formal and informal care. Note that in the model, as presented in Table 15, parents with moderate (severe) care needs have on average difficulty with 1.98 (3.60) ADLs, so parents with care needs in the model correspond to the middle and right-hand panels of this Figure.

Children are by far the dominant providers of informal care to single parents with two children. Of all the unpaid informal care providers helping such parents with ADLs or IADLs from 2002-2018[59], about half were children (31% daughters and 21% sons). The next most important source was non-relatives (i.e. all those unpaid individuals listed as

[59]I calculate the shares from these waves only because in 1998 and 2000 helpers were identified differently, so the data on composition by family member type are not comparable.

"other individuals"), with 15% of all helpers, followed by grandchildren with 11% and children-in-law with 9%.

A.3 Extra descriptive regressions and graphs

A.3.1 Care patterns in two-child, single parent families

Table 10: Predictors of care provision and the amount of care provided - two-child, single parent families

	Dependent variable:		
	I(Hours help p.m. > 0)		log(Hours help p.m)
	(1)	(2)	(3)
Kid is daughter	0.009	0.029	−0.102
	(0.024)	(0.029)	(0.263)
Parent is mother	0.047**		−0.111
	(0.016)		(0.171)
Daughter x Mother	0.034	0.036	0.387
	(0.023)	(0.026)	(0.225)
Kid is youngest	0.010	−0.019	0.019
	(0.010)	(0.014)	(0.088)
Kid lives less than 10 miles away	0.180***	0.204***	0.655***
	(0.011)	(0.016)	(0.090)
Sibling is a sister	−0.024*		−0.036
	(0.010)		(0.086)
Kid in couple	−0.030	−0.013	−0.480**
	(0.018)	(0.023)	(0.166)
Daughter x Kid in couple	0.033	0.017	0.169
	(0.023)	(0.027)	(0.190)
Kid went to college	0.003	0.049**	−0.078
	(0.012)	(0.016)	(0.096)
Kid cared in prev. period	0.319***	0.389***	0.312***
	(0.023)	(0.032)	(0.090)
Observations	7279	6574	1956
Adjusted R2	0.287	0.427	0.147
Parent x wave FEs	N	Y	N
Mean dep. var.	0.203	0.203	2.993

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: estimation via OLS. Sample: children of sick (ADL or IADL >0) single parents with exactly two children. Regression weights are HRS respondent-level weights. In all regressions, other controls are dummies for number of ADL and IADL difficulties, education of parent, total number of grandkids of the kid, quadratics in age for kid and parent, a dummy for parent being White Caucasian, the inverse hyperbolic sine of parent wealth and homeownership status of kid. Standard errors clustered at the parent level.

A.3.2 Gender care gaps across family type

Table 11 presents measures of care provision by sons and daughters in families of different types.

Table 11: Ratios of daughters' care to sons' care by family type

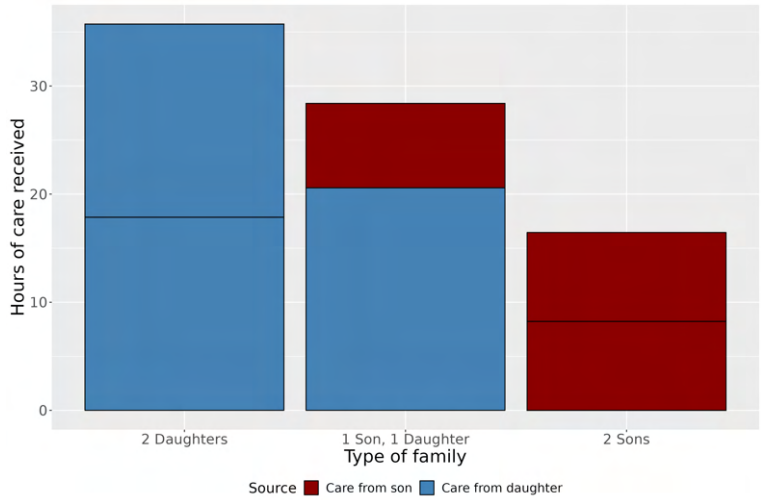
	I(Provides care)			Care hours		
	Son	Daughter	Ratio	Son	Daughter	Ratio
One-child family	0.16	0.25	1.57	10.00	23.15	2.32
Two-child daughter+son family	0.09	0.17	1.82	4.52	12.41	2.74
Two-child D+D family	-	0.15	-	-	10.36	-
Two-child S+S family	0.11	-	-	4.87	-	-

Notes: only sick parents (difficulties with at least one ADL or IADL) included. Means are weighted by HRS respondent-level weights. "I(Provides care)" is a dummy for whether the child provides any care, "Care hours" is a measure of mean care hours per month, averaging across all relevant children (including those providing no care). "Ratio" is daughters' mean outcome variable divided by sons' mean outcome variable - i.e. a measure of how much more care daughters provide than sons.

A.3.3 Care received by family composition

Figure 20 shows the mean child-provided care hours received by sick single parents of two children depending on the gender composition of their children. Parents with more daughters clearly receive higher overall child-provided care hours than parents with fewer daughters.

Figure 20: Mean kid care hours received by composition of kids



Notes: sick (ADLs or IADLs >0) single parents of two children only. Means presented are means of total kid care hours.

A.3.4 Caring and in-laws

Partnered children have two sets of parents who may need care, and a couple may only be able to live near only one of them. If couples choose between them according to how many other potential carers each set of parents has, care decisions are linked across nuclear families, which the model ignores. Here I provide some suggestive evidence on how important this omission is.

The HRS has no information on the families of respondents' children's spouses, so I instead use HRS respondents themselves, who report their own and their spouse's or partner's siblings. Table 12 regresses a dummy for whether a partnered respondent (R) lives within 10 miles of their mother, a proxy for being able to provide care to their mother, on whether R and R's spouse have a sister and on their numbers of siblings. The sample is partnered respondents whose mother and mother-in-law are both alive. I estimate the model for all respondents in Column 1 and for men only in Column 2 to show variation by gender.

Table 12: Caring and in-laws

	Dependent variable:	
	R lives within 10 miles of R's mother	
	(1)	(2)
R has sister	−0.045*	−0.072*
	(0.022)	(0.034)
R's total siblings	−0.004	−0.003
	(0.004)	(0.007)
R's spouse has sister	−0.008	0.021
	(0.022)	(0.032)
R's spouse total siblings	0.011*	0.012
	(0.004)	(0.006)
Observations	8416	4919
Adjusted R2	0.018	0.021
Mean dep. var.	0.347	0.363

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: estimation via OLS. Regression weights are HRS respondent-level weights. "R" stands for HRS Respondent. Column 1 is for all Respondents, Column 2 is for men only. Other controls are R's gender (Column 1 only), homeownership status, whether white Caucasian, education and a polynomial in age. Standard errors clustered at the respondent (R) level.

Considering Column 1, respondents are 4.5pp less likely to live within 10 miles of their

mother (mean 35%) if they have a sister, in line with the sister effect in Table 1. R is more likely to live near their mother if R's spouse has more siblings (1.1pp per sibling), which is consistent with the couple deciding to live near R's mother rather than the spouse's mother if they believe the latter will be well cared for anyway. However, the coefficient on R's spouse having a sister is close to zero and not significant, which is weak evidence against this explanation: if R's spouse has a sister, R's spouse's mother is presumably more likely to be cared for, so the couple should have more incentive, not less, to live near R's mother instead.

In Column 2, estimated on men only, the coefficient on R having a sister is larger, and much larger in magnitude than that on R's spouse having a sister. R's spouse having more siblings is still positively correlated with R living near their mother, but is no longer statistically significant.

As such, while links between the provision of care in different nuclear families through in-law relationships deserve further attention, and a spouse's number of siblings appears to have a small correlation with a respondent's location, this does not appear to be a first-order issue for the purposes of this paper, so I leave it aside.

B Children's location and parental health

To examine the effect of health shocks to parents on kids' location decisions I carry out an event study using the respondent-kid data in the HRS. I run the following regression:

$$Y_{ijt} = \alpha_{ij} + \kappa_t + \sum_{k \in -3,...2} \beta_k I(Period_{jt} = k) + X_{ijt}\gamma + \epsilon_{ijt} \tag{18}$$

In this regression, Y_{ijt} is the outcome variable, $Period_{jt}$ captures the period relative to the period where the parent j falls sick for the first time (i.e. in that period, $Period_{jt} = 0$). X_{ijt} is a set of time-varying controls[60]. I include child $\times$ parent fixed effects α_{ij} and wave fixed effects κ_t.

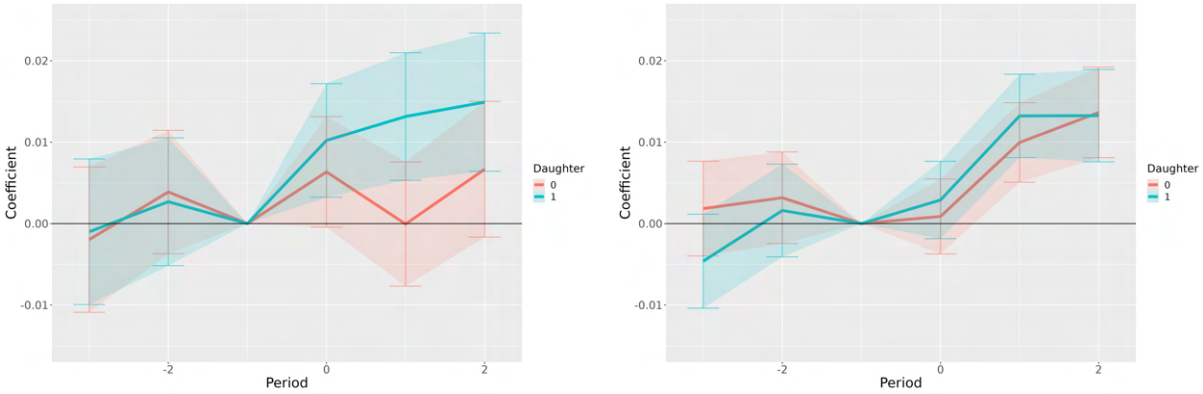
Figure 21 plots the results of this estimation. I estimate Equation 18 separately for sons and daughters. The height of each line at period k is the value of $\beta_k - \beta_{-1}$ estimated from the regressions.

In the left-hand panel the outcome variable is a dummy for whether child i lives within 10 miles of a parent j at t. In the right-hand panel, to avoid concerns about it being parents who choose which child to move close to, the outcome variable is a dummy for whether child i moves to live within 10 miles of their parent in t. Note that in my HRS descriptives sample in only 18.3% of cases where a parent lives more than 10 miles from a given child in $t-1$ and lives within 10 miles in t, it is the parent who has moved, not

[60]In this case, I control for polynomials in age of parent and child and couple status of parent and child.

the child. In other words, in roughly four fifths of cases it is the children who move close to the parent rather than the other way round[61].

Figure 21: Children's location before and after health shocks



(a) Outcome var: kid lives within 10 miles of parent

(b) Outcome var: kid moves to live within 10 miles of parent

Notes: estimation via OLS. Standard errors clustered at parent level. Confidence bands show 95% confidence intervals. Coefficients shown relative to β_{-1}.

The figures suggest that children are more likely to live closer to their parent after a parent falls ill, with this estimated association being larger for daughters. To put the magnitudes of the coefficients into context, in the HRS descriptives sample the proportion of kids who live within 10 miles of the relevant parent was 38.2% for sons and 38.9% for daughters.

Thus, there is some evidence to suggest that children do move back to live near their parents when their parents fall sick. This is particularly true of daughters.

C Other possible mechanisms of compensation

C.1 Compensation through rent-free accommodation

Some authors (e.g. Barczyk and Kredler (2018)) have argued that in addition to bequests children are compensated through rent-free accommodation when they are providing care to their parents.

HRS respondents are not directly asked whether their co-resident children pay any rent, but we can gain more insight by looking a generation upwards: while there are no data in the HRS on whether the children of survey respondents pay rent to the respondents when they live with the respondents, there is information on whether survey respondents themselves pay any rent when they live with their parents.

[61] If we consider only cases where the parent and child live within 10 miles of each other in t and not $t-1$, *and* parent has fallen sick for the first time in the data in t, then in 24.8% of these cases it is the parent who moves.

Table 13 presents these results. Column 1 shows proportions of tenure type for all respondents in the sample, where each observation is a respondent-wave combination. Column 2 shows tenure types for the subsample of respondents who provide more than 60 hours of care per month (I call these "intense caregivers", matching the threshold used to define *Intense care provision* in the structural model[62]). Column 3 restricts this further to those intense caregivers who co-reside with one of their parents.

Table 13: Tenure type by amount of care

	All respondents	"Intense caregivers"	IC+co-reside with parent
Own	0.79	0.80	0.72
Rent	0.17	0.14	0.13
Live rent-free	0.03	0.05	0.15
Other	0.01	0.01	0.01
Observations	154 506	987	285

Notes: data from HRS Waves 6 to 14 (2002-2018), using data for all respondents. "Intense caregivers"/"IC" refers to those respondents providing more than 60 hours of help to parents per month. All statistics are weighted means using HRS respondent-level weights.

The results in Column 2 suggest that only 5% of intense caregivers live rent-free. Even conditional on co-residence with a parent, only 15% of intense caregivers live rent-free. It is important to note that the HRS sample will not be fully representative of the sample of all carers so the average HRS respondent will be older, thus likely richer and less in need of rent-free accommodation, than the average carer. However, even with these caveats, the evidence on children receiving rent-free accommodation as compensation for care provision seems mixed.

This is important because if children do not receive rent-free accommodation in exchange for providing care, and do not receive substantively more by way of bequests, then this casts doubt on the suggestion that care hours are provided in exchange for a financial transfer of some kind.

C.2 Compensation through childcare

Finally, it might be argued that children provide care to their parents in exchange for childcare (i.e. care of the grandchildren) received from their parents. In other words, there is a dynamic contract between children and parents: in some period, the parent cares for their grandchildren, and in another period, the child provides care to the parent.

To investigate whether this channel of exchange is important I regress a dummy for whether a kid provides care to a parent, conditional on that parent being sick, on a set of

[62]HRS respondents are asked how many hours of personal care they provided to their parents since the last interview. I take this figure and divide it by 24 to arrive at the amount per month, given interviews are biennial.

explanatory variables, notably including a dummy for whether that kid is ever observed receiving childcare from that parent[63].

Table 14 reports the results. In Column 1, I regress the care dummy on the base set of explanatory variables. In Column 2, I add an interaction between the dummy for whether the kid is ever observed receiving childcare from the parent and the dummy for whether the kid lives within 10 miles of the parent. Columns 3 and 4 are the same as Columns 1 and 2 except parent-by-wave fixed effects are used.

Table 14: Childcare and caring for parents

	Dependent variable:			
	Kid provides care			
	(1)	(2)	(3)	(4)
Kid ever recd childcare	0.021***	0.037***	0.021***	0.038***
	(0.003)	(0.003)	(0.004)	(0.004)
Kid lives less than 10 miles away	0.141***	0.153***	0.161***	0.172***
	(0.003)	(0.004)	(0.004)	(0.005)
Kid ever recd childcare x Kid lives less than 10 miles away		−0.035***		−0.036***
		(0.006)		(0.007)
Observations	125 600	125 600	120 453	120 453
Adjusted R2	0.162	0.163	0.318	0.319
Parent x wave FEs	N	N	Y	Y
Mean dep. var.	0.104	0.104	0.100	0.100

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: estimation via OLS. Sample: children of sick parents (at least one ADL or IADL difficulty) in HRS waves 4 to 14 (1998-2018). Regression weights are HRS respondent-level weights. In all regressions, other controls are dummies for number of ADL and IADL difficulties, education of kid and parent, number of total kids of the parent, dummies for the kid's number of own children, whether the kid has a sister, whether the parent is in a couple, whether the kid is in a couple, gender of kid and parent and their interaction, the interaction of kid gender with kid couple status, quadratics in age for kid and parent, a dummy for the parent being White, the inverse hyperbolic sine of parent wealth, homeownership status of kid, and dummies for being the eldest and the youngest child. In Columns 3 and 4, parent x wave fixed effects absorb the parent-level controls. Standard errors clustered at the parent level.

The results in Columns 1 and 3 suggest a substantive association between whether a kid ever received help with childcare from a parent and whether the kid provides care to the parent: kids who received childcare are (using the results from Column 3) 2.1pp more likely to provide care, relative to a mean probability of providing care of 10.0%. However, the inclusion of the interaction term in Columns 2 and 4 shows that this association is driven by kids who live further than 10 miles from their parent in the current period. For

[63]In particular, a kid is treated as receiving childcare from a parent for a given wave if the parent says that they or their spouse spent at least 100 hours caring for their (great-)grandchildren via that child. Note that although the regression sample only includes sick parents, the "ever observed receiving childcare" variable is constructed from all waves for that kid, including waves when the parent is not sick.

those kids who live further than 10 miles, those who ever received childcare are (using the results from Column 4) 3.8pp more likely to provide care than those who did not. For those kids who live within 10 miles of their parents, the difference is only $3.8-3.6 = 0.2$pp. One explanation for this is that within the group of kids who live more than 10 miles from parents, those who received childcare from parents are more likely to live closer (e.g. 20 miles away) than those who did not. As such, while this basic analysis does not rule out a childcare-eldercare exchange channel, there seems to be little indication that kids provide care to parents in exchange for parents providing childcare for the kids.

D Model discussion

D.1 Move order specifications

Across various model specifications, I allow for four different move order specifications:

1. *Random sequential moves.* At the beginning of each period, the move order for that period is determined by an *iid* fair coin flip, so in any given period Child A and Child B are equally likely to be the first mover. This is the main move order assumption and is used for all three specifications in Tables 5 and 6.

2. *Fixed move order.* Child A (the elder child) always moves first and Child B second. This could be motivated by the elder child typically having left home first, and so having been the first to make location and work decisions.

3. *Deterministic move order.* Which child moves first in a given period is a deterministic function of the children's choices in the previous period. The first mover in t is determined by the following tiebreaking conditions, applied in order until one child is selected:

   a) if child i was the only child to provide care in $t-1$, they move first in t;

   b) otherwise, if child i was the only child living within 10 miles of the parent in $t-1$, they move first;

   c) otherwise, if child i was the only child not working FT in $t-1$, they move first;

   d) otherwise, the elder child moves first.

   This always selects a unique first mover. It could be motivated by the idea that the child in the better position to provide care (having cared in the previous period, living nearby, or not working FT) makes the initial caregiving decision, and the other child then responds.

4. *Simultaneous moves.* Both children move at the same time. This treats the two children symmetrically, but the model may then have multiple equilibria and cannot be solved by backwards induction.

As move order specification 1 represents a compromise between a priori neutrality between the two children in terms of move order while still allowing for a unique equilibrium and relatively straightforward calculation of counterfactuals, I use it as the main specification, but in Appendix G I present and discuss how estimation results vary under the alternative specifications.

D.2 Bequest specifications

Across various model specifications, I allow for four different bequest specifications:

1. *Equal split.* If the parent dies at the end of period t, their wealth is split equally between their two children, regardless of care effort.

2. *Proportional bequest (binary).* If the parent dies at the end of period t, and if exactly one of the two children provided care in period t, that child receives the entire bequest and the other receives none. If both children provided care, or neither did, the bequest is split equally.

3. *Proportional bequest (hours).* If the parent dies at the end of period t, then each child's share of the bequest is equal to their share of aggregate child-provided care hours in t. If no care was provided by children in t, the bequest is equally split.

4. *Redistributive split.* If the parent dies at the end of period t, then the bequest is split in such a way as to equalise as far as possible both children's remaining lifetime resources (i.e. their remaining lifetime equivalised income, plus any equivalised bequests - children's post-parental-death income is discussed below). In some cases, child i's remaining income is so far below child j's that child i receives all the bequest and child j receives none.

Bequest specifications 1, 2 and 3 represent different assumptions about the role of strategic bequests in the model. Bequest specification 4 is a more altruistic version of the bequest decision, where bequest share does not vary with care effort but instead varies with the child's alternative resources.

D.2.1 Children's income and consumption after parental death

To model children's benefit from bequests it is necessary to make some assumptions about how they behave after the death of the parent. I assume that children work FT until they reach 65, after which they retire. I assume both children die 10 years (5 periods)

after the period in which the parent would have reached their maximum age of 100[64]. To avoid tracking children's choices after the parent's death, their income in this period depends only on their age, gender, education and marital status at the parent's death (see Appendix E.4 for details).

As for consumption, I assume that after the parent's death children allocate their remaining resources, including any bequest, optimally across their remaining periods. This is a departure from the assumption in the rest of the model that children consume exactly their income every period, but avoids imposing the alternative, less realistic, assumption that children must consume their entire bequest within one period.

D.3 $\omega_{warm}(\cdot)$ in full

Written out, the index $X_t^i\gamma$ in Equation 12 is

$$\begin{aligned} X_t^i\gamma = \gamma_{far} far_t^i + \gamma_{son} son^i + \gamma_{dad} dad + \gamma_{dad\times son} dad \times son^i \\ + \gamma_{youngest} I(i=B) + \gamma_{youngest\times son} I(i=B) \times son^i \\ + \gamma_{col} col^i + \gamma_{married} married^i + \gamma_{married\times son} married^i \times son^i \\ + \gamma_{medicaid} medicaid_t + \gamma_{lab\times cg} L_t^i. \end{aligned} \tag{19}$$

Here, far_t^i is a dummy for whether child i lives far (i.e. more than 10 miles) from their parent at time t; son^i is a dummy for whether child i is a son; dad is a dummy for whether the parent is a father; $I(i=B)$ is a dummy for whether child i is the younger of the two siblings; col^i is a dummy for whether child i has a college education; $married^i$ is a dummy for whether child i is married or partnered; $medicaid_t$ is a dummy for whether the parent is Medicaid-eligible in period t, i.e. has wealth at or below the Medicaid asset floor $\underline{a}$; and L_t^i is child i's labour supply in period t, normalised relative to part-time hours (0 if not working, 1 if working part-time, and the ratio of full-time to part-time hours if working full-time). As intense care enters with weight κ, the ratio of intense to moderate care hours, every term in $X_t^i\gamma$ scales proportionately with care intensity, while γ_{start} does not.

D.4 Unobserved heterogeneity

Any dynamic model of this kind faces an important problem of unobserved heterogeneity (Aguirregabiria, Collard-Wexler, and Ryan 2021). The only unobserved heterogeneity in

[64]Ideally, children's decisions after the parent's death would be generated endogenously within the model, so that e.g. a child who received a very large bequest could retire early. The problem with this approach is that child behaviour after the parent's death is not observed in this dataset, because (barring HRS exit interviews) there is no information on children's subsequent behaviour. This means that empirical CCPs cannot be recovered and future behaviour cannot be projected through the forward simulation procedure.

the model takes the form of *iid* preference shocks, independent across agents and across time. In particular, the model does not allow for permanent unobserved heterogeneity, in the form of, for instance, different unobserved levels of affection in the relationship between a particular child and a particular parent. This could create problems because of correlations between this unobserved heterogeneity and the child's endogenous state variables at the time of making location, work and care choices in period t. For instance, children who feel more affection for their parents may a) be more likely to live near their parent and b) be more likely to provide care when their parent falls sick in period t.

I take two steps to limit this problem. First, the model conditions on a rich set of state variables, including the parent's health, wealth and urban status and the child's gender, education, marital status and birth order, as well as the child's previous care, location and work choices. This means that much of the relevant heterogeneity is already captured by observables. Second, to avoid an initial conditions problem in which children have already selected into caregiving before they are first observed, I drop any family in which the parent is sick in the first period observed and either child provided care in the previous period. Selection into caregiving in the estimation sample is therefore generated within the model.

Remaining unobserved heterogeneity would most likely bias two sets of estimates. The first is the cost of caring when living far: if children with a stronger attachment to their parent, and so a stronger preference for providing care, are more likely to live close to them anyway, then the estimated cost of providing care when living far may be too high. Similarly, because persistent preferences for providing care generate similar observable patterns to state dependence (Hiedemann, Sovinsky, and Stern 2018), the estimated fixed cost of starting to provide care, and hence the degree to which a first caregiver is "locked in", may also be overestimated. However, the estimated gender differences in preferences are likely less exposed, because systematic differences in, e.g., levels of affection by gender will be absorbed by the gender terms in the net warm glow function. Moreover, heterogeneity common to both siblings does not affect the comparison between sons and daughters within a family, and the reduced-form gender gap in Table 1 is present even with parent-by-wave fixed effects.

A possible extension of the current model to address these issues would be to allow for a finite number of unobserved types, which can be incorporated into CCP estimation (Arcidiacono and Miller 2011), at the cost of making the model computationally much more complex.

E Parameters calculated outside the model

E.1 Care choices, costs and hours

As set out in the main text, children recorded as providing fewer than 8 hours of care per month, or with no information on hours, are treated as providing no care; children providing between 8 and 60 hours a month are treated as providing moderate care; and children providing more than 60 hours a month are treated as providing intense care.

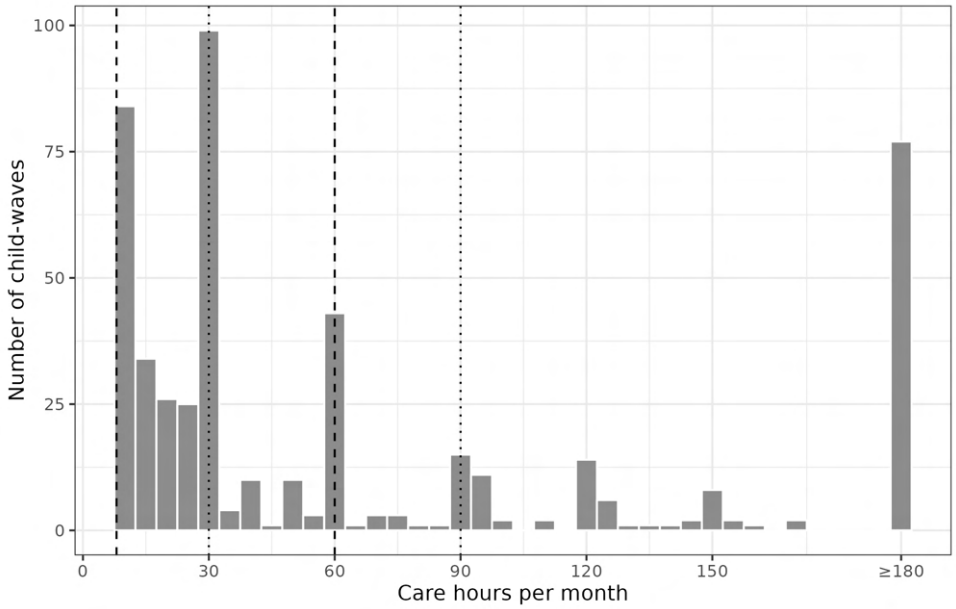
I assume that moderate care corresponds to 30 hours of care per month and intense care to 90 hours per month, i.e. roughly one and three hours of care a day, so that the ratio of intense to moderate care hours is $\kappa = 3$. It is the ratio κ which is relevant for the utility function rather than the raw hours figures. This choice of κ is motivated by balancing two competing considerations: basing care hours choices on values from the data while not allowing estimated warm glow preference parameters to be driven almost entirely by the demographics of intense caregivers, given that these parameters are scaled by κ for intense caregivers. Figure 22 presents the distribution of care hours in the data, for those children who provide at least 8 hours of care per month to parents with care needs. As discussed in the main text, there are large concentrations of care hours at multiples of 30, corresponding to respondents who measure care hours in terms of average hours a day, which is then subsequently multiplied by 30. In this way, 30 and 90 correspond to common hours choices, with 30 hours being the 35th-49th percentile and 90 hours being the 71st-74th. As a robustness check, I re-estimate the main specification with $\kappa = 4.8$, the ratio of the 75th to the 25th percentiles of the care hours distribution (96 and 20 hours respectively). Since κ also determines how much of a severely ill parent's care need moderate informal care covers, under this specification informal care offsets a smaller share of nursing home costs when the children provide only moderate care. The results are qualitatively similar and are presented in Appendix G.

For the interaction between care and work in $\omega_{warm}(\cdot)$ it is also necessary to impose a ratio of hours worked in full-time versus part-time work. I set this ratio to 2, so full-time workers work twice the hours of part-time workers.

Formal care costs come from the Genworth 2008 Cost of Care Survey (Genworth 2008), which reports costs for 90 regions: 40 metropolitan areas, which I treat as urban, and 50 rest-of-state or whole-state regions, which I treat as rural. I average across the regions in each group and convert to 2010 dollars using the CPI, so that formal care costs differ between parents living in urban and rural areas.

A parent with moderate care needs ($h_t = 1$) who receives no informal care pays for a home health aide (non-Medicare certified, licensed), at an average hourly rate of \$19.04 in rural and \$19.90 in urban areas. I assume the aide is hired for 30 hours per month, the same as moderate informal care, giving an annual cost of \$6.9k in rural and \$7.2k

Figure 22: Distribution of monthly care hours



Notes: children providing at least 8 hours of care per month to a parent with care needs, two-child families in the estimation sample. Hours above 180 per month are pooled in the final bar. Dashed lines: the 8- and 60-hour thresholds for moderate and intense care. Dotted lines: the assumed hours for moderate (30) and intense (90) care.

in urban areas. A parent with severe care needs ($h_t = 2$) who receives no informal care lives in a semi-private room in a nursing home, at an average annual cost of \$66.3k in rural and \$72.9k in urban areas. As a model period lasts two years, the cost per period is twice these annual figures. Section 3.2.4 sets out how these costs fall when the children provide some care.

E.2 Health and health transition probabilities

To estimate health transition probabilities I first must decide what conditions in the data match with belonging to the various health states in the model. I estimate care need in the data by a multi-step process:

1 - Using the estimation sample of two-child families, I regress whether a parent receives help with any ADL or IADL from any source – formal or informal – on a set of objective health measures in the HRS[65]. Although care receipt is endogenous, the existence of means-tested formal care through Medicaid suggests that care receipt from any source will be a reasonable indicator of actual care need.

2 - I use the regression to assign predicted care need to each parent in each wave, i.e.

[65]The explanatory variables are dummies for having no ADL difficulties and for having no IADL difficulties, and the numbers of ADL and IADL difficulties and their squares, each interacted with a dummy for memory disease and with a quadratic in age.

the probability of receiving any type of care that their objective health measures imply, according to the regression.

3 - I class as having moderate care needs ($h_t = 1$) all those between the 90th and 95th percentile of predicted care need among two-child family-waves and I class as having severe care needs ($h_t = 2$) all those above the 95th percentile.

Table 15 below presents statistics on objective health measures by assigned h_t. As expected, ADL difficulties, IADL difficulties and memory disease are very low for those with $h_t = 0$ and increase with h_t.

Table 15: Objective health statistics by model health state

	$h_t = 0$	$h_t = 1$	$h_t = 2$
# ADL difficulties	0.21	1.98	3.60
# IADL difficulties	0.14	2.39	3.83
Memory disease	0.01	0.14	0.38
Any care from a child	0.04	0.49	0.59
Intense care from a child	0.01	0.17	0.21

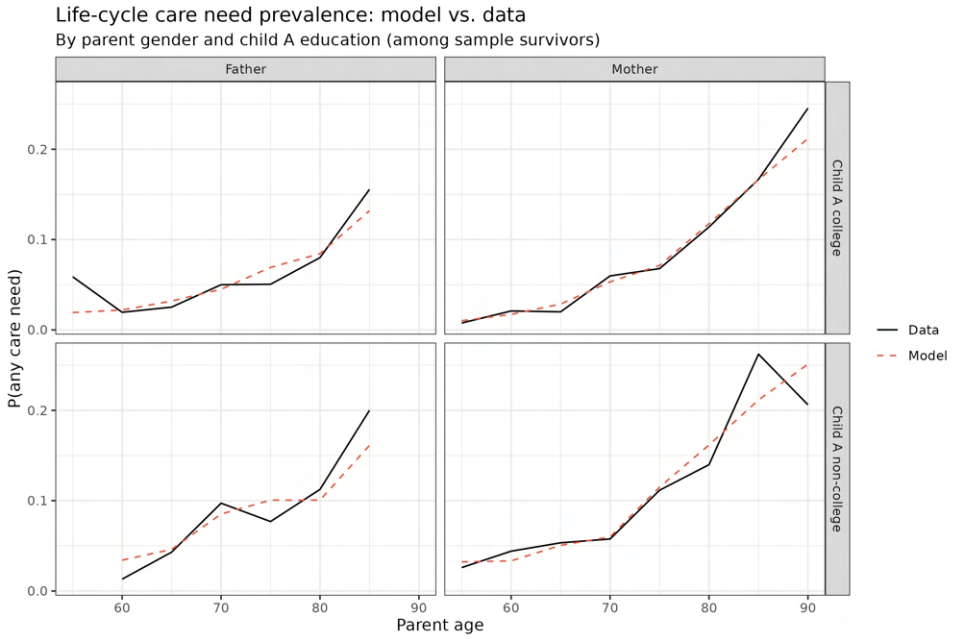
Notes: h_t=0, 1, 2 refer to no care needs, moderate care needs and severe care needs respectively. Each value in the table is the mean of the left hand variable for parents in the relevant health state in the estimation sample of two-child families. "Any care from a child" ("Intense care from a child") is the share of parent-waves in which at least one child reports providing at least 8 (more than 60) hours of care per month, before care provided to parents without care needs is set to zero for estimation.

Then, for the health transition probabilities themselves, I estimate separately for each current health state $h_t \in \{0, 1, 2\}$ a multinomial logit model of the parent's state two years later, which is either one of the three health states or death. The explanatory variables are whether any of the parent's children has a college education (a proxy for the parent's socioeconomic status, as discussed in Section 3) and the parent's gender, each interacted with a quadratic in the parent's age. The next-wave health state is classified in the same way as above for every parent interviewed in the next wave, including those whose family is no longer in the estimation sample; a parent who is not interviewed is classed as having died if their recorded year of death falls within the two years, and is otherwise excluded. I exclude the final wave (2018), for which the next state cannot be observed.

Figure 23 below presents the share of parents with care needs in the next wave, among those who survive to it, in the data and predicted by the model, separately for fathers and mothers and by whether any of their children went to college. Figure 24 shows transition probabilities between health states in the data and in the model, by the parent's age group, for each current health state (columns) and each next-wave state including death (rows).

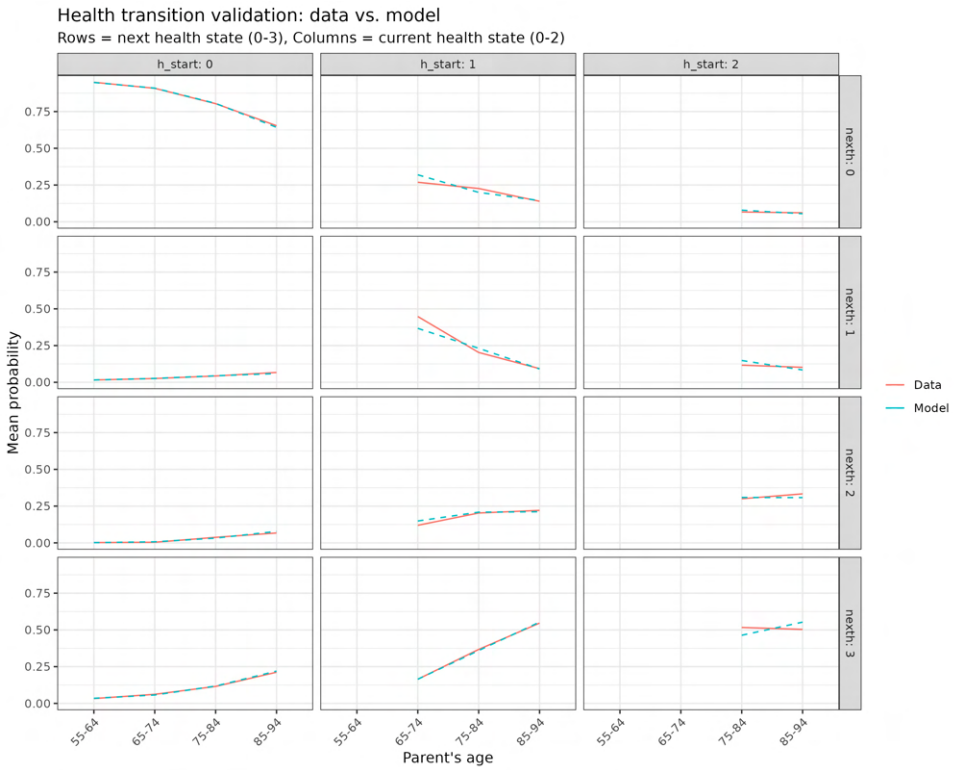
As the transition probabilities in Figure 24 suggest, care needs are persistent but there is some degree of recovery possible: of those parents with $h_t = 2$, 8% of them have no

Figure 23: Care need: data versus model



Notes: cells with data observations < 30 are not displayed.

Figure 24: Transitions between health states: data versus model



Notes: cells with data observations < 30 are not displayed.

care needs in $t+1$ and 10% have $h_{t+1} = 1$. In the data, for those who transition to being fully healthy, then by construction caregiving drops to 0; for those who transition to $h_{t+1} = 1$, of all the children who were providing care in t, 53% still provide care in $t+1$.

E.3 Income while the parent is alive

I estimate Equation 14 using the PSID Family File (*PSID* 2024), waves 1999-2019. The sample comprises household heads and spouses aged 21 to 65 with positive family income. Because the regression includes full-time work in the previous period, I also require each person to be observed in the previous wave. Regressions are weighted using PSID family weights.

The dependent variable is log total family income in the previous calendar year, deflated to 2010 dollars using the CPI, multiplied by two (since a model period lasts two years) and divided by the equivalence scale $\sqrt{1 + couple_t^i}$, where $couple_t^i$ indicates a spouse or partner present in that wave. The equivalence scale counts adults only, since the model does not track children's own children, and is the same scale used when splitting bequests. An individual is classed as not working if they worked at most 400 hours in the previous year, as working part-time if they worked between 401 and 1,499 hours, and as working full-time if they worked at least 1,500 hours. College education is defined as more than 12 years of schooling.

Table 16 presents the results.

E.4 Income after the parent's death

As discussed in Appendix D.2.1, after the parent dies children are assumed to work full-time until age 65 and then retire. For this period I estimate simpler income functions, separately for men and women, of the form

$$\log y_t^i = \zeta_0^{g,r} + \zeta_1^{g,r} age_t^i + \zeta_2^{g,r} (age_t^i)^2 + \zeta_3^{g,r} col^i + \zeta_4^{g,r} couple^i + \nu_t^i, \qquad r = \begin{cases} W, & \text{if } age_t^i \leq 65 \\ R, & \text{if } age_t^i \geq 66, \end{cases} \tag{20}$$

using the same PSID data, weights and variable definitions as above. Pre-retirement income is estimated on full-time workers aged 21 to 65 and retirement income on non-workers aged 66 and over with positive income. Table 17 presents the results.

E.5 Other parameters

To estimate the mean and variance of the wealth shock I consider two-year changes in the parent's net wealth (in 2010 dollars) in my estimation sample of two-child families,

Table 16: Estimated income process

	Dependent variable:	
	Log equivalised income	
	Men	Women
Constant	8.798***	8.179***
	(0.455)	(0.263)
Age	0.025	0.079***
	(0.021)	(0.012)
Age sq.	0.000	−0.001***
	(0.000)	(0.000)
College	0.576***	0.610***
	(0.061)	(0.042)
Couple	0.389***	0.734***
	(0.025)	(0.022)
Works (PT or FT)	0.170	0.785**
	(0.457)	(0.266)
Works × Age	0.035	−0.013
	(0.021)	(0.012)
Works × Age sq.	−0.001**	0.000
	(0.000)	(0.000)
Works × College	−0.073	−0.154***
	(0.061)	(0.043)
Works × FT in $t-1$	0.247***	0.133***
	(0.017)	(0.011)
Full-time	0.387***	0.227***
	(0.021)	(0.013)
Observations	44 102	51 283
Adjusted R2	0.359	0.348

* p < 0.05, ** p < 0.01, *** p < 0.001

Notes: estimation via OLS weighted by PSID family weights, on PSID Family File data, 1999–2019, heads and spouses aged 21-65. Standard errors clustered at the individual level.

Table 17: Estimated income process after the parent's death

	Dependent variable:			
	Log equivalised income			
	Men, working	Women, working	Men, retired	Women, retired
Constant	9.430***	9.455***	12.265***	9.082***
	(0.093)	(0.088)	(1.427)	(1.737)
Age	0.073***	0.069***	−0.033	0.046
	(0.005)	(0.004)	(0.037)	(0.045)
Age sq.	−0.001***	−0.001***	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
College	0.498***	0.428***	0.544***	0.463***
	(0.017)	(0.017)	(0.043)	(0.039)
Couple	0.268***	0.462***	0.212***	0.526***
	(0.025)	(0.018)	(0.051)	(0.041)
Observations	38 023	31 810	5070	7272
Adjusted R2	0.201	0.263	0.153	0.194

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: estimation via OLS weighted by PSID family weights, on PSID Family File data, 1999–2019. Working: full-time workers (at least 1,500 hours) aged 21–65. Retired: non-workers aged 66 and over with positive income. Standard errors clustered at the individual level.

for parents who are healthy in both waves and hence face no long-term care costs. I winsorise the wealth changes at the 10th and 90th percentiles to reduce the impact of measurement error. The mean and standard deviation of the winsorised wealth changes are −\$1.2k and \$76.3k respectively.

I set the Medicaid asset floor $\underline{a}$ to \$2,000, the resource limit for a single person for Medicaid eligibility in most states (Johns Hopkins University 2026).

I set the age gap between the parent and the elder child to 24 years and between the parent and the younger child to 29 years, the mean age differences in the estimation sample. The model starts when the parent is 55 and the parent's maximum age is 100[66].

Finally, I assume the discount factor β is equal to 0.93. Note that each period lasts 2 years so this corresponds to an annual discount factor of $\sqrt{0.93} = 0.964$.

[66]Few parents in their 50s have care needs, using the definition of care needs in the model (i.e. being above the 90th percentile for predicted probability of care receipt, as a function of observable health states): only 3.4% of parents aged 55-59 in the estimation sample meet this criterion. In principle, the model could be solved from a later starting age (for instance, the 25th percentile age of a parent first developing care needs in the data is 72). The reason I have kept the model age relatively early is that, as shown in Figure 23, it is from these early ages that care need starts to increase from near 0, so I am able to capture the full extent of this rise in care needs over the age distribution. Moreover, by modelling labour and location choices for periods before a parent falls ill, I am able to capture any pre-selection into, for instance, living near a parent or being outside of the labour force.

F Pseudo maximum likelihood estimation

F.1 Likelihood

Consider family n observed in period τ in state $s_{n\tau}$, with observed choices $d^A_{n\tau}$ and $d^B_{n\tau}$. Let $P^i_1(d^i \mid s)$ denote the probability that child i makes choice d^i as the first mover, and $P^i_2(d^i \mid s, d^j)$ the probability that they make choice d^i as the second mover, having observed their sibling's choice d^j, so that $I^i_t = s_t$ for the first mover and $I^i_t = (s_t, d^j_t)$ for the second mover, as in the definition of equilibrium in the main text. These are the choice probabilities implied by the equilibrium strategies σ^* at parameters θ. Let $D_{n\tau} = 1$ if child A is the first mover in family n in period τ, and $D_{n\tau} = 0$ if child B is. The likelihood contribution of family n in period τ is

$$\ell_{n\tau}(\theta) = \left[P^A_1(d^A_{n\tau} \mid s_{n\tau})\, P^B_2(d^B_{n\tau} \mid s_{n\tau}, d^A_{n\tau})\right]^{D_{n\tau}} \left[P^B_1(d^B_{n\tau} \mid s_{n\tau})\, P^A_2(d^A_{n\tau} \mid s_{n\tau}, d^B_{n\tau})\right]^{1-D_{n\tau}}, \tag{21}$$

and the likelihood is

$$L^*(\theta) = \prod_{n=1}^{N} \prod_{\tau=1}^{T_n} \ell_{n\tau}(\theta), \tag{22}$$

where T_n is the number of periods for which family n is observed.

Maximising Equation 22 directly raises two problems. First, it requires the full solution of the model, which would be prohibitively costly. Second, in the main specification (under which move order is determined by an *iid* fair coin flip every period) $D_{n\tau}$ is not observed.

With regard to this first problem, I avoid the full solution of the model by instead maximising the pseudo likelihood (Aguirregabiria and Mira 2007) of observed choices, in which the equilibrium choice probabilities are replaced by policy iteration operators evaluated at estimates $\hat{\sigma}$ of the children's strategies (Ko 2022)[67]. For the first mover, the policy iteration operator is given by:

$$\Psi^i_1(d^i \mid s, \hat{\sigma}, \theta) = \frac{\exp\left(W^i_1(s, d^i, \hat{\sigma})\, \theta\right)}{\sum_{d' \in F^i(s)} \exp\left(W^i_1(s, d', \hat{\sigma})\, \theta\right)}, \tag{23}$$

and $\Psi^i_2(d^i \mid s, d^j, \hat{\sigma}, \theta)$ is defined in the same way for the second mover, with $W^i_2(s, d^j, d^i, \hat{\sigma})$ in place of $W^i_1(s, d^i, \hat{\sigma})$. Here $F^i(s)$ is child i's choice set in state s, and $W(\cdot)\theta$ is the choice-specific value function implied by behaving according to $\hat{\sigma}$ in all future periods, which is linear in θ. Because $W(\cdot)$ does not depend on θ, within each pass it is computed only once, by forward simulation, as described in the main text. In other words, the policy iteration operator is an approximation of the true optimal decision rules which updates the policy

[67] In the first of the two estimation passes, $\hat{\sigma}$ is estimated directly from the data; in the second, it is given by the choice probabilities obtained by solving the model at the parameters estimated in the first pass.

function estimates $\hat{\sigma}$ by using these policy function estimates to generate implied choice-specific value functions and then using these choice-specific value functions to recover approximate optimal decision rules in each state.

With regard to the second problem, it is important to recognise that observed choices are informative about move order. For instance, if first movers tend to leave care to their siblings, then a family-period in which one child provides care and the other does not is more likely to be one in which the non-carer moved first.

I make use of this insight in the estimation by employing the EM algorithm[68]. Estimation proceeds in two passes, described in the next subsection, and within each pass the EM algorithm is run to convergence, holding $\hat{\sigma}$ and hence $W(\cdot)$ fixed. In particular, having constructed the choice-specific value functions and hence policy iteration operators by forward simulation, and starting from $q_{n\tau}^{(0)} = 1/2$, where $q_{n\tau}$ is the posterior probability that child A was the first mover in family n in period τ, at EM step m I use the following procedure:

M-step: choose $\theta^{(m)}$ to maximise

$$\sum_{n,\tau} \left[q_{n\tau}^{(m-1)} \left(\ln \Psi_1^A + \ln \Psi_2^B \right) + \left(1 - q_{n\tau}^{(m-1)} \right) \left(\ln \Psi_1^B + \ln \Psi_2^A \right) \right], \tag{24}$$

where each Ψ is evaluated at the observed choices of family n in period τ;

E-step: update the posterior probability that child A moved first,

$$q_{n\tau}^{(m)} = \frac{\Psi_1^A \Psi_2^B}{\Psi_1^A \Psi_2^B + \Psi_1^B \Psi_2^A}, \tag{25}$$

with each Ψ evaluated at $\theta^{(m)}$.

I repeat these steps until the largest change in any $q_{n\tau}$ is below 10^{-4}. The prior probability of $1/2$ on each move order cancels from the posterior.

F.2 First and second pass

I estimate the model parameters θ by carrying out two estimation passes. In the first pass, $\hat{\sigma}_1$ is estimated directly from the data: I estimate multinomial logit models of children's observed choices on the state variables (for the first mover) and on the state variables

[68] Holding the first-stage choice probabilities fixed, this is the standard EM algorithm applied to the pseudo likelihood. Arcidiacono and Miller (2011), who set out the EM algorithm for conditional choice probability estimators, also update the choice probabilities at every step, using the posterior weights. This is feasible in their setting because the value terms are comparatively cheap to recompute. Here $W(\cdot)$ is obtained by forward simulation, so I hold the choice probabilities fixed within the EM algorithm and update them once, between the first and second passes described below, using the model-implied choice probabilities obtained by solving the estimated model by backwards induction.

and the sibling's care choice (for the second mover). When the parent has no care needs, neither child can provide care, and a single model of each child's choices on the state variables is used for both roles. Since move order is not observed, each observation is given equal weight in the first- and second-mover roles. As a result, these first-stage estimates mix first- and second-mover behaviour. They yield a set of choice-specific value functions by forward simulation. I then maximise the pseudo likelihood using the EM algorithm outlined above, yielding a first estimate $\hat{\theta}_1$.

In the second pass, I solve the model by backwards induction at $\hat{\theta}_1$ to obtain model-consistent first- and second-mover choice probabilities $\hat{\sigma}_2$. I use these choice probabilities in place of the first-stage estimates to recompute $W(\cdot)$ by forward simulation. I then run the EM algorithm again, starting from $q_{n\tau}^{(0)} = 1/2$. The resulting $\hat{\theta}_2$ is my final estimate. As discussed in the main text, in principle this procedure could be repeated further, but I stop after the second pass for computational reasons.

Under the alternative specifications in which move order is fixed or is a deterministic function of past choices and birth order, move order is observed in the data, so $\ell_{n\tau}$ contains only the term for the observed order. Both passes are carried out as above, but without the EM step. Under simultaneous moves, $\ell_{n\tau} = P^A(d_{n\tau}^A \mid s_{n\tau})P^B(d_{n\tau}^B \mid s_{n\tau})$. As the model cannot then be solved by backwards induction, I use the first-pass estimate.

I compute standard errors by bootstrapping. Each replicate draws families from the estimation sample with replacement and repeats the whole procedure, from the first-stage estimates through both passes.

G Alternative specification results

In this Appendix I present estimation results from alternative specifications of the model.

G.1 Alternative specifications by move order

Tables 18 and 19 below present results for the care and non-care parameters respectively under alternative assumptions about move order. In particular, Column 1 reproduces the main results from Table 5; Column 2 presents estimated parameters under the assumption that move order is fixed, with the elder child (Child A) always going first; Column 3 presents estimated parameters under the assumption of a deterministic move order rule depending on children's past decisions; and Column 4 presents estimated parameters under the assumption of simultaneous moves[69].

The fixed move order and deterministic move order produce results that are essentially the same as the main estimation using random move order, for both the care parameters and the non-care parameters.

[69]For more detail on these move order specifications, see Appendix D.1.

Table 18: Estimation results - care parameters - alternative move order specifications

	(1) Main (equal beq.)	(2) Fixed MO	(3) Deterministic MO	(4) Simultaneous MO
Public good care params				
α_{h1} - P in moderate bad health	0.421	0.407	0.423	2.002
	(0.293)	(0.298)	(0.278)	(0.906)
α_{h2} - P in severe bad health	0.456	0.457	0.348	1.419
	(0.231)	(0.221)	(0.194)	(0.790)
Net warm glow care params				
$\delta_{h1,mod}$ - P moderate bad health, K moderate care	-0.198	-0.185	-0.189	-1.319
	(0.240)	(0.231)	(0.229)	(0.766)
$\delta_{h1,int}$ - P moderate bad health, K intense care	-0.064	-0.046	-0.052	-0.957
	(0.286)	(0.278)	(0.280)	(0.788)
$\delta_{h2,mod}$ - P severe bad health, K moderate care	-0.033	-0.032	0.040	-0.590
	(0.192)	(0.182)	(0.175)	(0.612)
$\delta_{h2,int}$ - P severe bad health, K intense care	-0.013	-0.014	0.064	-0.241
	(0.238)	(0.231)	(0.223)	(0.594)
γ_{son} - K is a son	-0.031	-0.029	-0.050	0.044
	(0.079)	(0.080)	(0.082)	(0.085)
γ_{start} - K just starting caring	-1.037	-1.038	-1.059	-0.920
	(0.128)	(0.127)	(0.125)	(0.180)
γ_{far} - K currently lives far	-0.647	-0.648	-0.646	-0.565
	(0.075)	(0.075)	(0.074)	(0.084)
$\gamma_{youngest}$ - K youngest sibling	0.038	0.030	0.030	0.013
	(0.050)	(0.053)	(0.052)	(0.056)
$\gamma_{youngest \times son}$ - K youngest & K son	-0.035	-0.038	-0.026	-0.023
	(0.089)	(0.090)	(0.089)	(0.103)
γ_{dad} - P is father	-0.228	-0.230	-0.228	-0.301
	(0.107)	(0.108)	(0.106)	(0.118)
$\gamma_{dad \times son}$ - P father & K is son	-0.029	-0.026	-0.017	-0.010
	(0.131)	(0.132)	(0.132)	(0.135)
γ_{col} - K has a college education	0.075	0.077	0.078	-0.009
	(0.058)	(0.057)	(0.059)	(0.074)
$\gamma_{medicaid}$ - P at/below Medicaid asset floor	-0.048	-0.046	-0.043	-0.145
	(0.052)	(0.051)	(0.052)	(0.065)
$\gamma_{married}$ - K is married	-0.119	-0.116	-0.117	-0.091
	(0.074)	(0.072)	(0.073)	(0.077)
$\gamma_{married \times son}$ - K married & K is son	-0.222	-0.226	-0.211	-0.257
	(0.070)	(0.072)	(0.071)	(0.049)
$\gamma_{lab \times cg}$ - labour hours × care hours	-0.151	-0.151	-0.150	-0.188
	(0.033)	(0.033)	(0.033)	(0.037)
Obj. function	2.4071	2.4071	2.4072	2.4990

Notes: "P" refers to the parent and "K" refers to the kid. Estimation via maximisation of the pseudo likelihood. Standard errors calculated using 20 bootstrap replications. "MO" stands for move order.

Table 19: Estimation results - other parameters - alternative move order specifications

	(1) Main (equal beq.)	(2) Fixed MO	(3) Deterministic MO	(4) Simultaneous MO
Consumption and labour				
θ_c - weight on consumption for K	2.083	2.083	2.082	1.094
	(0.090)	(0.089)	(0.090)	(0.092)
θ_{PT} - weight on PT work dummy for K	-1.851	-1.851	-1.850	-1.483
	(0.046)	(0.046)	(0.046)	(0.055)
θ_{FT} - weight on FT work dummy for K	-0.627	-0.626	-0.626	-0.122
	(0.087)	(0.086)	(0.086)	(0.073)
Location choice params				
ϕ_{far} - K currently lives far	-0.009	-0.009	-0.009	-0.025
	(0.014)	(0.014)	(0.013)	(0.016)
ϕ_{col} - K lives far, K college educ.	0.103	0.103	0.103	0.100
	(0.023)	(0.023)	(0.023)	(0.026)
ϕ_{son} - K lives far, K is son	0.042	0.041	0.042	0.037
	(0.016)	(0.016)	(0.016)	(0.014)
ϕ_{urban} - K lives far, P is urban	0.001	0.001	0.002	-0.005
	(0.020)	(0.020)	(0.020)	(0.021)
$\phi_{urban \times col}$ - K lives far, K college educ., P is urban	-0.036	-0.036	-0.036	-0.024
	(0.029)	(0.029)	(0.029)	(0.034)
ϕ_{move} - K moved location	-1.814	-1.814	-1.814	-1.796
	(0.023)	(0.023)	(0.023)	(0.025)
Obj. function	2.4071	2.4071	2.4072	2.4990

Notes: "P" refers to the parent and "K" refers to the kid. Estimation via maximisation of the pseudo likelihood. Standard errors calculated using 20 bootstrap replications. "MO" stands for move order.

The main outlier is the simultaneous-move specification. For the care parameters, the estimated public good care parameters are much larger, with the corresponding net warm glow parameters more negative. Also, the estimated public good parameter for severe care needs is smaller than that for moderate care needs, though the two are not statistically significantly different from each other. For the non-care parameters, the notable differences are that the coefficient on consumption is much smaller and correspondingly[70] the disutility of part-time and full-time work is much smaller too.

There are two key differences between the simultaneous-move and sequential-move specifications that likely drive these results. The first relates to the identifying variation used. Under the sequential-move specification, the public good parameters are identified by changes in child i's care behaviour given child j's observed choice, in cases where i is the second mover, and by changes in child i's care behaviour given their expectation of how child j will respond to i's own choice, whenever i is the first mover. In particular, if - other things being equal - i responds to j providing care by providing less care themselves, this points to a greater role for the public good motive.

In the simultaneous-move case, there is no conditioning on the sibling's behaviour, so the public good motive to provide care is identified solely from variation in i's beliefs about j's probability of providing care (given by j's choice probabilities). If i is more likely to provide care when j's demographics or state variables predict j providing less care (for instance, j is a brother, or j lives far away), then the public good motivation is greater.

This difference is important because in the sequential specification, the public good parameters must fit how second movers respond to their sibling's realised choice, observed directly in the data. In the simultaneous specification this discipline is absent: conditional on the state, the two children's choices are independent, so any (unmodelled) tendency for a child's care to depend on who their sibling is, rather than on what their sibling does, can only be captured by the public good term[71]. To the extent that sibling characteristics shift j's predicted probability of caring only modestly, fitting these patterns requires larger public good parameters, which are also much less precisely estimated.

The second difference between the simultaneous and sequential results is that, because the sequential model has a unique equilibrium and is solvable by backwards induction, it is possible to re-estimate the model in a second pass using model-generated choice probabilities as inputs, as described in Appendix F. The simultaneous model cannot be

[70] The estimated parameters move together in this way because the less an agent values consumption, the less incentive they have (other things being equal) to work, so the lower the disutility of work has to be to compensate.

[71] As discussed in Section 4.6, the model fit results in Figure 2 suggest that even controlling for sibling care effort, the sibling's characteristics still correlate with the reference child's care effort. These residual correlations could be explained by, for example, norms about who should be providing care, independently of who actually is providing care. In a simultaneous-move specification they could only be matched by a larger public good care parameter, even if they are not driven by the public good motive.

solved by backwards induction, so I do not do this, and instead present only first-pass estimates here. The large differences in the consumption and labour parameters, which have little direct connection to move order, likely reflect the fact that the simultaneous model is not re-estimated in a second pass with model-consistent choice probabilities. The objective values are not directly comparable, as the simultaneous estimates are first-pass only.

G.2 Alternative specifications by preference structure

Tables 20 and 21 present results for alternative specifications imposing different assumptions about preferences from the main specification. In Column 2, I present results for a specification where the public good utility terms are set equal to 0. In Column 3, the specification sets all child-gender-dependent utility terms to 0, both in the net warm glow from care and in the preference for living far from the parent. In Column 4, the specification sets all $h = 2$ utility terms (i.e. the health-dependent public good and net warm glow utility terms) equal to their $h = 1$ counterparts, thus removing any direct utility dependence on the severity of care needs.

The main specification approximately[72] nests these three models, and the model fit, as captured by the objective function at the solution, is worse in each case. The loss of fit is largest when gender differences in preferences are removed, smaller when the public good terms are removed, and smallest when the health-dependent terms are collapsed[73].

Beyond this, differences between the specifications are limited. Eliminating the public good utility parameters in Column 2 forces the net warm glow parameters to be positive to keep caregiving rates approximately constant. In Column 3, eliminating gender differences in preferences means that, for instance, $\gamma_{married}$ becomes more negative, to capture the greater disutility of caring for married sons, who are now not differentiated from married daughters. Unsurprisingly, in Column 4, collapsing the health categories means that the public good care parameter is estimated with more precision. The remaining non-care parameters are almost identical across columns, given that, apart from the location term in Column 3, the preference restrictions only directly affect the care decision.

[72]The nesting is only approximate because each final set of estimates derives from two sequential estimation passes, and each specification is re-estimated in the second pass using its own model-generated choice probabilities. The estimated parameters therefore do not multiply exactly the same value functions across specifications, which is what would be required for the nesting to be exact.

[73]Treating the differences in the objective function as likelihood ratio tests gives approximate p-values of 0.02 for the removal of the public good terms, below 0.001 for the removal of gender differences and 0.45 for the removal of health differences. These are only indicative: the objective is a pseudo likelihood based on estimated choice probabilities and the nesting is only approximate, so the test statistics do not follow the standard χ^2 distribution exactly.

Table 20: Estimation results - care parameters - alternative preference specifications

	(1) Main (equal beq.)	(2) No PG	(3) No gender diff.	(4) No health diff.
Public good care params				
α_{h1} - P in moderate bad health	0.421	-	0.476	0.408
	(0.293)		(0.293)	(0.199)
α_{h2} - P in severe bad health	0.456	-	0.461	$= h_1$
	(0.231)		(0.239)	
Net warm glow care params				
$\delta_{h1,mod}$ - P moderate bad health, K moderate care	-0.198	0.124	-0.217	-0.076
	(0.240)	(0.131)	(0.246)	(0.154)
$\delta_{h1,int}$ - P moderate bad health, K intense care	-0.064	0.231	-0.041	-0.011
	(0.286)	(0.251)	(0.236)	(0.219)
$\delta_{h2,mod}$ - P severe bad health, K moderate care	-0.033	0.273	-0.039	$= h_1$
	(0.192)	(0.106)	(0.198)	
$\delta_{h2,int}$ - P severe bad health, K intense care	-0.013	0.264	-0.007	$= h_1$
	(0.238)	(0.203)	(0.217)	
γ_{son} - K is a son	-0.031	-0.043	-	-0.031
	(0.079)	(0.076)		(0.076)
γ_{start} - K just starting caring	-1.037	-1.073	-1.064	-1.073
	(0.128)	(0.137)	(0.124)	(0.141)
γ_{far} - K currently lives far	-0.647	-0.653	-0.652	-0.646
	(0.075)	(0.076)	(0.074)	(0.076)
$\gamma_{youngest}$ - K youngest sibling	0.038	0.042	0.036	0.037
	(0.050)	(0.050)	(0.048)	(0.050)
$\gamma_{youngest \times son}$ - K youngest & K son	-0.035	-0.032	-	-0.032
	(0.089)	(0.092)		(0.091)
γ_{dad} - P is father	-0.228	-0.201	-0.240	-0.227
	(0.107)	(0.098)	(0.093)	(0.106)
$\gamma_{dad \times son}$ - P father & K is son	-0.029	-0.032	-	-0.024
	(0.131)	(0.130)		(0.129)
γ_{col} - K has a college education	0.075	0.085	0.078	0.078
	(0.058)	(0.058)	(0.056)	(0.057)
$\gamma_{medicaid}$ - P at/below Medicaid asset floor	-0.048	-0.032	-0.053	-0.044
	(0.052)	(0.050)	(0.049)	(0.051)
$\gamma_{married}$ - K is married	-0.119	-0.116	-0.209	-0.115
	(0.074)	(0.072)	(0.068)	(0.075)
$\gamma_{married \times son}$ - K married & K is son	-0.222	-0.222	-	-0.219
	(0.070)	(0.066)		(0.069)
$\gamma_{lab \times cg}$ - labour hours × care hours	-0.151	-0.150	-0.169	-0.152
	(0.033)	(0.033)	(0.032)	(0.033)
Obj. function	2.4071	2.4075	2.4089	2.4072

Notes: "P" refers to the parent and "K" refers to the kid. Estimation via maximisation of the pseudo likelihood. Standard errors calculated using 20 bootstrap replications.

Table 21: Estimation results - other parameters - alternative preference specifications

	(1) Main (equal beq.)	(2) No PG	(3) No gender diff.	(4) No health diff.
Consumption and labour				
θ_c - weight on consumption for K	2.083	2.084	2.085	2.084
	(0.090)	(0.090)	(0.090)	(0.090)
θ_{PT} - weight on PT work dummy for K	-1.851	-1.851	-1.851	-1.851
	(0.046)	(0.046)	(0.046)	(0.046)
θ_{FT} - weight on FT work dummy for K	-0.627	-0.627	-0.627	-0.627
	(0.087)	(0.087)	(0.087)	(0.087)
Location choice params				
ϕ_{far} - K currently lives far	-0.009	-0.009	0.012	-0.009
	(0.014)	(0.014)	(0.016)	(0.014)
ϕ_{col} - K lives far, K college educ.	0.103	0.103	0.095	0.103
	(0.023)	(0.023)	(0.024)	(0.023)
ϕ_{son} - K lives far, K is son	0.042	0.042	-	0.042
	(0.016)	(0.016)		(0.016)
ϕ_{urban} - K lives far, P is urban	0.001	0.002	-0.000	0.001
	(0.020)	(0.020)	(0.020)	(0.020)
$\phi_{urban \times col}$ - K lives far, K college educ., P is urban	-0.036	-0.036	-0.033	-0.036
	(0.029)	(0.029)	(0.029)	(0.029)
ϕ_{move} - K moved location	-1.814	-1.814	-1.815	-1.814
	(0.023)	(0.023)	(0.023)	(0.023)
Obj. function	2.4071	2.4075	2.4089	2.4072

Notes: "P" refers to the parent and "K" refers to the kid. Estimation via maximisation of the pseudo likelihood. Standard errors calculated using 20 bootstrap replications.

G.3 Other alternative specifications

The final two specifications that I estimate are i) a specification under which parents split their bequest in such a way as to, as far as possible, equalise the sum of post-death lifetime resources between their children, and ii) a specification under which $\kappa = 4.8$ rather than $\kappa = 3$ as in the main specification. Tables 22 and 23 present the results, with Column 1 reproducing the main specification.

For the redistributive bequest specification in Column 2, changes are very limited. This is unsurprising because under this bequest specification there is still no strategic bequest mechanism: all that has changed relative to the main specification is that the bequest is split unevenly, but not in a way that the children can manipulate, so children just have slightly different incentives to keep the overall bequest as high as possible depending on what share of that bequest they are guaranteed to receive. In Column 3, as most of the γ terms are now scaled by 4.8 rather than 3 for intense care, these parameters are estimated to be around $3/4.8 = 62.5\%$ of their original size. The model fit is worse, with the increase in the (minimised) objective function greater than in the case of removing public good care parameters in Table 20. Across both columns, non-care parameters hardly change.

H Additional model fit and counterfactual figures

H.1 Model fit and counterfactual figures in hours

In the main text I presented model fit figures for labour supply and care provision using binary indicators of both choices, i.e. a dummy for whether the child is working FT and a dummy for whether the child is providing any care. Here, I present the corresponding figures using hours. For the labour supply figures, PT work hours are normalised to 1 and FT work hours to 2. For the care figures, moderate care hours are normalised to 1 and intense care hours to 3.

Table 22: Estimation results - care parameters - alternative bequest and care-hours specifications

	(1) Main (equal beq.)	(2) Redistributive beq.	(3) $\kappa = 4.8$
Public good care params			
α_{h1} - P in moderate bad health	0.421	0.414	0.430
	(0.293)	(0.292)	(0.295)
α_{h2} - P in severe bad health	0.456	0.462	0.447
	(0.231)	(0.228)	(0.235)
Net warm glow care params			
$\delta_{h1,mod}$ - P moderate bad health, K moderate care	-0.198	-0.231	-0.354
	(0.240)	(0.238)	(0.242)
$\delta_{h1,int}$ - P moderate bad health, K intense care	-0.064	-0.167	0.066
	(0.286)	(0.283)	(0.277)
$\delta_{h2,mod}$ - P severe bad health, K moderate care	-0.033	-0.080	-0.152
	(0.192)	(0.194)	(0.197)
$\delta_{h2,int}$ - P severe bad health, K intense care	-0.013	-0.154	0.119
	(0.238)	(0.229)	(0.246)
γ_{son} - K is a son	-0.031	-0.024	0.009
	(0.079)	(0.082)	(0.049)
γ_{start} - K just starting caring	-1.037	-1.035	-1.106
	(0.128)	(0.127)	(0.123)
γ_{far} - K currently lives far	-0.647	-0.647	-0.423
	(0.075)	(0.075)	(0.064)
$\gamma_{youngest}$ - K youngest sibling	0.038	0.060	0.023
	(0.050)	(0.050)	(0.032)
$\gamma_{youngest \times son}$ - K youngest & K son	-0.035	-0.041	-0.039
	(0.089)	(0.092)	(0.063)
γ_{dad} - P is father	-0.228	-0.234	-0.140
	(0.107)	(0.109)	(0.069)
$\gamma_{dad \times son}$ - P father & K is son	-0.029	-0.023	-0.058
	(0.131)	(0.133)	(0.091)
γ_{col} - K has a college education	0.075	0.095	0.035
	(0.058)	(0.057)	(0.038)
$\gamma_{medicaid}$ - P at/below Medicaid asset floor	-0.048	-0.038	-0.014
	(0.052)	(0.052)	(0.033)
$\gamma_{married}$ - K is married	-0.119	-0.106	-0.083
	(0.074)	(0.072)	(0.044)
$\gamma_{married \times son}$ - K married & K is son	-0.222	-0.224	-0.172
	(0.070)	(0.072)	(0.045)
$\gamma_{lab \times cg}$ - labour hours $\times$ care hours	-0.151	-0.150	-0.105
	(0.033)	(0.033)	(0.022)
Obj. function	2.4071	2.4076	2.4079

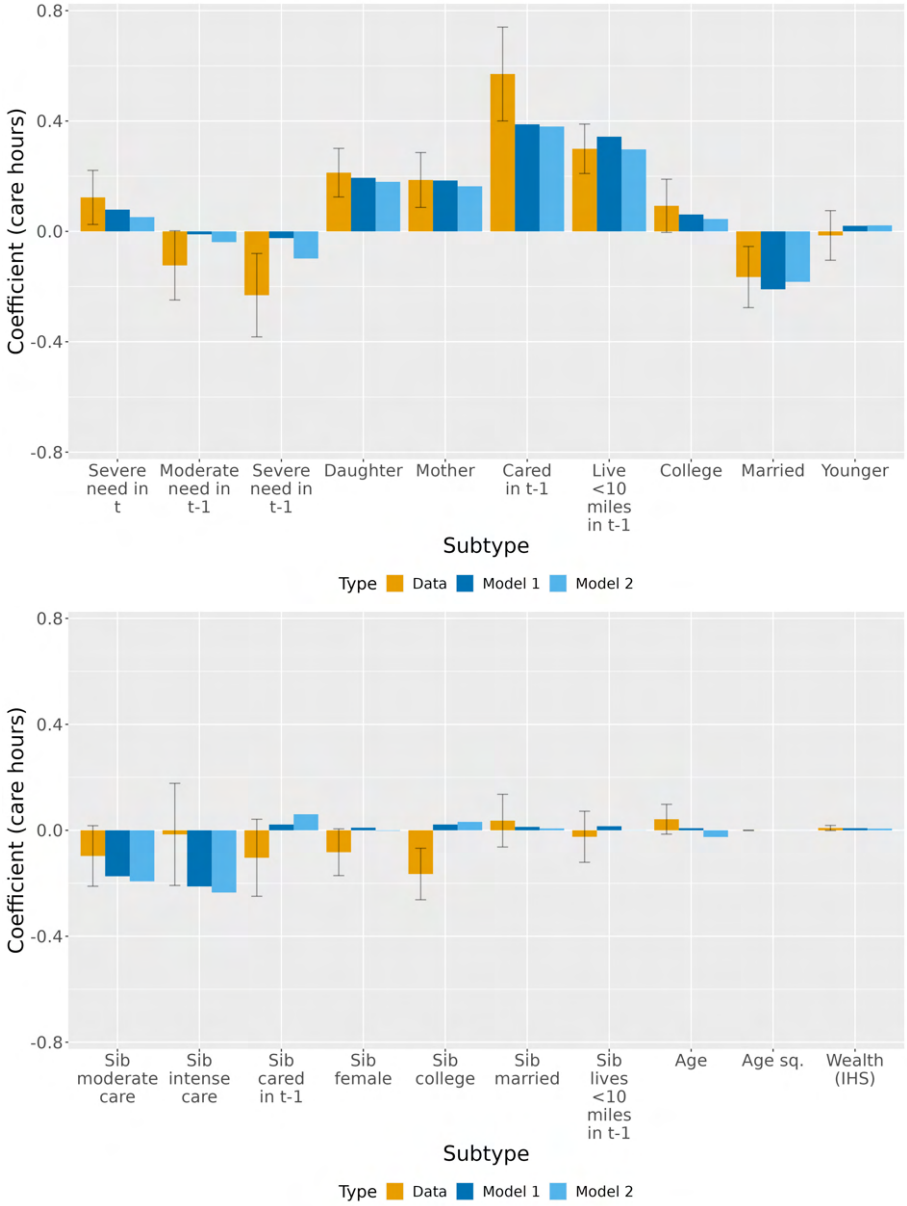
Notes: "P" refers to the parent and "K" refers to the kid. Estimation via maximisation of the pseudo likelihood. Standard errors calculated using 20 bootstrap replications.

Table 23: Estimation results - other parameters - alternative bequest and care-hours specifications

	(1) Main (equal beq.)	(2) Redistributive beq.	(3) $\kappa = 4.8$
Consumption and labour			
θ_c - weight on consumption for K	2.083	2.070	2.082
	(0.090)	(0.089)	(0.089)
θ_{PT} - weight on PT work dummy for K	-1.851	-1.845	-1.851
	(0.046)	(0.046)	(0.046)
θ_{FT} - weight on FT work dummy for K	-0.627	-0.616	-0.627
	(0.087)	(0.086)	(0.087)
Location choice params			
ϕ_{far} - K currently lives far	-0.009	-0.008	-0.012
	(0.014)	(0.014)	(0.014)
ϕ_{col} - K lives far, K college educ.	0.103	0.102	0.102
	(0.023)	(0.023)	(0.023)
ϕ_{son} - K lives far, K is son	0.042	0.041	0.043
	(0.016)	(0.016)	(0.016)
ϕ_{urban} - K lives far, P is urban	0.001	0.002	0.001
	(0.020)	(0.020)	(0.020)
$\phi_{urban \times col}$ - K lives far, K college educ., P is urban	-0.036	-0.036	-0.036
	(0.029)	(0.029)	(0.029)
ϕ_{move} - K moved location	-1.814	-1.814	-1.813
	(0.023)	(0.023)	(0.023)
Obj. function	2.4071	2.4076	2.4079

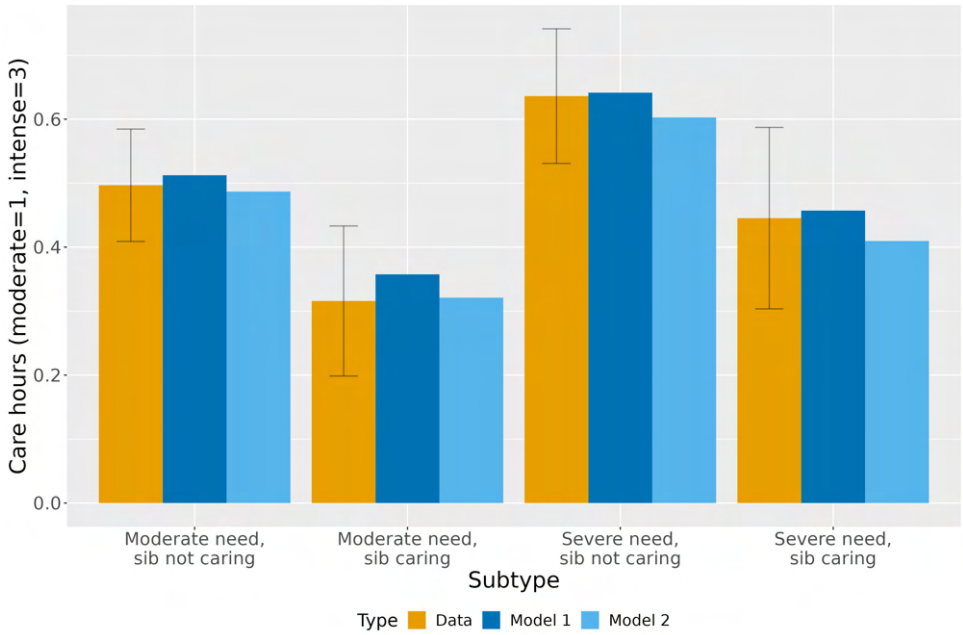
Notes: "P" refers to the parent and "K" refers to the kid. Estimation via maximisation of the pseudo likelihood. Standard errors calculated using 20 bootstrap replications.

Figure 25: Results from regression of care hours on observables - data versus model



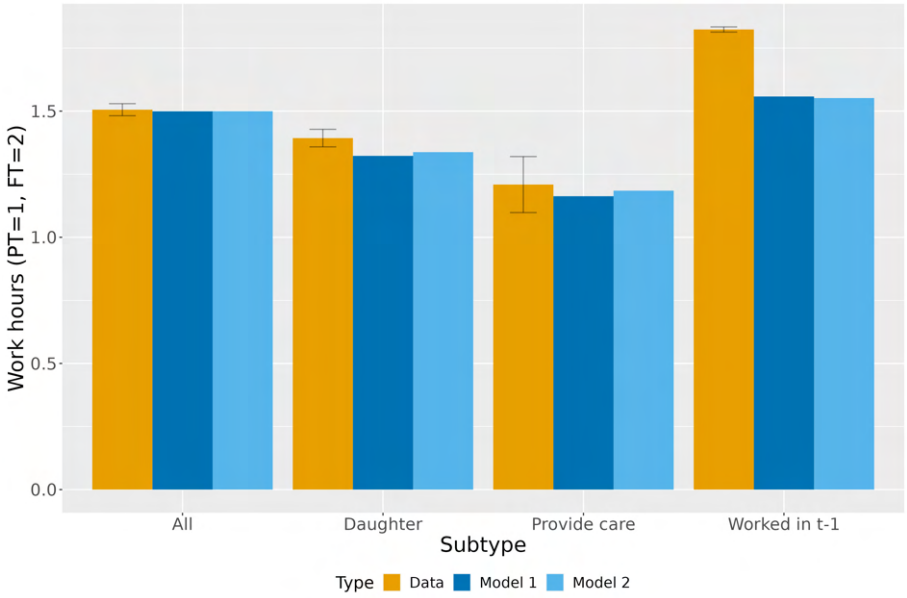
Notes: estimation via OLS; the dependent variable is normalised care hours (moderate care = 1, intense care = 3). Standard errors clustered at parent level.

Figure 26: Care hours by sibling care rate and parental health



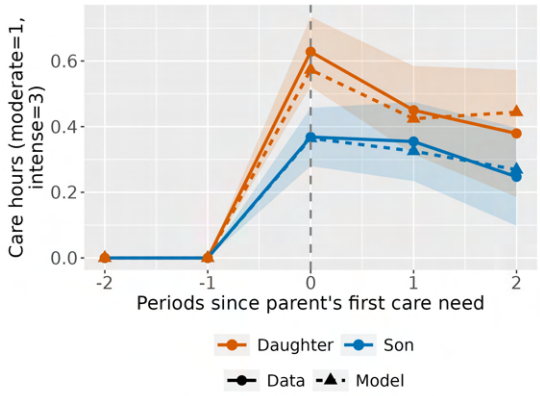
Notes: care hours normalised so that moderate care = 1 and intense care = 3. Standard errors clustered at parent level.

Figure 27: Hours worked - data versus model

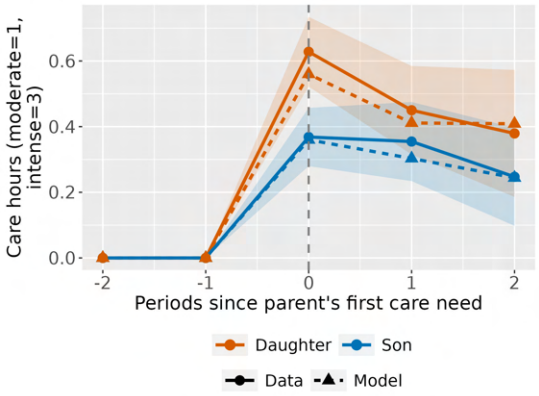


Notes: work hours normalised so that PT work = 1 and FT work = 2. Standard errors clustered at parent level.

Figure 28: Care hours around the onset of care needs



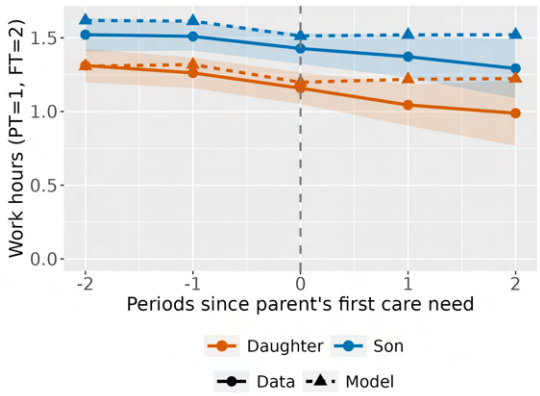
(a) Data vs Model 1



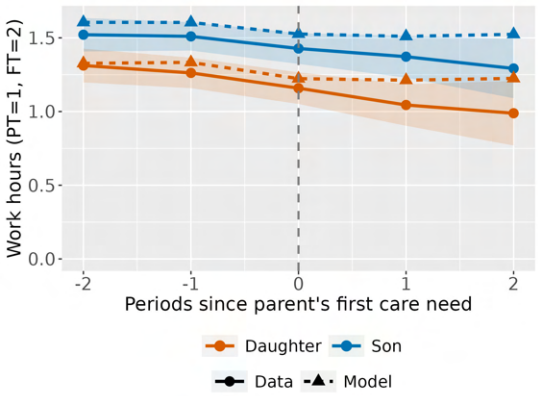
(b) Data vs Model 2

Notes: figures show mean normalised care hours (moderate care = 1, intense care = 3) in periods around the event of the parent developing care needs ($h \geq 1$) for the first time. Shaded bands are 95% confidence intervals.

Figure 29: Work hours around the onset of care needs



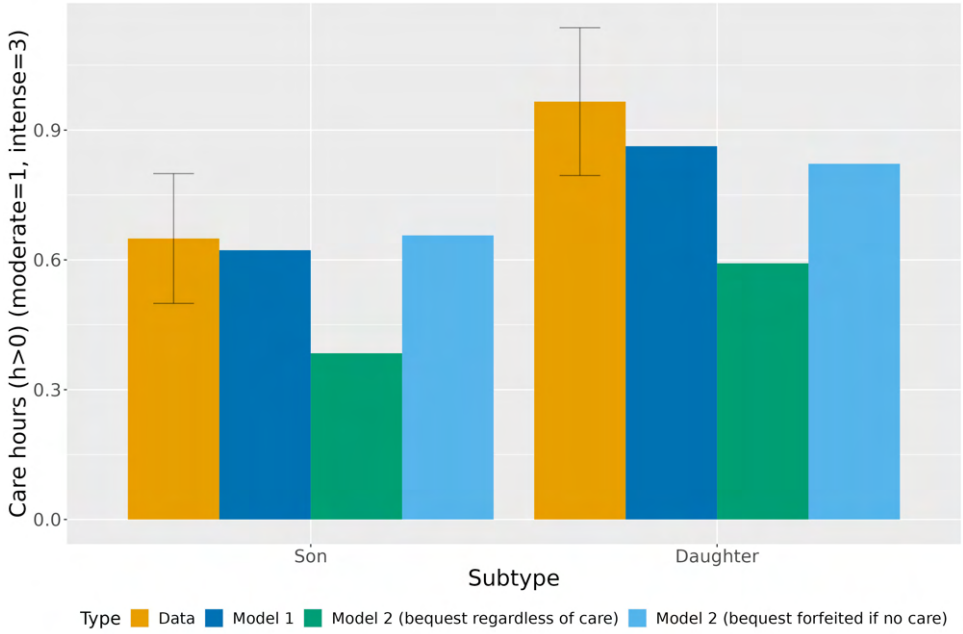
(a) Data vs Model 1



(b) Data vs Model 2

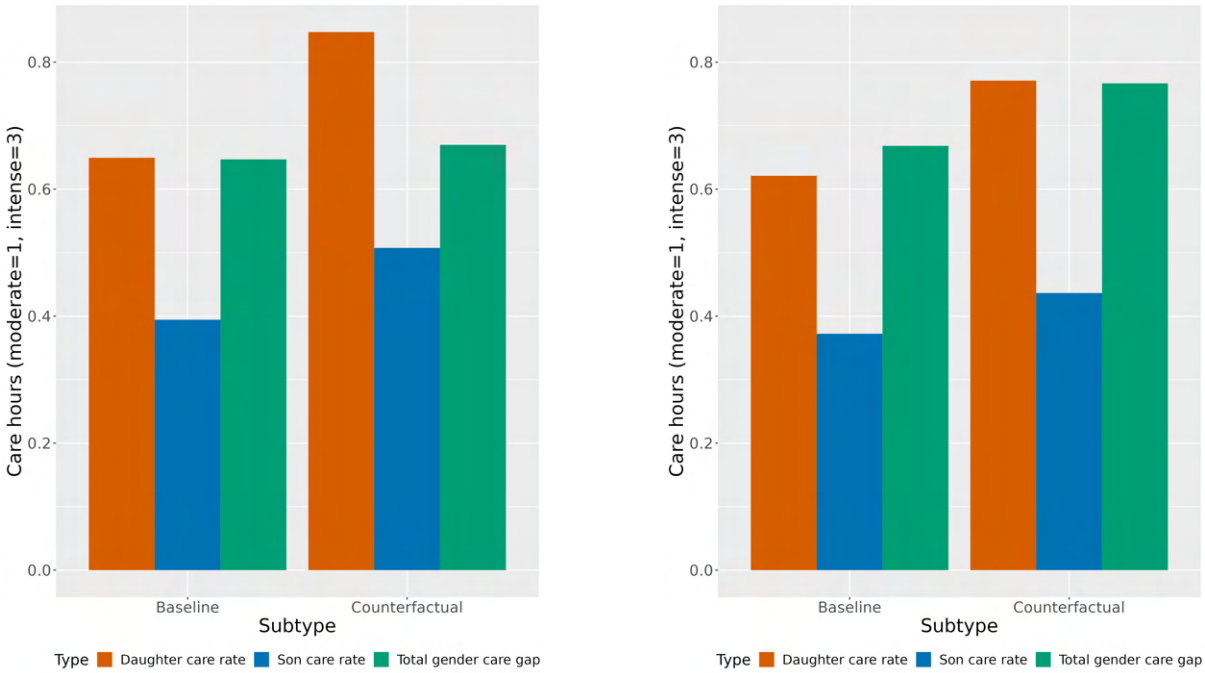
Notes: figures show mean normalised work hours (PT work = 1, FT work = 2) in periods around the event of the parent developing care needs ($h \geq 1$) for the first time. Shaded bands are 95% confidence intervals.

Figure 30: One-child families: care hours in data versus estimated models



Notes: care hours normalised so that moderate care = 1 and intense care = 3. Standard errors clustered at parent level.

Figure 31: Gender care gap in hours: cooperation, son-daughter families only



(a) Model 1, baseline vs counterfactual

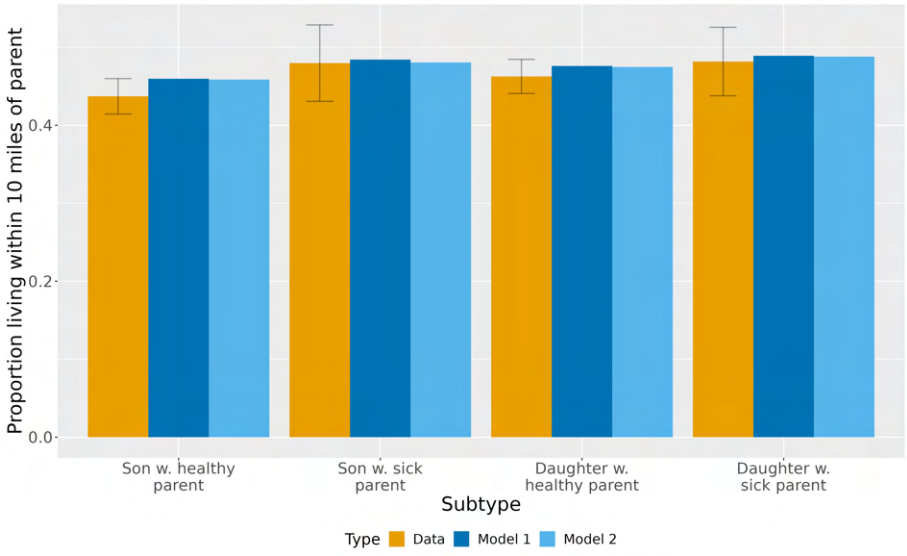
(b) Model 2, baseline vs counterfactual

Notes: "care hours" is mean normalised care hours (moderate care = 1, intense care = 3) of children with a sick parent. "Total gender care gap" is how much larger, in percentage terms, daughters' mean care hours are than sons'.

H.2 Model fit - location

The main text discusses the model's ability to match the change in child location by gender when a parent becomes sick. Figure 32 shows the proportion of children living within 10 miles of their parent, by child gender and parental health status. The fit is good, though there is not much variation to match in the data.

Figure 32: Proportion of children living near parents - data versus model



Notes: standard errors clustered at parent level.